

Consensus Recommendations for the Diagnostic Investigation of Dogs with Suspected Glomerular Disease

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Background: The International Renal Interest Society (IRIS) offers guidelines for chronic kidney disease and acute kidney injury. As dogs with glomerular disease may present differently and require different treatment than those with whole nephron or tubular disease, the IRIS Canine Glomerulonephritis (GN) Study Group was convened to formulate guidelines for these cases. The Diagnosis Subgroup was asked to make recommendations for diagnostic evaluation of such cases.

Objective: To seek consensus among renal specialists for the evaluation of dogs with proteinuria because of suspected glomerular disease.

Methods: After reviewing the literature, subgroup members discussed and wrote the draft paper and recommendations, which members of the IRIS Canine GN Study Group voted upon by electronic secret ballot, with comments noted. Consensus was declared if votes showed strong or general agreement from 85% of the respondents.

Results: Diagnostic tests were categorized as essential, recommended, or potentially helpful, with prioritization dependent on case characteristics, eg, for cases with uncomplicated proteinuria versus complicated with hypoalbuminemia, azotemia, or both. Consensus was reached with 86–100% agreement on all questions posed. All cases should have basic examinations including blood pressure measurement, blood, and urine testing, and a search for infectious diseases relevant to their environs. The majority ranked imaging (chest radiographs, abdominal ultrasonogram) and renal biopsy procured and interpreted by experienced personnel as essential evaluations in complicated cases, but a few respondents deemed these to be essential in uncomplicated cases as well.

Conclusions and Clinical Importance: Strong consensus about recommendations for diagnostic evaluation of dogs with suspected glomerular protein loss was attained. These guidelines help clinicians characterize disease processes for more informed therapeutic decision-making.

Key words: Canine; Glomerulonephritis; Glomerulopathy; Protein-losing nephropathy; Proteinuria.

We define primary glomerulopathies as those nephropathies that arise with involvement of glomeruli in the processes that initiate renal injury, regardless of their pathologic mechanism(s), whereas secondary glomerular changes occur after renal tubular or whole nephron damage. In contrast, in human medicine, the term “primary” glomerular disease connotes disease which primarily involves the kidney and “secondary” glomerular disease indicates a disease in which kidney involvement is part of a systemic disorder.^{1–3}

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Abbreviations:

ANA	antinuclear antibody
CKD	chronic kidney disease
GN	glomerulonephritis
GWAS	genome-wide association study
Hx	history
IF	immunofluorescence microscopy
IRIS	International Renal Interest Society
LM	light microscopy
PE	physical examination
PLN	protein-losing nephropathy
SDS-PAGE	sodium dodecyl sulfate-agarose gel electrophoresis
TEM	transmission electron microscopy
UPC	urine protein/creatinine ratio
USG	urine specific gravity

der.^{1–3} In any case, diagnostic evaluation of suspected glomerular diseases in dogs is intended to expose the clinical pathological features that will serve as the basis for categorizing dogs with glomerular diseases in ways that will facilitate the formulation of appropriate therapeutic recommendations for these patients.

Proteinuria is an acknowledged hallmark of glomerular lesions in dogs, but proteinuria is not always attributable to renal, much less glomerular, disease. Consequently, when proteinuria is detected, it first must be properly assessed to determine its potential clinical significance. This is a crucial step, the description of which is provided elsewhere and is beyond the scope of this discussion.^{4–6} However, proteinuria requires investigation of 3 key elements (localization, persistence, and magnitude), and often it is the

identification of persistent renal proteinuria (see Additional Comments below) by this investigation that causes a suspicion of glomerular disease to arise. Nonetheless, persistent renal proteinuria is not always a marker of glomerular disease. Renal proteinuria, albeit of low magnitude, can be caused by tubular lesions alone, and even when proteinuria is largely of glomerular origin, the glomerular lesions are not necessarily caused by an intrinsic glomerular disease. As repeatedly shown by the results of studies of canine chronic kidney disease (CKD) initiated by partial renal ablation (ie, the remnant kidney model), maladaptive compensatory responses of the surviving nephrons in dogs with CKD typically induce subsequent glomerular changes manifested in part by proteinuria of mild-to-moderate magnitude.⁷⁻⁹

In addition, the clinicopathologic manifestations of various glomerular diseases in dogs are diverse and each ranges over a wide spectrum of severity. Consequently, recommendation of a single set of guidelines for diagnostic evaluation of all dogs with suspected glomerular disease, regardless of its evident manifestations, is not appropriate. That is, what is reasonable to recommend for investigation of glomerular disease in an otherwise healthy dog with proteinuria alone differs from what should be done for a dog with proteinuria, azotemia, hypoalbuminemia, hypertension, and edematous extremities (for example).

Table 1. Descriptions of tiers recommended for grouping dogs with glomerular diseases.^a

Tier I—Persistent renal proteinuria without hypoalbuminemia or azotemia

Tier I-A—Persistent subclinical renal proteinuria that is not accompanied by any discernible renal-related signs or sequelae

Tier I-B—Persistent renal proteinuria with hypertension as the only discernible renal-related sign or sequella, with or without evidence of target organ damage

Tier II—Renal proteinuria associated with hypoalbuminemia, but not azotemia

Tier II-A—Persistent renal proteinuria with hypoalbuminemia, with or without any of its associated complications or sequella (mainly edema and thromboembolic events), but without hypertension or azotemia

Tier II-B—Persistent renal proteinuria with hypoalbuminemia, with or without any of its associated complications/sequella (mainly edema and thromboembolic events), plus hypertension (with or without evidence of target organ damage), but without azotemia

Tier III—Renal proteinuria associated with renal azotemia

Tier III-A—Renal proteinuria with renal azotemia, but not hypertension or hypoalbuminemia

Tier III-B—Renal proteinuria with renal azotemia and hypertension (with or without evidence of target organ damage), but without hypoalbuminemia

Tier III-C—Renal proteinuria with renal azotemia and hypoalbuminemia with or without any of its associated complications/sequella (mainly edema and thromboembolic events), which often (but not always) are accompanied by hypertension (with or without evidence of target organ damage)

^aApply tier classification criteria after initial patient stabilization including correction of dehydration, if present.

Recommendation 1:

To facilitate matching diagnostic recommendations to each patient's clinicopathologic circumstances, we recommend separating dogs with glomerular diseases into several different tiers based on the clinical manifestations of their disease when the investigation is conducted.

96% of voting consensus members agreed with Recommendation 1 and 57% of these voters expressed "strong agreement."

Recommendation 2:

Tiers are recommended for grouping dogs with glomerular diseases into categories based on major clinical manifestations of the disease to appropriately match diagnostic recommendations to each dog's clinicopathologic circumstances (Table 1).

86% of voting consensus members agreed with Recommendation 2, and 47% of these voters expressed "strong agreement."

Recognizing that the importance or utility of particular tests or diagnostic procedures is not equal and varies with the contextual circumstances, we here offer prioritized recommendations for the appropriate diagnostic evaluation of suspected glomerular diseases in dogs. In principal, each test or procedure is intended to serve at least one of three broad purposes:

- To identify an underlying disease or condition (eg, infectious, inflammatory, immune-mediated, vascular, neoplastic, endocrine, toxic, or genetic causes) that might be causing the glomerular injury;
- To detect and assess the severity of various glomerular disease sequella (eg, azotemia, hypertension, hypoalbuminemia, hypercoagulopathy, nitrogen and fluid balance) particularly those that might require independent medical management; and
- To characterize the pathologic changes in the kidneys as a means to better inform the attending clinician about the diagnosis, prognosis, and therapeutic options for the glomerulopathy itself.

Recommendation 3:

Particular diagnostic testing or procedures should be assigned a specific hierarchy of importance dependent on the contextual circumstances including clinical severity, resources, expertise, finances, and client willingness (Table 2).

100% of voting consensus members agreed with Recommendation 3, and 59% of these voters expressed "strong agreement."

Table 2. Description of terms used to categorize diagnostic tests and procedures in a hierarchy reflecting their relative importance or utility in the specified situation.

Essential—considered the minimum diagnostic assessment(s) with the highest priority regardless of financial or coexistent clinical circumstances

Recommended—suggested to be what should *always* be done, resources permitting

Potentially helpful—diagnostic assessments that might be performed in specific circumstances to be completely thorough or for investigational or academic analyses

Guidelines for Diagnostic Testing

Recommendations for diagnostic testing are summarized in Boxes 1, 2, and Table 3. Many of the recommendations, particularly those in Box 1, also are components of investigations of proteinuria that are needed to localize its origin and assess its persistence and magnitude, and therefore might already have been performed.

Recommendation 4:

A select group of diagnostics should be recommended in all cases with suspected glomerular disease (Box 1).

95% of voting consensus members agreed with recommendation 4 and 43% of these voters expressed “strong agreement.”

Recommendation 5:

Additional diagnostics are recommendations for dogs with suspected glomerular disease associated with high magnitude (UPC ≥ 3.5) or progressive proteinuria, hypertension, hypoalbuminemia, and/or azotemia (Box 2).

95% of voting consensus members agreed with Recommendation 5 and 43% of these voters expressed “strong agreement.”

Recommendation 6:

Diagnostic priorities should be based on canine glomerular disease tier (Table 3).

86% of voting consensus members agreed with Recommendation 6 and 42% of these voters expressed “strong agreement.”

Additional Comments

Repeating UPC Determinations

Repeated UPC determinations are needed in most situations, regardless of whether the initial observation

Box 1

Consensus Diagnostic Recommendations for All Cases with Suspected Glomerular Disease

Comprehensive history (Hx) including signalment, family, environs, exposures, etc.

Consider breed predispositions, eg, for glomerular disease, neoplasia, Cushing's disease

Consider environs or travel-associated infectious diseases that might cause glomerular disease, eg, Lyme disease, heartworm, Ehrlichiosis, Leishmaniasis

Consider drug or diet exposures that can cause hypertension or glomerular disease, eg, phenylpropanolamine, steroids, raw-food diet, sulfonamides, tyrosine kinase inhibitors

Complete physical examination (PE) including body condition score, retinal and rectal examinations

Blood pressure measurements

Readings on at least 2 occasions on different days are generally required¹⁰

A reliable elevated blood pressure measurement found on a single occasion may be adequate to initiate treatment if found in conjunction with target organ damage consistent with hypertension¹⁰

Blood and urine testing

CBC (including platelets)

Biochemical profile (including BUN, creatinine, phosphorus, calcium, sodium, potassium, albumin, globulin, glucose, alanine transaminase, alkaline phosphatase, bilirubin, cholesterol, and if possible, enzymatic CO₂)

Urinalysis (including urinary sediment evaluation)

Urine protein/creatinine ratio (UPC, at least 2 readings, see Repeating UPC Determinations below)

Urine culture, if warranted, eg., if microscopic pyuria, hematuria, or bacteriuria, USG < 1.025, azotemia, or suspected hyperadrenocorticism, diabetes, etc. exists, with possible occult infection

If living in an endemic area or travel history warrants, rule out the common diseases associated with glomerulonephritis characteristic for the area, eg, Lyme disease, heartworm, and Ehrlichiosis in endemic areas in the United States and Ehrlichiosis and Leishmaniasis in southern Europe/Mediterranean regions (see Searching for Infectious Disease below).

Appropriate, problem-specific investigation of any concomitant extrarenal diseases or abnormalities identified by the minimum evaluations (see Investigating Concurrent Extrarenal Abnormalities and Diseases below).

If proteinuria is of high magnitude, progressive, or nonresponsive to treatment, a more comprehensive diagnostic investigation is warranted⁴ (see Box 2).

Consider saving appropriate blood, serum, or urine samples^a for possible future testing, as recommended in Table 3.

^aThe following are examples of types of samples that might be saved for later needs:

- Whole blood in EDTA (purple top tube), eg, a sample taken before antimicrobial treatment is initiated may be useful for future PCR testing for certain infectious diseases; a DNA test may be available in some predisposed breeds^{6,11,12} or may be banked for future study.
- Serum, eg, for paired antibody testing for certain infectious diseases or for auto-antibody testing.
- Urine, eg, for a Bence Jones protein test or for SDS-PAGE testing, to help differentiate glomerular from tubular proteinuria.

Box 2**Additional Diagnostics Recommended for Dogs With Suspected Glomerular Disease Associated with High Magnitude (UPC ≥ 3.5) or Progressive Proteinuria, Hypertension, Hypoalbuminemia, and/or Azotemia**

If any of the above complications exists:

Complete abdominal ultrasound examination including evaluation of renal and adrenal architecture, presence of effusion, organomegaly, etc.

Thoracic radiographs (eg, search for infiltration, effusion, etc.)

Perform a more comprehensive evaluation for infectious diseases (see Searching for Infectious Disease below)

If hypertensive:

Assess for extrarenal causes of primary or secondary blood pressure elevations. In the absence of hypoalbuminemia and dehydration, hypertension may rarely be the primary cause of proteinuria. Examples of extrarenal causes of hypertension include hyperadrenocorticism, pheochromocytoma, hyperaldosteronism, adverse drug effects, fluid and/or salt overload, etc.

Consider echocardiography to check for concentric left ventricular hypertrophy

If hypoalbuminemic, azotemic, or both:

Rule out neoplasia (imaging, possibly lymph node or bone marrow aspirate)

Rule out concurrent or other causes of hypoalbuminemia, eg, liver disease, gastrointestinal losses, or malnutrition

Characterize the cause, time course, and stability of identified azotemia as extrarenal (ie, prerenal or postrenal), acute kidney injury, chronic kidney disease, or acute-on-chronic kidney disease. Stage or grade the kidney disease with the appropriate IRIS classification scheme.¹³

Renal biopsy is recommended, especially if the proteinuria is substantial (UPC is ≥ 3.5), unresponsive to treatment, or progressive despite institution of standard therapy and/or if administration of immunosuppressive drug therapy has been instituted or is contemplated and when the kidney disease is not end-stage. It cannot be overemphasized that experienced personnel at all stages of procuring, preparing, and then interpreting the renal biopsy, not only by light microscopy but also electron microscopy and immunofluorescence, is paramount to get useful information. See Evaluating Renal Biopsy Findings below.

Potentially helpful, investigational, or academic considerations:

Antithrombin testing

Thromboelastography

SDS-PAGE analysis on urine

DNA banking for future GWAS studies for predisposed breeds

demonstrated minimal or excessive proteinuria. When the magnitude of proteinuria is relatively mild (eg, UPC < 2.0), repeated UPC determinations are recommended to verify persistence of the proteinuria and establish the likelihood of an intrinsic glomerular disease.⁴ The greater the magnitude of proteinuria, the more certain it becomes that a glomerular disease is present, and, therefore, repeated UPC determinations are unnecessary merely to verify the presence of glomerular disease when the magnitude of proteinuria is

large. However, serial assessments of UPC are important to establish a reliable estimate of the dog's prevailing magnitude of proteinuria as a baseline for comparison to subsequent values to evaluate response (s) to treatment. When the magnitude of proteinuria is large, day-to-day variability in the UPC values obtained from a given dog also increases, making it necessary to average the values obtained on several days to obtain a reliable estimate of the dog's prevailing UPC value.¹⁴ As an alternative to determining the UPC on each of several samples and averaging the values, one can just mix equal volumes of the several samples together, and then determine the UPC value for the mixture.¹⁵

Searching for Infectious Disease

A variety of infectious diseases can cause renal proteinuria in dogs. Infectious agents potentially associated with renal proteinuria¹⁶ are listed in alphabetical order (Appendix 1) with preferred diagnostic testing methods and laboratory noted, when appropriate; otherwise, multiple diagnostic laboratories are suitable (Appendix 2).

The search for infectious diseases must be guided by clinical judgment. Clinicians must be cognizant of the infectious diseases particular to each patient's environs, including where it may have lived or traveled previously, and test accordingly. For instance, in Lyme endemic areas, consensus recommendations (Box 1) would include a SNAP-4DxPlus or similar screening test for heartworm antigen and antibodies against Lyme, *Anaplasma*, and *Ehrlichia* spp., and Leptospirosis antibody testing. Additional recommendations (Box 2) would include testing for other infectious diseases in the area, such as Babesiosis, Bartonellosis, and in cases with acute presentations, Rocky Mountain Spotted Fever.

As many nonclinical, nonproteinuric dogs have natural exposure antibodies toward *Borrelia burgdorferi*, Lyme seropositivity may represent a nonspecific coincidence in some cases and serve merely as a marker for tick and wildlife exposure. In such cases, testing for co-infections often is warranted. Elution studies have proven that the entity called "Lyme nephritis" is associated with Lyme-specific antigen-antibody complexes causing an immune-mediated glomerulonephritis; however, there is no readily available test to prove that a particular individual's proteinuric nephropathy is caused by its Lyme seropositive status. The magnitude of a dog's quantitative C6 antibody value does not predict illness, but is recommended only to rule out Lyme nephritis, if the value is very low, or to use as a baseline for future comparisons after treatment, if the value is elevated.¹⁷

Leptospirosis is associated primarily with tubular proteinuria rather than glomerular disease; however, evident hypoalbuminemia in these cases may mimic glomerular disease, and thus a diagnosis of Leptospirosis needs to be excluded in endemic areas.¹⁸ Potential cross-reactivity between antibodies against

Leptospira and *Borrelia* spp. may be problematic and require additional specific confirmatory testing.¹⁸

Investigating Concurrent Extra-Renal Abnormalities and Diseases

Glomerular diseases are often secondary to disease processes primarily located in other organ systems, and appropriate problem-specific investigation of any concurrent extrarenal diseases or abnormalities identified by the minimum evaluations is crucial. The intent is to identify any disease process that might be incriminated as an underlying secondary cause of a glomerulopathy. This is especially important if effective treatment of the primary disease can reduce or stop its provocation of glomerular injury. General categories of conditions promoting secondary glomerular injury are infectious (see Appendix 1) and noninfectious inflammatory or vascular conditions, certain endocrine disorders (particularly hyperadrenocorticism), neoplasia (especially disseminated malignancies), and immune-mediated disorders. As for infectious diseases, the search for underlying diseases with other pathogenesises must be guided by similar clinical judgment and consideration of general diagnostic principles. Clues to the presence of an identifiable underlying disease are usually present among the findings of the minimum evaluations listed above. The rational search strategy is not to “test for everything” but to investigate the evident clues in a focused and problem-specific manner until their consequential potential explanations (ie, diagnoses of underlying diseases) are either ruled out or ruled in. A more vigorous and exhaustive search is appropriate for patients that have more serious complications of their nephropathies (ie, those in Tiers II and III) than those that do not (ie, those in Tier I). Absence of an identifiable underlying disease is a common occurrence, despite exhaustive diagnostic searching, so it frequently is necessary to decide when to stop unproductive searching. For example, performing diagnostic imaging studies (eg, thoracic radiographs, abdominal ultrasound exam) has higher priority for patients in Tiers II and III than in Tier I (see Table 3). In addition, even if the initial search was unproductive and stopped, it is wise to revisit diagnostic efforts from time to time; some conditions (eg, neoplasia) develop new manifestations as they progress and become more readily detected later in their course. Finally, recognition of some concurrent diseases is important to the management strategy of the nephropathy even if they are not truly an underlying cause of the glomerular disease. Examples include dermatologic or gastrointestinal disorders that are managed in part by diets or drugs that might be suboptimum or contraindicated for the nephropathy. Cardiovascular problems might cause or contribute to difficulties in maintaining adequate fluid balance or distribution.

Testing for Immune-Mediated Disorders. Consideration of performing specific tests related to particular immune-mediated disorders occurs in 2 greatly different clinical settings. The first, and most important of

these, is when a dog that has renal proteinuria also exhibits concurrent extrarenal clinical signs or laboratory test findings that might be explained by an immune-mediated disease. Examples include fevers of undetermined origin, especially those with a waxing and waning course; lameness or joint swelling that might be attributable to noninfectious (nonerosive or erosive) polyarthritides; anemia accompanied by changes (eg, spherocytes, poikilocytes) suggestive of a hemolytic process; thrombocytopenia; and certain dermatologic lesions. In this setting, performing tests for various autoantibodies (eg, antinuclear antibody [ANA], Coombs', Rheumatoid factor) is part of an appropriately focused and problem-specific investigation of a suspected underlying disease (see above). Diagnoses of immune-mediated diseases typically are based on a weight-of-the-evidence strategy (ie, multiple findings that are compelling only when they are taken together and support one another) rather than on any single finding considered in isolation. And, when such a disease is diagnosed in a dog with a proteinuric nephropathy, the possibility that the glomerular disease actually is part of a multisystemic immune-mediated disorder (eg, systemic lupus erythematosus) becomes highly relevant, both diagnostically and therapeutically. The second setting in which tests for autoantibodies might be considered is when a dog has renal proteinuria but does not exhibit any signs or test abnormalities (besides the proteinuria itself) that might be explained by an immune-mediated disease. In this setting, a positive test result must be interpreted with great caution. Absent grounds for a weight-of-the-evidence diagnosis of an immune-mediated disease affecting some other organ system, the implications of such a finding are uncertain at best. There is no published or anecdotal evidence available to date that indicates that the results of any test for the presence of autoantibodies (eg, ANA, Coombs') discriminate among dogs with proteinuria, but without extrarenal manifestations of a possible immune-mediated disease in any way, that reliably differentiates those that have an immune-complex glomerulonephritis from those that do not.

Evaluating Renal Biopsy Findings

It is of highest priority that the renal biopsy procedure, from collection to in-depth evaluation, should be performed by experienced personnel. Dogs with suspected glomerular disease are likely to benefit the most from a renal biopsy before the disease has progressed to an advanced stage (IRIS CKD Stage IV), because secondary fibrosis and interstitial nephritis may mask the original glomerular disease. Therapeutic intervention at advanced stages is unlikely to ameliorate the clinical features, and the risk of biopsy-associated complications (eg, bleeding) increases. Before the procedure is performed, hypertension needs to be adequately controlled, and coagulation, including platelet number and function, must be adequate as well (discontinue antithrombotics for at least 3 days prior, and evaluate buccal

Table 3. Diagnostic priorities based on canine glomerular disease tier.

	Tier I Uncomplicated Renal Proteinuria			Tier II Renal Proteinuria with Hypoalbuminemia without Azotemia			Tier III Renal Proteinuria with Azotemia		
	Essential	Recommended	Potentially helpful	Essential	Recommended	Potentially helpful	Essential	Recommended	Potentially helpful
Hx, PE, BPM CBC, Chem UA $\pm$ culture UPCs ^a	X			X			X		
Investigate “evident” extrarenal disease		(X)			(X)			(X)	
Abdominal ultrasound	^b	X		X			X		
Chest radiographs	^b		X		X			X	
Work-up for hypertension	(X)			(X)			(X)		
Work-up for infectious diseases ^a	X			X			X		
Work-up for hypo- albuminemia				X			(X)		
Classify and work-up for azotemia ^a							X		
Renal biopsy ^a		^b	X		X			X	
AT, TEG, SDS-PAGE, save samples for DNA, etc.			X			X			X

(X) if appropriate; Hx, history; PE, physical examination; BPM, blood pressure measurement; Chem, biochemical profile; UA, urinalysis; AT, antithrombin; TEG, thromboelastography; SDS-PAGE, sodium dodecyl sulfate-agarose gel electrophoresis.

^aSee text.

^bA small minority of respondents made this recommendation.

mucosal bleeding time and coagulation times, PT/PTT, if there is any concern about adequacy of hemostasis). The renal biopsy findings required to formulate appropriate therapeutic recommendations for dogs with glomerular diseases generally include ultrastructural (ie, transmission electron microscopic, TEM) and immunostaining (ie, immunofluorescence microscopic, IF) observations, as well as light microscopic (LM) observations obtained with specialized techniques including 3 micron thin sections and appropriate special stains.¹⁹ Moreover, the findings must be evaluated and interpreted by experienced veterinary nephropathologists to be of optimal diagnostic utility. Light microscopy alone usually cannot reliably determine whether immune-mediated glomerular disease exists, and decision-making concerning the use of immunosuppressive protocols is hampered. It is essential that renal biopsies of dogs with glomerular disease contain adequate samples of cortex and are properly processed for LM, TEM, and IF evaluations using suitable fixatives or preservatives, and submitted to a pathology laboratory that routinely performs and interprets evaluations of canine kidney

specimens. The centers that perform these evaluations typically provide renal biopsy kits that contain the materials and instructions needed to obtain and submit suitable samples; however, for optimum results, the center must be contacted and kits obtained before biopsy samples are obtained. Centers currently available to provide such services include:

International Veterinary Renal Pathology Service, a joint collaboration of Texas A&M and The Ohio State University

For renal biopsy samples: International Veterinary Renal Pathology Service, Department of Veterinary Biosciences, The Ohio State University, Columbus, OH 43210; Contact: Dr Rachel Cianciolo—rachel.cianciolo@cvm.osu.edu.

For SDS-PAGE on urine samples: International Veterinary Renal Pathology Service, Texas A&M University, College Station, TX, USA; Contact: Dr George E. Lees—glees@cvm.tamu.edu; Dr Mary B. Nabity—mnabity@cvm.tamu.edu.

In Europe: Utrecht Veterinary Nephropathology Service, Utrecht University, Utrecht, The Netherlands;

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Appendix 1: Infectious diseases associated with renal proteinuria

- *Adenovirus* I antigen/antibody—PCR/ELISA
- *Anaplasma* spp. antigen—cytology, PCR
- *Anaplasma phagocytophilum/platys* antibody—in-house SNAP-4DxPlus (IDEXX), AccuPlex4 (Antech); Quant at reference labs
- *Babesia* spp. antigen—cytology, PCR (NCSU)
- *Babesia canis/gibsoni* antibody—IFA
- *Babesia microti* antibody—Protatek
- *Bartonella* spp. antigen—BAPGM culture and PCR at Galaxy Diagnostics
- *Bartonella* spp. antibody—Western blot (National Veterinary Lab), IFA (Galaxy Diagnostics)
- *Borrelia burgdorferi* natural exposure antibody—in-house SNAP-4DxPlus and Lyme C6Quant (IDEXX), AccuPlex4 (Antech), Multiplex (Cornell University), Abaxis Lyme (Abaxis)
- *Borrelia* spp. (relapsing fever group)—whole cell ELISA, IFA
- *Brucellosis* antigen—culture, PCR
- *Brucellosis* antibody—AGID, RSAT, TAT, FA, ELISA
- *Dirofilaria* antigen—in-house SNAP-4DxPlus (IDEXX), Solo Step CH (HESKA), AccuPlex4 (Antech)
- *Ehrlichia* spp. antigen—cytology, PCR
- *Ehrlichia canis/chaffeensis/ewingii* antibody—in-house SNAP-4DxPlus (IDEXX), AccuPlex4 (*E. canis*, Antech), Quant at reference labs
- *Fungal* (systemic) antigen/antibody—cytology, culture, AGID, Blastomyces urine antigen test
- *Hepatozoon* spp. antigen—PCR (Auburn)
- *Leishmaniasis* antigen/antibody—PCR; ELISA, FA, WB
- *Leptospira* spp. antigen—PCR (IDEXX)
- *Leptospira* spp. antibody—State laboratories
- *Mycoplasma* spp. antigen—PCR (IDEXX)
- *Rickettsia rickettsia* (RMSF) antigen—PCR or DFA (on tissue)
- RMSF antibody—IFA, LA
- *Trypanosomiasis*—cytology, PCR, fast dipstick test, FA, RIP

Appendix 2: Laboratories (in alphabetical order)

Antech Diagnostics, www.antechdiagnostics.com

Auburn University Molecular Diagnostics

Department of Pathobiology, College of Veterinary Medicine

252A Greene Hall

Auburn University

Auburn, AL 36849-5519

334-844-2648

www.vetmed.auburn.edu/index.pl/molecular_diagnostics

www.vetmed.auburn.edu/canine_hepatozoonosis

Galaxy Diagnostics

Animal Health Division

2 Davis Drive

Research Triangle Park

Durham, NC 27709

919-354-1055

www.galaxydx.com

HESKA Veterinary Diagnostic Laboratories, www.heska.com

IDEXX Laboratories, www.idexx.com

National Veterinary Laboratory, Inc.

P.O. Box 239

1 Tice Road

Franklin Lakes, NJ 07417

201-891-2992

www.natvetlab.com

NCSU – College of Veterinary Medicine

ATTN: Vector Borne Disease Lab

1051 William Moore Drive

Room 462A

Raleigh, NC 27607

919-513-8279

www.cvm.ncsu.edu/vhc/csds/ticklab.html

Protatek Reference Laboratory

574 East Alamo Drive, Suite 90

Chandler, AZ 85225

480-545-8499

www.protatek.com/RefLab/index.html

Consensus Recommendations for Standard Therapy of Glomerular Disease in Dogs

IRIS Canine GN Study Group Standard Therapy Subgroup, S. Brown, chair, J. Elliott, T. Francey, D. Polzin, and S. Vaden

Standard therapy forms the basic foundation for care of dogs with glomerular disease, as it is herein recommended for use in all affected animals regardless of causation of the disease. Consensus recommendations target the evaluation and management of proteinuria, inhibition of the renin-angiotensin-aldosterone system, modification in dietary intake with special consideration for those nutrients with renal effects, diagnosis and treatment of systemic hypertension, and evaluation and management of body fluid volume status in dogs with glomerular disease.

Key words: Amyloidosis; Chronic renal failure; Glomerulonephritis; Protein losing nephropathy.

The goal of the present report was to provide consensus recommendations for the management of canine glomerular disease, regardless of the inciting cause. As such, these recommendations should be viewed as forming the basis of standard, or routine, care of dogs with glomerular disease.

Canine glomerular disease is here considered to be primary if the process that initiates renal injury originates in the glomerulus. In most renal diseases, regardless of the site of the initiating renal injury, there are pathologic changes that occur in glomeruli after injury in other compartments (eg, tubulointerstitial disease or generalized loss of nephrons), and these changes are termed secondary glomerular disease. The recommendations outlined below apply to canine glomerular diseases in general, both primary and secondary.

In the management of dogs with glomerular disease, some recommendations are standard therapeutic considerations for all dogs with glomerular disease, regardless of the cause or severity. Although the present recommendations are considered a standard part of treatment in all canine glomerular diseases, their institution in a patient requires considerable prudence on the part of the veterinarian and individualized therapeutic plans, with adjustments based on severity of the glomerular disease and other patient

Abbreviations:

ACEi	angiotensin-converting enzyme inhibitor
ARB	angiotensin-receptor blocker
ATE	arterial thromboembolism
BP	systemic arterial blood pressure
CKD	chronic kidney disease
GFR	glomerular filtration rate
IRIS	International Renal Interest Society
NS	nephrotic syndrome
PUFA	dietary polyunsaturated fatty acids
RAAS	renin-angiotensin-aldosterone system
TOD	target organ damage
UPC	urine protein-to-creatinine ratio
VTE	venous thromboembolism

factors. The severity of a glomerular disease is usually reflected in the magnitude of proteinuria,¹ assessed as the urine protein-to-creatinine ratio (UPC), especially early in the disease process.

Recommendation 1:

For the purposes of standard therapy recommendations, the magnitude of proteinuria, as assessed by serial measurement of the UPC, should be used to make decisions about therapeutic intervention in dogs with glomerular disease.

100% of voting consensus members agreed with Recommendation 1 and 70% of these voters expressed “strong agreement.”

There is evidence in spontaneous renal disease that dogs, including those with primary glomerular disease, exhibit more adverse outcomes if the UPC exceeds 2.0² and in an experimental model of secondary glomerular disease³ that intervention, which reduces the UPC below 0.5, improves renal structural outcomes. These data provided support for the following recommendation.

Recommendation 2:

Intervention with standard therapies should be considered whenever renal proteinuria is causing

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the UPC to persistently exceed 0.5 in a dog with glomerular disease, whether the glomerular injury is primary or secondary. In general, a reduction in the UPC to <0.5 (or a reduction in the UPC of 50% or more) should be considered as evidence of therapeutic success.

89% of voting consensus members agreed with Recommendation 2 and 50% of these voters expressed "strong agreement."

Inhibition of the Renin-Angiotensin-Aldosterone System (RAAS)

Because hemodynamic forces influence the trans-glomerular movement of proteins, it follows that altering renal hemodynamics would be effective in reducing proteinuria. The RAAS has been the major target system for this approach to reducing proteinuria (Fig 1). Agents that target RAAS include an angiotensin-converting enzyme inhibitor (ACEi; eg, enalapril, benazepril), angiotensin-receptor blocker (ARB; eg, losartan, telmisartan), and aldosterone-receptor blocker (eg, spironolactone). Although renin inhibitors (eg, aliskirine) are being used in people, they have not been used to any great extent in dogs. The mechanisms of agents interfering with the RAAS have not been fully elucidated, but it appears that they reduce proteinuria greater than would be expected on the basis of their antihypertensive effects alone.

Angiotensin-Converting Enzyme Inhibitors (ACEis)

An ACEi may, in part, reduce proteinuria and preserve renal function by decreasing efferent glomerular arteriolar resistance leading to decreased (normalized) glomerular transcapillary hydraulic pressure.^{3,4} In addition, proposed mechanisms for RAAS inhibitors to reduce proteinuria include reduced loss of glomerular heparan sulfate, decreased size of the glomerular capillary endothelial pores, improved lipoprotein metabolism, slowed glomerular mesangial growth and proliferation, and inhibition of bradykinin degrada-

tion.⁵ Enalapril significantly reduced proteinuria and delayed the onset or the progression of azotemia in dogs with glomerulonephritis.⁶ In large part because of their established effects,^{3,6} interference with the RAAS, particularly with ACEi, is considered part of standard care of dogs with glomerular disease.

Typically, an ACEi is given once daily initially, but more than half of the dogs will eventually need twice-daily administration and perhaps additional dosage escalations (Table 1).⁶ Further dosage escalations may be required as described below. Although the serum creatinine concentration should be monitored, it seems to be uncommon for dogs to have severe worsening of azotemia (ie, >30% increase from baseline) attributable to ACEi administration alone. Dogs that are dehydrated may be at highest risk for worsening of azotemia after initiating ACEi therapy. In people, the renoprotective effects of ACEi are independent of the baseline renal function and ACEi slowed progressive disease even in patients with severe renal failure.⁷ However, although ACEi administration is appropriate, caution is warranted when administering an ACEi to a dog in International Renal Interest Society (IRIS) Chronic Kidney Disease (CKD) stage 4.

Benazepril and its active metabolite, benazeprilat, are largely eliminated by the biliary route with a smaller fraction being excreted in the urine. The clearance is not affected in dogs with impaired renal function.⁸ On the other hand, enalapril and its active metabolite, enalaprilat, are primarily eliminated by the kidney. Thus, dogs in late IRIS CKD stage 3 or stage 4 may achieve a similar antiproteinuric effect with a lower dosage of enalapril. However, the pharmacokinetics of ACEi is complicated and the effects of disease on the pharmacodynamics of these drugs are not necessarily predictable. Although interdrug differences are known, there are currently no published studies in dogs with glomerular disease to support the recommendation that one ACEi is superior in its pharmacodynamic action to another.

Angiotensin-Receptor Blockers (ARBs)

The ARBs in clinical use block the angiotensin II type 1 receptor. Several ARBs have been studied extensively in people with glomerular disease (eg, losartan, irbesartan, telmisartan) and lead to a reduction in proteinuria similar to that which is seen with ACEi. There are limited data on the use of ARBs in dogs with glomerular disease and it is anticipated that the addition of information to the literature could alter our selection or method of use of these agents. The ARB that seems to be used most commonly in dogs is losartan. Even though dogs do not appear to produce one of the major active metabolites of losartan, there is good evidence that losartan exerts pharmacodynamic effects in dogs.⁹ Furthermore, there is anecdotal evidence that losartan has an antiproteinuric effect in dogs, particularly in combination with ACEi.

In human patients, losartan resulted in an average reduction in proteinuria of 35% from baseline during a

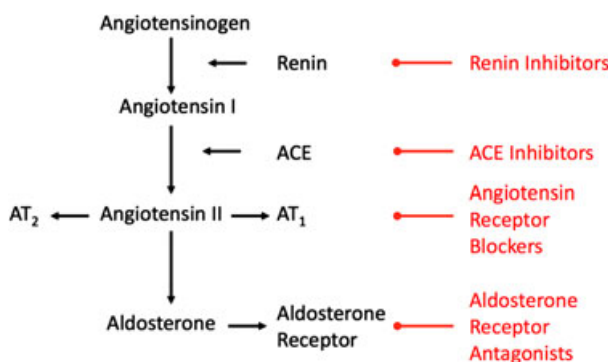


Fig 1. The renin-angiotensin-aldosterone system and approaches to its inhibition.

Table 1. Dosages of common inhibitors of the renin-angiotensin-aldosterone system used in the management of proteinuria in dogs with glomerular disease.

Drug	Indication	Initial Dose	Escalating Dose
Benazapril	Angiotensin converting enzyme inhibitor ^a	0.5 mg/kg PO q24h	Increase by 0.5 mg/kg/d to a maximum of 2 mg/kg PO per day. Can give q12h
Enalapril	Angiotensin converting enzyme inhibitor ^a	0.5 mg/kg PO q24h	Increase by 0.5 mg/kg/d to a maximum of 2 mg/kg PO per day. Can give q12h
Ramipril	Angiotensin converting enzyme inhibitor ^a	0.125 mg/kg PO q24h	Increase by 0.125 mg/kg/d to a maximum of 0.5 mg/kg PO per day. Usually give q24h
Imidapril	Angiotensin converting enzyme inhibitor ^a	0.25 mg/kg PO q24h	Increase by 0.25 mg/kg/d to a maximum of 5 mg/kg PO per day. Usually give q24h
Telmisartan	Angiotensin receptor blocker	1.0 mg/kg PO q24h	Increase by 0.5 mg/kg once daily up to 2 mg/kg/d
Losartan	Angiotensin receptor blocker ^b	0.125 mg/kg/d in azotemic dogs 0.5 mg/kg/d in nonazotemic dogs	0.25 mg/kg/d in azotemic dogs 1 mg/kg/d in nonazotemic dogs
Spironolactone	Aldosterone-receptor blocker ^c	1–2 mg/kg PO q12h	

^aACEi and ARBs are antiproteinuric drugs that generally have weak antihypertensive effects.

^bConcurrent administration of an ACEi is generally recommended.

^cReserved for the management of proteinuria in dogs that have increased serum aldosterone concentrations and either have failed or are intolerant of ACEi and ARBs.

3.4-year follow-up period; much of this reduction was in the first 6 months of treatment.^{10,11} In irbesartan-treated patients, every 50% reduction in proteinuria during the first 12 months of treatment reduced the risk of a negative renal outcome by more than half.^{10,11} Telmisartan, an ARB that is more lipophilic and has a longer half-life than losartan, was shown to be more effective in reducing proteinuria than was losartan in patients with diabetic nephropathy.¹¹ Telmisartan also has a higher affinity for the angiotensin-1 receptor and dissociates more slowly when compared with losartan and thus, its blocking effects appear to be insurmountable and persist in vivo for longer than would be predicted from its plasma kinetics. In normal dogs, telmisartan (1 mg/kg PO q24h) more effectively blocks the pressor response to intravenous administration of angiotensin I than enalapril (0.5 mg/kg PO q12h), warranting further study of this ARB in proteinuric dogs with glomerular disease (Hanford C, Coleman A, Schmiedt C, Brown S: Unpublished observations, 2013).

Combined Therapy with ACEi and ARB

Blockade of the angiotensin II type 1 receptor with an ARB may give rise to a compensatory increase in renin activity, and therefore an incomplete block of the RAAS.¹² Furthermore, an ACEi may incompletely block the formation of angiotensin II, particularly within the kidney. Hence, there may be an added benefit to combined therapy with an ACEi and an ARB because of the inability of monotherapy with either class of drug to provide complete RAAS block-

ade.^{10,11} Blockade may be more complete when ACEi is combined with an ARB, although it is still only 75–80% complete. Although there have not been any studies in dogs, studies in people have suggested that these drugs may be additive or perhaps even synergistic in reducing proteinuria.¹³ Because the dosage of each individual drug can be reduced during combined therapy, adverse effects may be less likely.

However, the approach of combining these 2 agents must be used cautiously in light of a recent study published in people where elderly patients prescribed this combination had a higher risk of kidney failure and death.¹⁴ There is limited information available related on the use of ARBs in dogs with glomerular disease. Experience (Vaden S: Unpublished observations, 2013) with its use in 11 dogs with proteinuria and glomerular disease suggests that it can be used safely in combination with ACEi (enalapril) and calcium channel blockade (amlodipine) and is associated with a reduction in proteinuria in most animals (7 of 11 dogs; typically reducing the UPC by ~25%), although its use is associated with a modest increase in serum creatinine (6 of 11 dogs) or potassium (4 of 11 dogs) in some animals. Controlled studies are needed in dogs to determine if the antiproteinuric effects of ACEi and ARBs are optimized by combination therapy or monotherapy with individualized dosage escalation.

Aldosterone-Receptor Blockers

Serum aldosterone increases over time (ie, aldosterone escape) in people treated even with maximal

dosages of ACEi and ARB. Prolonged hyperaldosteronism may have adverse effects on the heart, systemic blood vessels, and kidneys. Aldosterone receptor blockers have been shown to reduce proteinuria and stabilize kidney function in an additive fashion to ACEi and ARB in people.¹⁵ People who have high aldosterone concentrations are more likely to have a reduction in proteinuria in association with administration of an aldosterone receptor blocker. Eplerenone may be the drug of choice in people because of its relative lack of affinity for androgen and progesterone receptors, producing fewer endocrine side effects. However, endocrine side effects of spironolactone appear to be less problematic in dogs, making a rationale for use of eplerenone unclear in veterinary medicine. Although spironolactone has been used most commonly in veterinary medicine, there are little published data or anecdotal information supporting efficacy of this drug in dogs in the management of glomerular disease. Spironolactone would be expected to offer benefit only if serum aldosterone concentrations are increased. This drug could be tried in animals that have high serum aldosterone concentrations and persistent proteinuria in spite of treatment with an ACEi, ARB, or both with the understanding that its efficacy in reducing proteinuria has not been established.

Monitoring RAAS Inhibition

Recommendation 3:

The UPC, urinalysis, systemic arterial blood pressure (BP), and serum albumin, creatinine, and potassium concentrations (in fasting samples) should be monitored at least quarterly in all dogs being treated for glomerular disease.

95% of voting consensus members agreed with Recommendation 3, and 50% of these voters expressed "strong agreement."

In dogs with glomerular disease, introduction of a new drug or dosage modifications are important indications for frequent monitoring (Fig 2). One to 2 weeks after an ACEi or ARB is added or changed, the UPC, serum creatinine, serum potassium, and BP should be evaluated to verify that the recent change in treatment has had the desired therapeutic effect (ie, reduction in UPC), not resulted in a severe worsening of renal function (ie, >30% increase in serum creatinine), a concerning increase in serum potassium concentration, or hypotension (an unlikely occurrence with these drugs). Dogs with marked azotemia, late IRIS CKD stage 3 or stage 4, should be monitored more carefully.

Day-to-day variations in the UPC occur in most dogs with glomerular proteinuria, with greater variation occurring in dogs with UPC >4.¹⁶ Changes in urinary protein content are most accurately assessed by determining trends in the UPC over time. Because there is greater day-to-day variation in dogs with UPC

>4, consideration should be given to either averaging 2–3 serial UPC or measuring a UPC in urine that has been pooled from 2 to 3 collections.¹⁷ In 1 study, demonstration of a significant difference among serial values for UPC in proteinuric dogs required a change of at least 35% at high UPC values (near 12) and 80% at low UPC values (near 0.5).¹⁶ The reliability of the UPC is adversely impacted by changes in the glomerular filtration rate (GFR), which may occur as a result of disease progression or the hemodynamic effects of RAAS inhibitors. Thus, a reduction in UPC near these reported magnitudes, without an increase in the serum creatinine concentration, is required to indicate improvement or response to treatment.

Managing Hyperkalemia

Hyperkalemia appears to be a common side effect of RAAS inhibition in dogs with renal disease. Dogs with serum potassium concentrations of >6 mmol/L should be monitored closely. Pseudohyperkalemia, because of the high potassium content of some blood cells, may occur in dogs with glomerular disease. Before modifying treatment, pseudohyperkalemia should be eliminated as a cause by measuring the potassium concentration in lithium heparin plasma. Because of the cardiotoxic effects of potassium, treatment should be modified in dogs with serum potassium concentrations >6.5 mmol/L. In dogs with plasma potassium concentrations of >6 mmol/L, an ECG should be evaluated for cardiac conduction disturbances. True hyperkalemia can be managed by reducing the ACEi or ARB drug dosage, discontinuing spironolactone administration, feeding diets that are reduced in potassium, or administering an intestinal potassium binder (eg, kayexelate). Potassium-reduced home-prepared diets that were formulated by a veterinary nutritionist have been shown to effectively correct hyperkalemia in dogs with CKD.¹⁸

Making Therapeutic Adjustments of RAAS Inhibitors

Recommendation 4:

An ACEi should be initial treatment for most dogs with proteinuria. The initial choice of drug and starting dose may vary, but can be gradually increased to achieve a therapeutic target. The ideal therapeutic target is a reduction in the UPC to <0.5 without inappropriate worsening of renal function. However, as this ideal target is not achieved in most dogs, a reduction in UPC of 50% or greater is the recommended alternate target.

95% of voting consensus members agreed with Recommendation 4, and 60% of these voters expressed "strong agreement."

As previously mentioned, it is appropriate to adjust the target UPC to reflect the expected day-to-day

variations in this parameter (Fig 2). The degree to which worsening of renal function is tolerated will in part depend on the IRIS stage of the canine CKD. Dogs with stage 1 and 2 CKD can have an increase in serum creatinine of up to 30% without modifying treatment. The goal in dogs with stage 3 CKD would be to maintain stable renal function; if renal function deteriorates, therapeutic adjustments may be indicated. Dogs with stage 4 CKD are generally intolerant of worsening of renal function and any deterioration may have clinical consequences. Whereas RAAS inhibitors can be used in this subset of patients, renal function should be monitored closely and therapeutic adjustments made as needed to maintain baseline renal function.

If the target reduction in UPC is not achieved, the plasma potassium concentration is <6 mmol/L, and any changes in renal function fall within the tolerable limit, then dosages may be increased every 4–6 weeks, starting with a change to 0.5 mg/kg q12h. If the target reduction in UPC is not achieved when a dosage of 2 mg/kg/day of an ACEi is reached, a reasonable next step would be to add an ARB. Higher ACEi dosages may be used but only with caution. Alternatively, an ARB can be used as monotherapy if dogs appear to be intolerant of an ACEi.

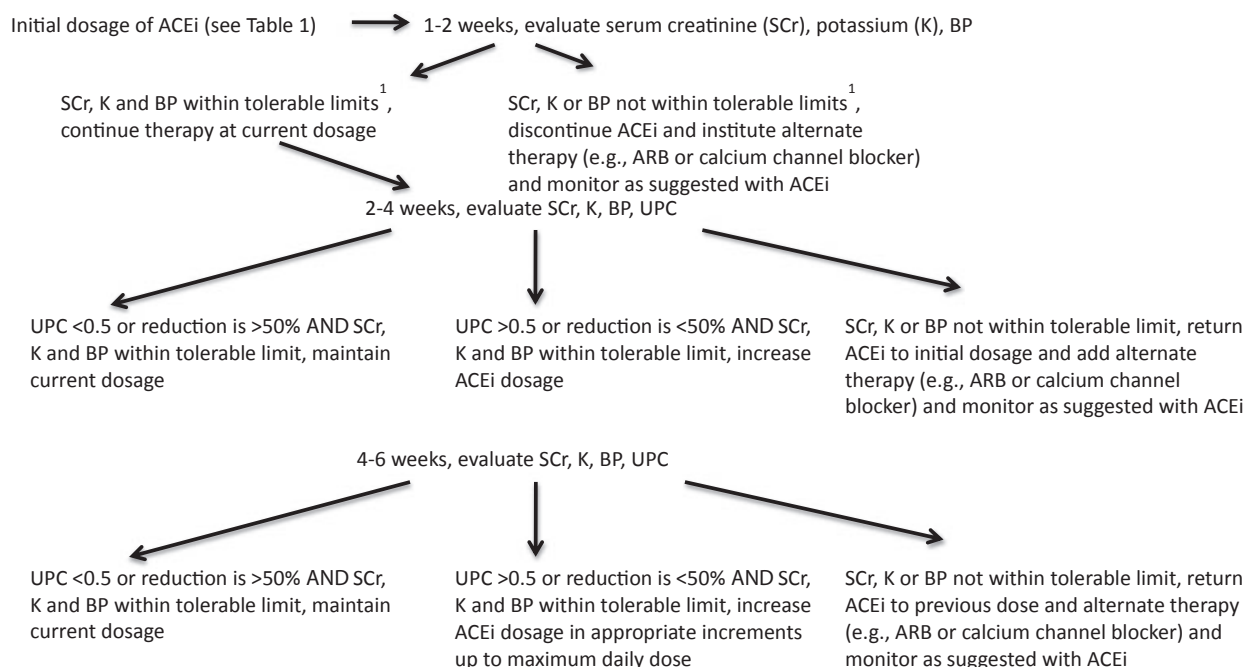
Dietary Therapy in Dogs with Glomerular Disease

Nutrition plays a central role in the management of kidney diseases in veterinary medicine. In dogs, the kidney is susceptible to progressive injury, which has been linked to the magnitude of proteinuria² and the extent of proteinuria may be modified by adjustments in dietary intake in models of CKD^{19–21} including those characterized by marked proteinuria,^{22,23} and the same is true in spontaneous CKD.⁶ Furthermore, alterations in dietary intake can affect the progression of CKD in induced models of CKD,^{20,21,25,26} genetic models of CKD²³ and in spontaneous CKD.²⁴ Substantially less is known about the effects of dietary modification in proteinuric dogs with unstable kidney function, although diet did effect the magnitude of proteinuria and renal hemodynamic response to an acute reduction in renal function in 1 laboratory study of induced CKD in dogs.²¹

Dietary Polyunsaturated Fatty Acids (PUFA)

In laboratory studies of canine kidney disease characterized by secondary glomerular injury, dietary supplementation with n-3 PUFA altered the long-term

Making Adjustments to RAAS Inhibition Therapy in Dogs with Glomerular Disease



1. Tolerable limits: SCr change generally considered tolerable in CKD stage 1 or 2 if $<30\%$ above baseline; in CKD stage 3 $<10\%$ but in stage 4 no increase in SCr may be tolerable. K tolerable when <6.0 mmol/L. Systolic BP should be AP2 (160–179 mmHg) or lower; BP decline acceptable as long as systolic BP >120 mmHg.

Fig 2. Protocol for making adjustments in renin-angiotensin-aldosterone (RAAS) inhibitor therapy.

course of renal injury,²⁰ the hemodynamic response to acute reduction in renal function,²¹ and the magnitude of proteinuria. Dogs with spontaneous CKD exhibit alterations in vasoactive urinary eicosanoid excretion; these changes were interpreted to support a role for glomerular hyperfiltration in progressive canine renal injury.²⁷ Interestingly, short-term studies in dogs with naturally occurring kidney disease indicate that supplementation with n-6 PUFA led to increased GFR.²⁷

In laboratory studies, dietary supplementation with fish oil (specifically docosahexaenoic acid and eicosapentaenoic acid) lowered glomerular pressure, decreased renal eicosanoid series-2 excretion, and provided renoprotection.²¹ Dietary supplementation with lesser amounts of these same PUFA (providing approximately 0.6% n-3 PUFA on a dry weight basis) reduced the dietary n-6 : n-3 PUFA ratio from 50 : 1 to 5 : 1, lowered glomerular capillary pressure, altered urinary excretion of eicosanoids and, in chronic studies, delayed the progression of CKD.²⁸ This latter dietary maneuver is believed to be of long-term benefit to delay progression of renal injury, although the benefits of n-3 PUFA supplementation alone remains to be established in dogs with spontaneous CKD. Although a diet with high n-3 PUFA content prolonged survival,²⁴ there were other potentially beneficial dietary modifications in this study.

In dogs with CKD, dietary supplementation with n-6 PUFA may increase GFR in the short term and dietary supplementation with n-3 PUFA may offer renoprotection in the long term. As these PUFA act competitively, it is not possible to achieve both affects simultaneously in the same animal. Until further information is available, the presently recommended approach is to utilize n-3 PUFA supplementation (specifically docosahexaenoic acid and eicosapentaenoic acid) for dogs with glomerular disease. Although the source and type of PUFA vary, commercially available “kidney diet” preparations that are supplemented with docosahexaenoic acid and eicosapentaenoic acid, providing n-6/n-3 ratios close to 5 : 1, are preferred for dogs with glomerular disease.

The effects of n-3 PUFA supplementation as monotherapy in spontaneous canine glomerular disease have not been well studied. In people with proteinuric renal diseases, specifically IgA nephropathy and lupus nephritis,²⁹ there is evidence of benefit in slowing progression, reducing proteinuria, or both with n-3 PUFA supplementation²⁹ or of a synergistic effect³⁰ of co-administration of n-3 PUFA with drugs that interfere with the RAAS. However, this is controversial, with some studies showing no benefit in people.²⁹

The efficacy of adding n-3 PUFA to diets already containing these fatty acids or the addition of n-3 PUFA to diets with a high n-6/n-3 ratio has not been evaluated in dogs with spontaneous glomerular diseases, but would seem an appropriate consideration. One concern is that supplementation with large amounts of n-3 PUFA alters the function of a variety of nonrenal tissues, including immunologic and hemostatic functions. However, hemostatic and

immunologic dysfunction were not reported in dogs with induced kidney disease fed a diet containing 4% n-3 PUFA, and an n-6/n-3 ratio of less than 0.5, for 20 months.²⁰ Higher levels of n-3 PUFA were shown to be safe and more efficacious than routine n-3 PUFA-supplemented diets in the treatment of canine osteoarthritis.³¹ On the basis of studies to date, where dietary supplementation with n-3 PUFA is chosen, a dosage of 0.25–0.50 g/kg body wt of docosahexaenoic acid and eicosapentaenoic acid should be considered. As PUFA within cell membranes are subject to oxidative damage, the addition of PUFA to the diet increases an animal's antioxidant (eg, vitamin E) requirements. Furthermore, tubular hypermetabolism in CKD³² and the regenerative phase of ongoing kidney damage are believed to expose the kidney to oxidative injury. In this setting, high PUFA diets may enhance oxidative injury and dietary antioxidants have been shown to increase survival in laboratory models of canine CKD.²⁸ Appropriately, commercially available “renal diets”, which are PUFA supplemented, contain a variety of supplemental antioxidants. Augmenting dietary PUFA content further would be expected to increase antioxidant requirements. Supplementation with vitamin E was utilized in one long-term laboratory study (1.1 IU of supplemental vitamin E/g of added fish oil).²⁰

Recommendation 5:

Dogs with glomerular disease should be fed a diet with a reduced n-6/n-3 PUFA ratio, approximating 5 : 1. Where dietary supplementation with n-3 PUFA by the owner is used to alter this ratio, a dosage of 0.25–0.50 g of n-3 PUFA/kg body wt, containing eicosapentaenoic acid and docosahexaenoic acid, appropriate for a typical canine diet.

95% of voting consensus members agreed with Recommendation 5, and 40% of these voters expressed “strong agreement.”

The PUFA within a diet or stored oil may also be oxidized. Although pet food manufacturers generally add specific antioxidants to diets to address this concern (eg, BHA, BHT, and ethoxyquin), care should be taken to utilize these diets according to recommended expiration dates. Free oils or capsules intended for addition to diets should be stored at 20°C or lower and used in a timely manner.

Dietary Protein

The generally accepted benefits of modification in dietary protein content in canine glomerular disease are to reduce intraglomerular pressure, the magnitude of proteinuria, and the rate of generation of uremic toxins. There is evidence that reduction in protein intake reduces proteinuria²² and that dietary modification, which includes protein restriction,²³

slows progression in genetic models of proteinuric CKD. Furthermore, it has been established that dietary modification, including the use of lower protein intake, is beneficial in spontaneous canine CKD.²⁴

Recommendation 6:

Modified protein diets should be fed to dogs with glomerular disease.

95% of voting consensus members agreed with Recommendation 6, and 80% of these voters expressed "strong agreement."

Dietary Sodium

Although normal dogs are apparently not as salt-sensitive as people or some inbred strains of rats, it is likely that dogs with kidney disease,³³ particularly those with nephrotic syndrome (NS), are salt-sensitive. Furthermore, salt restriction enhances the antihypertensive efficacy and renal hemodynamic effects of some antihypertensive agents in dogs³⁴ and cats,³⁵ particularly those that interfere with the renin angiotensin system.

Recommendation 7:

Based on evidence from laboratory studies, the study group recommends feeding diets formulated to contain reduced sodium chloride content to dogs with glomerular disease.

95% of voting consensus members agreed with Recommendation 7, and 25% of these voters expressed "strong agreement."

Antithrombotic Therapy in Dogs with Glomerular Disease

Thromboembolism is a recognized complication of proteinuria in dogs and humans.^{36–38} The prevalence of thromboembolism in dogs with proteinuric kidney disease (glomerular diseases) has been reported to be as high as 25%.³⁸ In humans with NS, thromboembolism is primarily attributed as venous thromboembolism (VTE) rather than arterial thromboembolism (ATE) with a prevalence of VTE reported to be as high as 42% and ATE as high as 5.5%.³⁹ Although both VTE and ATE have been described in dogs, clinical observations suggest that VTE may also be the more common cause for thromboembolism in dogs.^{36,38} The ratio of proteinuria to serum albumin has been reported to be predictive of VTE in humans with NS.³⁷ Although the relationship between the magnitude of proteinuria and development of thromboembolism has not been well established in dogs, it is generally accepted that the risk of thromboembolism increases as the concentrations of antithrombin III and serum albumin decline consequent to proteinuria.

Virchow postulated 3 main causes of thrombosis: stasis of the blood, changes in the vessel wall, and changes in the composition of blood. Factors in the first and third of Virchow's groups are linked to development of VTE.⁴⁰ The mechanisms underlying the thrombotic diathesis associated with proteinuria (NS) are poorly understood, but loss of antithrombotic substances into the urine is often invoked. However, this explanation probably grossly oversimplifies the circumstances. Decreased levels of factors IX and XI, and of antithrombin III have been recognized in humans with NS. Similarly, reduced levels of antithrombin III have also been recognized in dogs with severe proteinuria. Although there may be loss of small molecular weight factors into the urine, there is also an increase in plasma procoagulant cofactors, factors V and VIII, and fibrinogen levels.³⁹ In addition, platelet reactivity is increased (in vitro), but appears to be flow-dependent.⁴¹ Inactivity of human patients with NS has been linked to increased risk of VTE,³⁹ and venous stasis associated with volume depletion may also promote thrombus formation. Thus, the procoagulant state associated with VTE in the NS appears to be multifactorial in origin.

The pathophysiology of ATE is thought to differ from that of VTE, with changes in vessel walls and platelet activation being of greater importance. However, several studies have provided evidence that altered vessel wall activity may also promote VTE.⁴⁰ The recent observation that microalbuminuria is independently associated with increased risk for VTE in humans tends to support a possible role for vessel wall involvement in the development of VTE and ATE in glomerular diseases when viewed in the context of microalbuminuria being a marker for generalized endothelial dysfunction.⁴⁰ However, evidence of an association between endothelial dysfunction and microalbuminuria is lacking in the dog, largely because of the lack of robust methods for assessing endothelial cell function in vivo in dogs. Nonetheless, these observations may provide a pathophysiologic justification for the use of aspirin in preventing VTE in dogs with glomerular disease.

Recommendations for prophylaxis of VTE in humans as well as dogs include administration of heparin and vitamin K antagonists.^{42,43} Limited evidence suggests that antiplatelet agents may provide some protection against VTE in hospitalized human patients.⁴³ However, the American College of Chest Physicians Evidence-Based Clinical Practice Guidelines on Prevention of Venous Thromboembolism does not recommend the use of aspirin alone as prophylaxis against VTE, primarily because more effective methods of thromboprophylaxis are readily available.⁴³ In support of this position, they cite a number of trials that report no significant benefit from aspirin VTE prophylaxis, or found that aspirin was inferior to other thromboprophylaxis modalities.

Options for thromboprophylaxis available for clinical use in dogs are limited with heparin and aspirin predominating.⁴⁴ Very limited studies are available to

support the value of antithrombotic therapy in dogs, and no studies specifically address the issue of antithrombotic therapy in dogs with glomerular disease. Thus, there is little conclusive evidence on which to base recommendations for prophylaxis of VTE in dogs, regardless of cause.⁴² Such evidence is difficult to derive and would require large well-coordinated multicenter randomized controlled clinical trials enrolling sufficient patients at risk and to treat them for long enough to determine whether there was a benefit in reducing the incidence rate of this sporadic complication of glomerular disease.

Although coumarin derivatives (warfarin) and heparin are the anticoagulants most commonly used to manage VTE in humans, they are not recommended for long-term use in dogs with glomerular disease because of issues with administration, monitoring, and potential complications. Hemorrhagic complications are common in humans managed with coumarin derivatives and careful and frequent monitoring of drug dosage is essential with these drugs.⁴⁴ One recent study suggested that the antithrombotic effect of heparin may be most effective in dogs with immune-mediated hemolytic anemia when heparin dosage is adjusted according to factor Xa assay results, whereas standard (unadjusted) doses of heparin may in fact promote thrombosis.⁴⁵ In addition, as heparin acts through enhancement of the inhibitory actions of antithrombin, reduced levels of antithrombin found in many dogs with NS suggest that heparin is likely to be less effective in dogs most likely to develop thromboembolism.⁴⁴

There are no evidence-based clinical studies on which to base recommendations for determining when intervention is justified or for which drugs to use for prophylaxis of VTE or ATE in dogs with glomerular disease. The current recommendation for thromboprophylaxis in dogs with glomerular disease remains low-dose aspirin (0.5–5 mg/kg daily).^{36,42} However, the optimum aspirin protocol for limiting TE in dogs is unknown and appropriate assessments of the impact of low dosages of aspirin on canine platelets are limited. In one study, platelet aggregation was significantly reduced in healthy dogs in response to 0.5 mg/kg, PO, q12h.⁴⁶ However, a recent abstract reported no effect of aspirin at this dosage on platelet function measured by aggregometry and a shear-based function test (PFA100®).⁴⁷ Instead, the authors reported 1 mg/kg to be the minimum dosage to alter platelet function in 73 healthy dogs when assessed with these methods. A more recent study found that administration of 1 mg/kg/d of aspirin failed to consistently suppress platelet function, suggesting that dosages of 1 mg/kg or less per day may not consistently impact platelet-mediated thrombosis/thromboembolism.⁴⁸ Studies documenting effectiveness of doses between 1 and 5 mg/kg of aspirin in dogs have not been published.

Preliminary evidence suggests that clopidogrel (Plavix) may be effective in reducing platelet activity, at least in normal dogs, at an oral dose of approximately 1.1 mg/kg every 24 h.⁴⁹ However, evidence that it is superior to aspirin in prophylaxis of VTE in dogs

is lacking. In a clinical trial of dogs with spontaneous primary immune-mediated hemolytic anemia, a condition commonly complicated by development of fatal venous thromboembolism, clopidogrel alone or in combination with 0.5 mg/kg/d of aspirin, was safe but had similar short-term survival compared with aspirin alone at the same dosage in a small group of animals treated with standard immunosuppressive therapy.⁵⁰ Evidence of safety and effectiveness in preventing VTE or ATE in dogs with spontaneous glomerular disease is not available. Although clopidogrel may be safe and possibly effective in dogs, it is substantially more expensive than low-dose aspirin.

Recommendation 8:

The study group recommends daily administration of low-dose aspirin (1–5 mg/kg daily) for thromboprophylaxis in dogs with proteinuric glomerular disease. Clopidogrel may be similarly effective to aspirin and may be used instead of aspirin; however, there is no compelling evidence that it is superior to low-dose aspirin in dogs with glomerular disease.

95% of voting consensus members agreed with Recommendation 8, and 40% of these voters expressed “strong agreement.”

Although evidence supporting efficacy of aspirin as prophylaxis for VTE in dogs is poor (expert opinion and uncontrolled clinical observations), it is broadly applied to proteinuric dogs and appears to be safe provided that the dogs are well hydrated and normotensive. Previous recommendations for aspirin dosage in dogs with NS have included dosages as low as 0.5 mg/kg; however, more recent studies in normal and dogs with immune-mediated hemolytic anemia (Stiller A, Armstrong PJ, Polzin DJ, Smith SA: unpublished observations, 2013) have suggested that aspirin at 0.5 mg/kg is unlikely to modify platelet function. In addition, the recommendation to administer aspirin to dogs with glomerular disease is based on studies supporting an apparent beneficial effect of a thromboxane synthetase inhibitor in ameliorating proteinuria in dogs with glomerular disease.^{51–53} Direct evidence that aspirin therapy influences renal function or pathology in dogs with spontaneous glomerular disease is lacking.

Systemic Hypertension and Target Organ Damage (TOD) to the Kidneys

Systemic hypertension is problematic because chronically sustained elevations of BP produce injury to tissues⁵⁴; the rationale for treatment of hypertension in dogs with glomerular disease is to minimize or prevent this injury in the kidney, eyes, brain, or cardiovascular system. Damage that results from the presence of sustained high BP is referred to as TOD. In the kidney, TOD is generally manifest as an enhanced rate of decline of renal function, mortality, increased fre-

quency of uremic crises, or increased magnitude of proteinuria.^{3,4,54–58} Although hypertension appears to be more prevalent in advanced CKD, it may be present in any IRIS stage. Ocular, central nervous system, and cardiovascular TOD is observed in many dogs with hypertension and effective treatment is generally believed to reduce the likelihood of further TOD in these tissues.⁵⁴

Measurement of BP

Diagnosis and management of hypertension in dogs with acute or chronic glomerular disease should be based on measurement of the patient's BP.⁵⁴ The choice of measurement device depends on operator experience and preference. Results of all BP measurements, rationale for excluding values, the final (mean) result, and interpretation of the result by the veterinarian should be noted. The animal's position and attitude, cuff size and site, and cuff site circumference (cm), and values obtained should be carefully considered and noted in the animal record.

Treatment of Hypertension

Recommendation 9:

The initial assessment of a dog with glomerular disease suspected of having systemic hypertension should include recognizing conditions that may be contributing to an increase in BP, identifying and characterizing TOD, and determining if there are any seemingly unrelated concurrent conditions that may complicate antihypertensive therapy.

100% of voting consensus members agreed with Recommendation 9, and 60% of these voters expressed "strong agreement."

Because hypertension is often a silent, slowly progressive condition requiring vigilance and life-long therapy, it is important to be absolutely certain about the diagnosis: a high BP measurement in a dog with glomerular disease could represent either secondary or artifactual (anxiety-induced or "white-coat") hypertension or a combination of both.⁵⁴ White-coat hypertension is not an indication for treatment in dogs. A decision to use antihypertensive drugs should be based on the integration of all clinically available information and multiple measurements of BP.

Decision to Institute Treatment for Hypertension

In people, any reduction in BP that does not produce overt hypotension lowers the risk of TOD. This finding is consistent with results of a laboratory study in dogs,⁵⁸ but remains to be confirmed in dogs with spontaneous glomerular disease. The ACVIM Hypertension Consensus Panel⁵⁴ and International Renal Interest Society (IRIS)⁵⁹ recommend that BP be

Table 2. Staging of blood pressure (BP; mmHg) in dogs and cats based on risk for future target organ damage.^a

Systolic	Diastolic	Risk of Future Target Organ Damage	BP Stage
<150	<95	None or minimal	AP0
150–159	95–99	Low	AP1
160–179	100–119	Moderate	AP2
≥180	≥120	High	AP3

^aWhere reliable measurements (see text) lead to different categories based on separate consideration of the patient's systolic and diastolic BP, the patient's BP stage should be taken as the higher risk.

categorized on the basis of risk of future TOD (Table 2). While there are interbreed differences in BP in dogs, at this time, only the difference (20 mmHg higher values for each category) for Sight Hounds mandates separate categorization at this time.^{59–61}

Recommendation 10:

Dogs with glomerular disease are presumed to have TOD and the general consensus is to institute treatment in a patient wherein reliable measurement of BP indicates that systolic BP (SBP) exceeds 160 or diastolic BP (DBP) exceeds 100 mmHg (AP2 or higher—see Table 2 for definitions).

90% of voting consensus members agreed with Recommendation 10, and 45% of these voters expressed "strong agreement."

Antihypertensive Therapy

Recommendation 11:

Antihypertensive therapy must be individualized to the patient and its concurrent conditions. Regardless of the initial level of BP, the goal of treatment should be to maximally reduce the risk of future TOD (Stage AP0: SBP<150, DBP <95 mmHg, or both) and to significantly lower the magnitude of proteinuria (UPC <0.5 is the primary goal; UPC reduced by 50% or more if the primary goal is not achievable). Certainly, a minimal goal of antihypertensive therapy is to achieve a reduction in stage of risk for TOD (Table 2). Except in a hypertensive crisis wherein severe ocular or central nervous system TOD is present, this is not an emergency and BP lowering should be achieved with a gradual, persistent reduction in BP achieved over several weeks.

85% of voting consensus members agreed with Recommendation 11, and 40% of these voters expressed "strong agreement."

Although available evidence suggests that sodium restriction alone generally does not reduce BP,³³ high salt intake may produce adverse consequences in some settings^{33,62} but not others.⁶³ Although the BP of normal dogs is minimally affected by variations in NaCl intake,³³ it is likely that dogs with kidney disease, particularly those with NS, are salt-sensitive. Furthermore, salt restriction enhances the antihypertensive efficacy of some antihypertensive agents, particularly those that interfere with the RAAS. This evidence serves as the basis for recommendation 7 to feed a diet with modified sodium chloride content to dogs with proteinuric kidney disease, regardless of IRIS stage.

Clinical experience indicates that RAAS inhibitors reduce BP slightly (~10–15%) and are antiproteinuric. For a variety of reasons, these agents will already be used in dogs with glomerular disease (including those classified as AP0 or AP1). If hypertension is identified in a dog with glomerular disease that is not receiving an ACEi or an ARB, institution of an ACEi is an appropriate first step. The starting dosage should be at or above the lower end of the recommended range (Table 3). The upper limit of our recommended dosage range for the ACEi is controversial as some experts will stop at or below this dosage, while others will increase the ACEi dosage even further. Most consider the addition of a calcium channel blocker (ie, amlodipine), or an ARB, to be an appropriate next consideration. If an antihypertensive agent of choice is only partially effective, the usual approach is to increase the dosage or add an additional drug.

Recommendation 12:

In dogs with glomerular disease and either severe systemic hypertension (AP3) or a hypertensive emergency (eg, systolic BP above 200 mmHg or evidence of ocular or neurologic target organ damage), co-administration of two agents with different mechanisms of action (generally an ACEi plus amlodipine) is recommended.

100% of voting consensus members agreed with Recommendation 12, and 55% of these voters expressed "strong agreement."

Proteinuric dogs with significant hypertension will usually require more than two antihypertensive agents. In this setting, the general consensus is to add the calcium channel blocker, amlodipine, to RAAS inhibition. Some uncommon disease conditions may be best addressed with the addition of specific classes of agents in addition to RAAS inhibitors plus amlodipine, such as alpha- and beta-blockers or surgical excision for pheochromocytoma or an aldosterone receptor blocker or surgical excision of an adrenal tumor in animals with hypertension associated with hyperaldosteronism. Diuretics are not commonly used in dogs with CKD, but may be useful, especially in patients with concurrent hypertension and NS (see below).

The benefits of BP lowering on TOD within the kidney (ie, slowing of progressive decline in GFR and reduction in UPC) are directly dependent on the degree of proteinuria in people⁵⁶ and cats,⁵⁷ leading to

Table 3. Oral agents for antihypertensive therapy in dogs with glomerular disease^a

Class	Drug (Examples of Trade Name)	Usual Oral Dosage
ACEi/ARB		
Angiotensin-converting enzyme inhibitor	Benazepril (Lotensin; Fortekor)	see Table 1
	Enalapril (Vasotec; Enacard)	see Table 1
	Ramipril	see Table 1
	Imidapril	see Table 1
Angiotensin receptor blocker	Telmisartan	see Table 1
	Losartan (Cozaar)	see Table 1
CCB		
Calcium channel blocker	Amlodipine (Norvasc)	0.1–0.75 mg/kg q24h
Other Agents		
β blocker	Atenolol (Tenormin)	0.25–1.0 mg/kg q12h
	Prazosin (Minipress)	0.5–2 mg/kg q8–12h
α ₁ blocker	Phenoxybenazime (Dibenzyline)	0.25 mg/kg q8–12h or 0.5 mg/kg q24h
	Hydralazine (Apresoline)	0.5–2 mg/kg q12h (start at low end of range)
Direct vasodilator	Acepromazine (PromAce)	0.5–2 mg/kg q8h
Diuretics		
Thiazide diuretic	Hydrochlorothiazide (HydroDiuril)	2–4 mg/kg q12–24h
Loop diuretic	Furosemide (Lasix)	1–4 mg/kg q8–24h
Aldosterone receptor blocker	Spironolactone (Aldactone, Prilactone)	see Table 1

^aAgents that interfere with the renin-angiotensin-aldosterone system (ie, ACEIs, ARBs, or aldosterone blockers) are used to manage dogs with glomerular disease that exhibit proteinuria, hypertension, or both. Although the end-points differ when managing proteinuria (UPC is the end-point) and hypertension (BP and the UPC are used to assess the efficacy of antihypertensive therapy), the dosing and follow-up are essentially the same when approaching either problem or both together in a canine patient.

our consensus that achievement of BP control is more important in proteinuric dogs.

Besides the antiproteinuric effects of agents that interfere with the RAAS, other classes of agents may be considerations for the future. The CCB blocker in common use clinically (ie, amlodipine) is an L-type calcium channel blocker, which appear to be less antiproteinuric because of preferential dilatation of the afferent arteriole in dogs,^{55,64} forming the basis for the recommendation of co-administration of RAAS inhibitors. There are alternative CCB, which are L/T (eg, efonidipine)⁶⁵ and L/N blockers (eg, clinidipine)⁶⁶ both of which are more likely to effectively dilate the efferent arteriole than amlodipine. Furthermore, the L/N blockers appear to be sympatholytic.⁶⁶ Both of these newer classes of calcium channel blockers hold promise because of their potential for renoprotection but are, as of yet, not tested in the clinical setting in dogs with glomerular disease.

An exception to the above graded approach to BP reduction is dogs with AP3 and evidence of severe or progressing neural or ocular TOD, which is generally taken to constitute an emergency. In this setting, more aggressive, combination therapy with a RAAS inhibitor plus amlodipine is an appropriate first step.

Monitoring Antihypertensive Therapy

Ocular and neural TOD may result in a hypertensive crisis, necessitating rapid lowering of BP.⁵⁴ In most other situations, hypertension is not an emergency and 2–4 weeks should be allowed between dosage adjustments. There has been concern about acute exacerbation of azotemia with ACEi, although this is an unusual complication and modest increases in serum creatinine concentration (<30%) may occur and are generally tolerable in IRIS CKD stages 1–2 and early stage 3.

Recommendation 13:

A dog with glomerular disease in IRIS CKD stages 1 or 2 should be evaluated 3–14 days following any change in antihypertensive therapy. In unstable patients and those with IRIS stage 3 or 4 CKD, this recheck should be conducted in a shorter timeframe, typically 3–5 days. Patients deemed to be hypertensive emergencies and hospitalized patients, particularly those receiving fluid therapy or pharmacological agents with cardiovascular effects, should be assessed daily. The purpose of these short-term assessments is to determine if there are any findings that are unexpected (eg, new or worsening TOD) or adverse (eg, marked worsening of azotemia or systemic hypotension).

100% of voting consensus members agreed with Recommendation 13, and 55% of these voters expressed “strong agreement.”

Treatment should be adjusted downward if an increase in serum creatinine of >30% or a BP <120/60 mmHg (combined with clinical findings of weakness or syncope) is observed.⁵⁴

Re-evaluation at 1–4 month intervals is recommended, with chosen interval depending on stability of BP and stage of disease (more frequent if BP or other conditions are unstable, late stage 3 or stage 4 CKD) and level of elevation (more frequent in AP3) is appropriate. Follow-up evaluations, which are employed to assess efficacy of treatment and make adjustments if appropriate, should include measurement of BP, serum creatinine concentration, urinalysis with UPC, funduscopic examination, and other specific assessments depending on the individual circumstances (eg, TOD, causes of secondary hypertension, and concurrent conditions) of the patient. In dogs, a key predictive parameter for antihypertensive efficacy is the effect of treatment on the magnitude of proteinuria (UPC)^{1–3,20,55,64} and a benefit is predicted if the antihypertensive regimen used is antiproteinuric (ie, reduces UPC by at least 50%, preferably to <0.5). The frequency and nature of re-evaluations will vary depending on the BP substage, stability of BP, other aspects of the health of the patient, and frequency of dosage adjustment to antihypertensive therapy. As signs of progression of TOD can be subtle, BP should be closely monitored over time in patients receiving antihypertensive therapy, even when hypertension is seemingly well controlled.

Fluid and Diuretic Therapy in Dogs with Glomerular Disease

Alterations of fluid homeostasis are common in small animals with glomerular diseases and they include excesses, deficits, and intercompartmental maldistribution.^{38,67} These alterations may require interventions as a supportive measure for anesthesia (renal biopsy, unrelated surgery), as a compensation for increased losses such as those associated with gastrointestinal complications, or to decrease severe interstitial fluid accumulation in nephrotic patients. The restoration of disturbed fluid homeostasis can, however, be challenging in dogs affected with glomerular disease. Attempts to treat these animals commonly result either in exacerbation of peripheral edema and systemic hypertension with fluid administration or in worsening azotemia and uremic crisis with diuretic therapy. This difficulty is likely a consequence of our poor understanding of the mechanisms underlying edema formation and fluid maldistribution in glomerular diseases and of our limited ability to evaluate precisely and accurately the various clinical parameters of fluid balance in individual animals.

Mechanisms of Edema Formation in Glomerular Diseases and Patient Evaluation

Edema formation in nephrotic patients has been explained classically as being attributable to decreased

plasma colloid osmotic pressure resulting from the hypoalbuminemia associated with glomerular protein loss. According to this explanation, leakage of plasma water into the interstitium results in hypovolemia, activates the RAAS, causes sodium and fluid accumulation, and thus worsens edema formation.⁶⁸ However, this “underfill theory” has been challenged based on clinical observations and results of studies in models of NS, which indicate a lack of evidence of hypovolemia in most patients and a lack of efficacy of colloid therapy or RAAS blockade for the resolution of nephrotic edema.⁶⁹ Experimentally, severe hypoalbuminemia or even analbuminemia alone, is rarely sufficient to cause severe edema since the colloid osmotic pressure of the interstitium decreases in parallel to that of the plasma, maintaining a similar transcapillary oncotic gradient.⁷⁰ Clinically, except when suffering from additional fluid losses, most patients do not appear to be in a volume-contracted state.

Evidence of inappropriate renal tubular sodium retention and excessive volume expansion as a major causative mechanism for edema formation is accumulating from rodent models of NS.^{70,71} Although this “overflow theory” cannot necessarily be extrapolated to dogs, it could explain some of the controversial clinical observations such as the worsening of edema sometimes observed with colloid use, the high prevalence of systemic hypertension, and the poor response of hypertension to RAAS blockade alone.⁷² Furthermore, the “vascular hyperpermeability theory” suggests an additional contributor to interstitial fluid accumulation in some forms of systemic vascular and glomerular inflammation. A recent review of the clinical manifestations of NS did not provide conclusive evidence for or against these 3 theories, possibly indicating a combination of potential mechanisms and their respective dominance in certain defined conditions.⁶⁷

Fluid Volume Status: Patient Assessment

Recommendation 14:

As data on the prevailing mechanisms of nephrotic edema in the canine species are lacking, careful assessment of the hydration, and of the vascular volume of individual dogs with glomerular disease should be a priority both before and during fluid therapy.

90% of voting consensus members agreed with Recommendation 14, and 45% of these voters expressed “strong agreement.”

Nephrotic patients are, per definition, overhydrated. However, this interstitial fluid accumulation can be associated with either vascular volume contraction or expansion.⁷³ The evaluation of the volume status of an individual animal with NS should therefore focus primarily on the intravascular volume and aim at differentiating hypovolemic-underfilled from hypervo-

lemic-overfilled dogs. Non-nephrotic animals with glomerular disease should be evaluated similarly. However, as our ability to predict iatrogenic edema formation with fluid therapy is limited, these dogs should be considered at risk with fluid therapy being prescribed conservatively, rather than assuming that they are volume contracted.

Recommendation 15:

The evaluation of the fluid status of a dog with glomerular disease should be based on anamnestic data (eg, serial body weight) and a complete physical examination with special emphasis on skin turgor, mucous membrane color and moisture, capillary refill time, temperature of the extremities, heart rate, pulse quality, and systemic BP.

95% of voting consensus members agreed with Recommendation 15, and 55% of these voters expressed “strong agreement.”

More objective techniques for measurement of body composition such as isotope dilution or bioimpedance spectroscopy could help the assessment of fluid distribution, but they are rarely available clinically and they remain to be validated for dogs in this setting. Indirect markers of vascular expansion (eg, natriuretic peptides) and echocardiographic parameters have been suggested for the evaluation of the volume status in nephrotic people⁷³ and they could prove useful for assessment of dogs in the future. A clinical study in children with nephrotic edema indicated that volume-contracted patients had a higher BUN, higher BUN/creatinine ratio, higher urine osmolality, and lower fractional excretion of sodium compared with volume-expanded patients. They also had significantly higher renin, aldosterone, and antidiuretic hormone levels.⁷⁴ Whether these simple markers could prove useful for evaluation of dogs with nephrotic edema remains to be evaluated.

Nephrotic Syndrome: Indications for Intervention

Treatment of nephrotic edema and effusion is addressed primarily through treatment of the underlying etiology and reduction in proteinuria. Specific treatment of edema and drainage of effusions are reserved for the symptomatic support of animals with significant respiratory distress, abdominal discomfort, or other severe complications caused by excessive fluid accumulation.

Recommendation 16:

Because there is no rationale for diuretic intervention in dogs with glomerular disease and mild peripheral edema, use of diuretics in dogs with edema should be limited to situations where organ function is critically impaired (eg, ascites or pleural effusion that impairs respiration).

90% of voting consensus members agreed with Recommendation 16, and 35% of these voters expressed “strong agreement.”

Overfilled, stable dogs with NS typically should not require fluid therapy either, even for short elective anesthesia such as required for ultrasound-guided renal biopsy, for example. Unstable nephrotic animals necessitating anesthesia, however, require a thorough global evaluation and the need for intervention has to be assessed individually, taking into consideration volume, oncotic, and osmotic parameters.⁷³ If deemed absolutely necessary, fluid administration should remain very conservative and be re-evaluated frequently. There are no data to support the combination of diuretics and fluids. Severe pleural and abdominal effusions are more efficiently removed by thoraco- and abdominocentesis, respectively.

Underfilled dogs with or without edema may require therapeutic volume expansion for acute clinical manifestations (eg, vomiting, diarrhea), peri-operatively, or for worsening azotemia. As the rate of complications is high in animals with glomerular disease treated with fluids, the exact indication and the specific goals of fluid therapy need to be assessed critically and the fluid rates chosen very conservatively. Careful monitoring

and re-evaluation of the response and safety of the treatment guide the fluid prescription (see Fig 3).

Non-nephrotic dogs with IRIS CKD stages 3–4 commonly benefit from subcutaneous fluid administration with improved appetite and activity. It is therefore likely that a similar benefit could be expected in dogs with glomerular disease reaching these advanced stages of CKD. The fluid and sodium load associated with this treatment should, however, be carefully considered, especially in hypertensive and volume-expanded, overfilled dogs or those with NS. Careful titration of the fluid dose, monitoring of the resorption of the administered fluid, and evaluation of potential adverse effects should be performed when this treatment is chosen.

Nephrotic Syndrome: Diuretic Therapy

When intervention is indicated, the loop diuretic furosemide is usually recommended as the initial choice for the treatment of overfilled, nephrotic humans.^{70,73,74} To promote natriuresis and diuresis, this diuretic needs to reach a threshold concentration at its site of action in the thick ascending limb of the loop of Henle, through active secretion into the proxi-

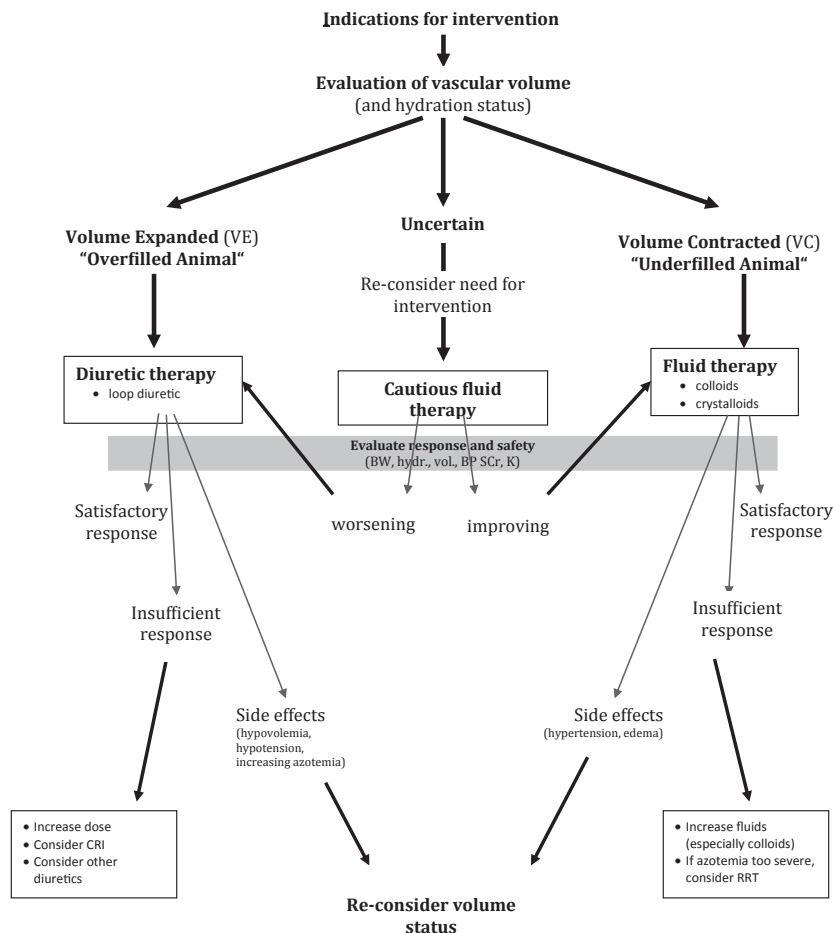


Fig 3. Algorithm for treatment of fluid imbalances in dogs with glomerular disease (BW, body weight; hydr., hydration; vol., volemia; BP, blood pressure; SCr, serum creatinine; K, potassium; CRI, constant-rate infusion; RRT, renal replacement therapy).

mal tubule. Pharmacokinetic alterations are common in patients with glomerular diseases, including decreased renal diuretic delivery and secretion, increased protein binding of the diuretic in the urine, increased renal metabolism, and increased sodium and water reabsorption in downstream segments.⁷⁰ Altered threshold of action and decreased maximal diuretic response often necessitate a careful titration with higher doses and more frequent administration in nephrotic patients.⁷⁵ Administration as a constant-rate infusion and titration to effect have been recommended for refractory cases to overcome the short half-life of loop diuretics and to avoid significant postdose sodium retention by the kidney. Increased diuretic efficacy has been shown in normal Greyhounds when furosemide was administered as a CRI as opposed to intermittent bolus injections.⁷⁶ However, no data are available for nephrotic dogs.

Furosemide is routinely co-administered with IV albumin in nephrotic humans with severely decreased serum albumin concentration aiming to overcome diuretic resistance and to avoid hypovolemia.⁷⁷ However, furosemide alone was shown to be equally safe and effective in a study involving nephrotic children⁷⁴ and there are no data to support this strategy in nephrotic dogs.

Humans commonly require a combination approach with distally acting diuretics, such as spironolactone or thiazides.⁷⁵ This sequential nephron blockade needs to be performed with caution, because a combination with thiazide diuretics can induce a marked kaliuresis and the use of potassium-sparing diuretics, such as spironolactone, can result in severe hyperkalemia, especially when combined with an ACEi or ARB for the treatment of proteinuria. Although anecdotal accounts provide some support for the efficacy of spironolactone in dogs with NS, neither single-drug nor combination protocols have been evaluated in dogs with glomerular disease.

Recommendation 17:

When indicated in a dog with glomerular disease, diuretic therapy with furosemide or spironolactone may be instituted. Furosemide may be the first choice drug in dogs with pulmonary edema or hyperkalemia and spironolactone for dogs with pleural or abdominal effusion. Furosemide may be administered at an initial dosage of 1 mg/kg q6–12h, with incremental increases of 0.5–1 mg/kg q6–12h or conversion into continuous IV infusion at a rate of 2–15 µg/kg/min after an initial loading dose of 2 mg/kg in animals with insufficient response. Spironolactone may be started at 1 mg/kg q12–24h and titrated in increments of 1 mg/kg q12–24h to a maximum of 4 mg/kg q12–24h.

76% of voting consensus members agreed with Recommendation 17, and 25% of these voters expressed “strong agreement.”

Goals of diuretic treatment should include a slow progressive decrease in the edema to a clinically acceptable condition. Overaggressive treatment should be avoided because the resulting vascular volume depletion can lead to worsening azotemia, venous stasis, and thromboembolic disease. Treatment efficacy should be assessed clinically by serial body weight measurements. Electrolytes, especially potassium, should be monitored closely in all animals with glomerular disease treated with diuretics, initially within 1 week of initiating treatment. As successful diuresis can cause exaggerated intravascular volume depletion and acute decompensation of renal function, physical examination and serum creatinine should be monitored even more closely, as early as 1–2 days after initiation of diuretic therapy in critical patients.

The use and efficacy of additional measures to mobilize edema is controversial and based purely on pathophysiological considerations.⁷⁸ Dietary sodium restriction may seem logical and useful in the light of inappropriate renal sodium retention in the pathogenesis of nephrotic edema. Mild and regular exercise may help eliminate excessive fluid accumulation and decrease risk of thromboembolism.⁷⁹ Although sometimes recommended for dogs with severe edema, recommendations to institute cage rest in dogs with NS are not supported by evidence and this approach could potentially prove detrimental by worsening vascular stasis and thrombotic risk.

Nephrotic Syndrome: Fluid Therapy

Recommendation 18:

Fluid replacement therapy should be used with great caution in dogs with glomerular disease because they are predisposed to fluid overload. Intravenous fluid therapy is indicated for hemodynamic stabilization (eg, a patient with dehydration, poor tissue perfusion) of the patient. Colloids or plasma/albumin should not be administered solely on the basis of decreased oncotic pressure, serum albumin concentration, or total protein concentration, or for mobilizing edema. The end-point of treatment should be the patient's response to treatment (ie, correction of tissue hypoperfusion, hypotension, hypovolemia, dehydration). Judicious administration of colloids should be considered when crystalloid fluid support has failed to correct the patient's hemodynamic dysfunction.

95% of voting consensus members agreed with Recommendation 18, and 30% of these voters expressed “strong agreement.”

Early evaluation of the clinical response should guide changes in fluid type and administration rate. Further monitoring should include at least twice-daily physical examination, with special attention to hydration and blood volume; twice daily monitoring

of body weight, BP, and urine output; and daily measurement of serum creatinine, BUN, and electrolytes.

Although the pathophysiologic basis of the “underfilled theory” supports the use of colloids in dogs with NS, there are limited clinical data on their utility in dogs with glomerular disease. Recent studies in people have suggested that colloids should not be routinely administered for mobilizing edema^{80,81} and may have adverse effects on the kidney.⁸⁰ It is appropriate to consider the judicious administration of colloids whenever crystalloid fluid support has failed to correct the patient’s hemodynamic dysfunction (ie, tissue hypoperfusion, systemic hypotension, hypovolemia, dehydration).

When the fluid status cannot be established with certainty, a very conservative approach should be chosen, with critical re-evaluation of the real or perceived needs of the animal. Initially, a similar fluid therapy protocol as for hypovolemic dogs seems safe and probably appropriate for most animals. It should, however, be monitored even more closely for adverse effects and complications, with physical examinations, urine production, and BP monitoring every 6 h. The trend of the clinical status will either confirm the treatment choice or, in case of worsening edema and systemic hypertension, a change in strategy using diuretics should be considered to improve salt or water excretion or both.

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Consensus Recommendations for Immunosuppressive Treatment of Dogs with Glomerular Disease Based on Established Pathology

IRIS Canine GN Study Group Established Pathology Subgroup

Consensus Recommendations for Immunosuppressive Treatment of Dogs with Glomerular Disease Based on Established Pathology

IRIS Canine GN Study Group Established Pathology Subgroup, G. Segev, chair, L.D. Cowgill, R. Heiene, M.A. Labato, and D.J. Polzin

The purpose of this report was to provide consensus recommendations for the use of immunosuppressive therapy in dogs with active glomerular diseases. Recommendations were developed based on comprehensive review of relevant literature on immunosuppressive therapy of glomerular disease in dogs and humans, contemporary expert opinion, and anecdotal experience in dogs with glomerular disease treated with immunosuppression. Recommendations were subsequently validated by a formal consensus methodology. The Study Group recommends empirical application of immunosuppressive therapy for dogs with severe, persistent, or progressive glomerular disease in which there is evidence of an active immune-mediated pathogenesis on kidney biopsy and no identified contraindication to immunosuppressive therapy. The most compelling evidence supporting active immune-mediated mechanisms includes electron-dense deposits identified with transmission electron microscopic examination and unequivocal immunofluorescent staining in the glomeruli. For diseases associated with profound proteinuria, attendant hypoalbuminemia, nephrotic syndrome, or rapidly progressive azotemia, single drug or combination therapy consisting of rapidly acting immunosuppressive drugs is recommended. The Study Group recommends mycophenolate alone or in combination with prednisolone. To minimize the adverse effects, glucocorticoids should not be used as a sole treatment, and when used concurrently with mycophenolate, glucocorticoids should be tapered as quickly as possible. For stable or slowly progressive glomerular diseases, the Study Group recommends mycophenolate or chlorambucil alone or in combination with azathioprine on alternating days. Therapeutic effectiveness should be assessed serially by changes in proteinuria, renal function, and serum albumin concentration. In the absence of overt adverse effects, at least 8 weeks of the rapidly acting nonsteroidal drug therapy and 8–12 weeks of slowly acting drug therapy should be provided before altering or abandoning an immunosuppressive trial.

Key words: Canine; Formal consensus; Immunosuppression; Kidney biopsy; Proteinuria.

There are many causes and pathogenic mechanisms associated with glomerular diseases in dogs.^{1–3} The etiopathogenesis may be linked to genetic mutations, infectious agents, inflammatory, neoplastic, and immune-mediated diseases.⁴ In the majority of instances, the underlying cause remains unknown, and the disease is classified as idiopathic. There has been a dramatic improvement in treatment outcomes in human patients over the last 3–4 decades after establishment of the World Health Organization classification scheme for human glomerular disease, based on light, immunofluorescent, and electron microscopic imaging, and subsequent controlled clinical trials of pathologic entities based on this classification scheme.⁵ To date, data are available on the efficacy and improved outcomes attending use of immunosuppressive therapy across all types of primary and secondary

Abbreviations:

CBC	complete blood count
CKD	chronic kidney disease
GBM	glomerular basement membrane
IRIS	International Renal Interest Society
PCR	polymerase chain reaction
UPC	urine protein to creatinine ratio
WSAVA RSP	World Small Animal Veterinary Association Renal Standardization Project

glomerular diseases in human patients.⁶ Classification of human glomerular diseases based on the patient's clinical and pathological characteristics has proven useful in enhancing the therapeutic response in affected individuals.

As a consequence of the World Small Animal Veterinary Association Renal Standardization Project (WSAVA-RSP), more comprehensive light histopathologic imaging,^a as well as electron microscopic and immunofluorescent microscopic imagery, has become readily available for use in dogs with glomerular disease. Examination of kidney biopsies using these advanced imaging techniques allows greater insight into the morphologic features of glomerular pathology and foundation for an improved and more discriminating classification system for glomerular diseases of dogs. Establishment of more specific pathological diagnoses holds the potential to improve conventional therapeutic and prognostic recommendations for dogs with glomerular disease. When kidney biopsy establishes evidence of an immune-mediated contribution to the glomerular disease, use

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of immunosuppressive/anti-inflammatory therapy may be rational and indicated. However, despite the logic of immunosuppressive strategies for these diseases, compelling evidence supporting this therapeutic practice in dogs is lacking. Currently, any use of immunosuppressive drug therapy, which may portend detrimental effects, should be founded on substantive criteria to objectively balance potential benefits against predictable adverse effects.⁷

Materials and Methods

The recommendations presented in this report were developed with consideration of a comprehensive review of pertinent literature on immunosuppressive therapy of glomerular disease, expert opinion, and review of a selected group of dogs with glomerular disease treated with immunosuppressive therapies. Recommendations were subsequently validated by a formal consensus methodology (see, "Development of Clinical Guidelines for Management of Glomerular Disease in Dogs" in this supplement). The Pubmed Medline database was searched for relevant studies published from 1975 to 2013 (literature search date) using the following MeSH terms and text words: immunosuppression, kidney, glomerular disease, chronic kidney disease, cyclosporine, mycophenolate mofetil, azathioprine, chlorambucil, cyclophosphamide, glucocorticoids, and dog. The search included combinations of terms and text words at the discretion of the authors to detect potentially relevant citations. The literature search was not limited to certain study designs or languages.

Use of Immunosuppressive Drugs in Dogs with Glomerular Disease

Recommendation 1:

The empirical application of immunosuppressive/anti-inflammatory therapy is recommended for dogs with severe, persistent, or progressive glomerular disease in which there is renal biopsy-supported evidence of an active immune pathogenesis and no identified contraindication to immunosuppressive therapy.

100% of voting consensus members agreed with Recommendation 1 and 60% of these voters expressed "strong agreement."

The rationale underlying the recommendation to initiate immunosuppressive therapy is based on the prediction that suppression of humoral or cell mediated immunity and the associated glomerular inflammatory response will favorably influence the progression, severity, and clinical outcome of the disease. Currently, there is limited experience or evidence available to predict which specific therapy will influence positively or negatively the clinical outcome, specific patterns of glomerular injury, or immunologic activity. Until such evidence becomes available, recommendations remain anecdotal and consensus-based and should be applied cautiously on a case-by-case and trial basis. The treatment should be discontinued or modified if it fails to achieve treatment goals or if the treatment promotes intolerable adverse effects (see below). Before applying immuno-

suppressive protocols, a systemic search using PCR and serology-based diagnostics is indicated to identify any underlying or concurrent infectious disease (see Recommendations for Therapy for Dogs with Serology Positive Glomerular Disease).

Evidence Supporting an Immune-Mediated Pathogenesis in Canine Glomerular Disease

The most definitive criteria implicating an immunopathogenesis in glomerular disease is identification of components of the immune system associated with active and ongoing pathologic injury in the glomerulus. By extension, a documented immunopathogenesis should be established before the use of immunomodulatory therapy. Regrettably, there are no readily available serologic or peripheral markers or noninvasive diagnostic modalities in dogs to characterize and establish these criteria. The evidence must come from the glomerulus itself and is evident only with specialized light, immunofluorescent, and electronmicroscopic imaging of appropriately processed kidney biopsies. An immunopathogenesis is most compelling when there is consistent and multifaceted coherence of findings across these imaging modalities, which correspond with the clinical features in the animal. When there is a lack of correspondent information or an incomplete set of diagnostic imaging, decisions supporting an immunopathogenesis, and the justification to use immunosuppressive therapy, must be prioritized from a hierarchy of information from the available imaging modalities and observations.

Recommendation 2:

Evidence of an active immune-mediated pathogenesis is supported by the following findings:

The most compelling evidence supporting active immune-mediated mechanisms promoting the glomerular injury identified in the kidney biopsy include:

- By Electron Microscopy: Clear and definitive identification of electron-dense deposits in subendothelial, subepithelial, intramembranous, or mesangial locations in the glomerulus (Fig 1).
- By Immunofluorescent Microscopy: Positive and unequivocal immunofluorescent staining for IgG, IgM, IgA, light chains, and/or complement in an immune-complex (intermittent) or antiglomerular basement membrane (GBM) (linear) pattern of deposition in peripheral capillary loops and/or the mesangial compartment (Fig 2).

Probable evidence supporting active mediated immune mechanisms promoting the glomerular injury identified by light microscopy in the kidney biopsy include:

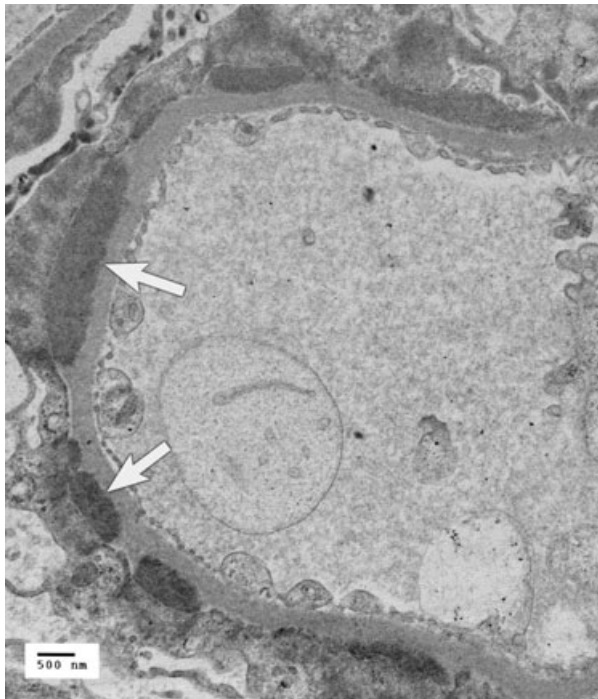


Fig 1. Transmission electron micrograph (15,000 $\times$) of a canine glomerular capillary loop demonstrating deposition of multifocal 0.25–2.0 micron nodular, subepithelial electron-dense deposits (arrows) consistent with immune complexes (Courtesy of Drs George E. Lees and Fred J. Clubb, Texas Veterinary Renal Pathology Service).

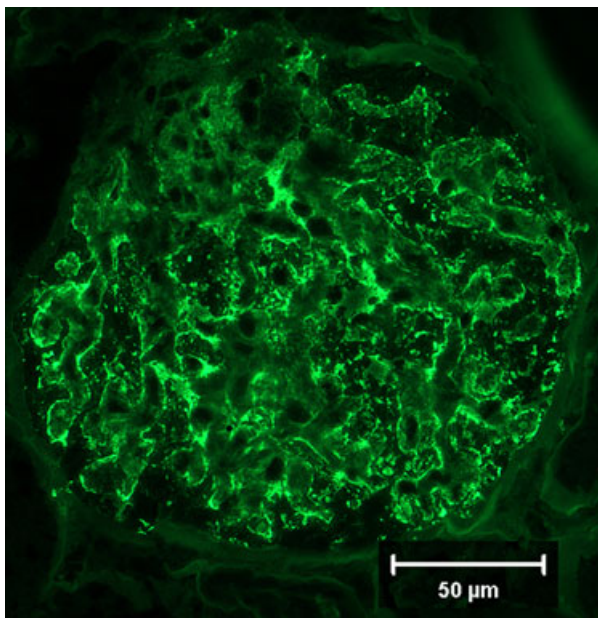


Fig 2. Immunofluorescent image of a canine glomerulus (40 $\times$) immunostained for IgG demonstrating extensive global coarse granular capillary wall staining and segmental fine granular mesangial staining documenting immune deposits and predicting an immune-driven pathogenesis to the glomerular injury (Courtesy of Dr George E. Lees, Texas Veterinary Renal Pathology Service).

- Convincing identification of “red granular” staining on the capillary walls on Masson’s trichrome-stained sections (probable immune deposits) (Fig 3).
- Convincing identification of “spikes” along the glomerular basement membrane on Jones Methenamine silver-stained sections (Fig 4).
- Identifiable “holes” within the glomerular basement membrane on Jones Methenamine silver-stained sections (Fig 4).

91% of voting consensus members agreed with Recommendation 2, and 60% of these voters expressed “strong agreement.”

As a caveat to these recommendations, it is possible to identify intramembranous electron-dense material with electron microscopy that represents residual immunological material deposited in the GBM during a previous immunological insult that is no longer associated with active or ongoing immunological events or susceptible to immunosuppressive therapy. Note also that if the EM findings provide evidence for chronicity, complete or rapid response to treatment may not be expected. Similarly, it is possible to identify “spikes” or “holes” on silver-stained sections using light microscopy that represent residual pathological reactions and remodeling of the GBM to a previous immunological insult that has resolved and is no longer active or susceptible to immunosuppressive therapy. Pathologic immune-product deposition also must be distinguished from nonspecific immunoglobulin entrapment in the

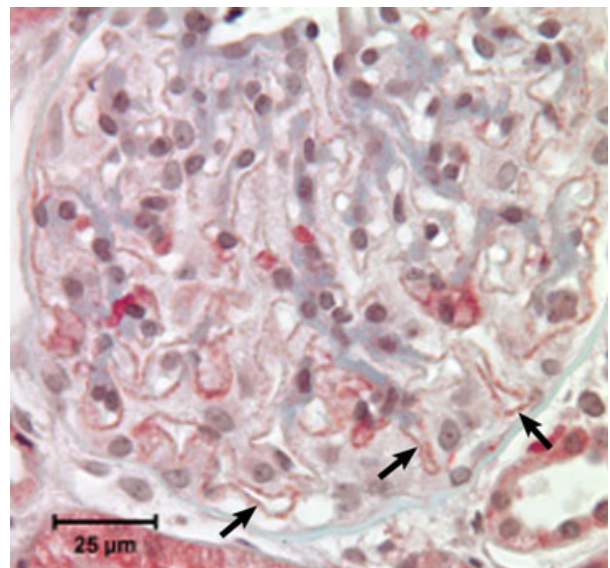


Fig 3. Light histologic image of a Masson’s Trichrome-stained canine glomerulus (100 $\times$) demonstrating extensive red granular deposits along the subepithelial aspect of the capillary walls (black arrow) consistent with immune deposits on the glomerular basement membrane (Courtesy of Dr George E. Lees, Texas Veterinary Renal Pathology Service).

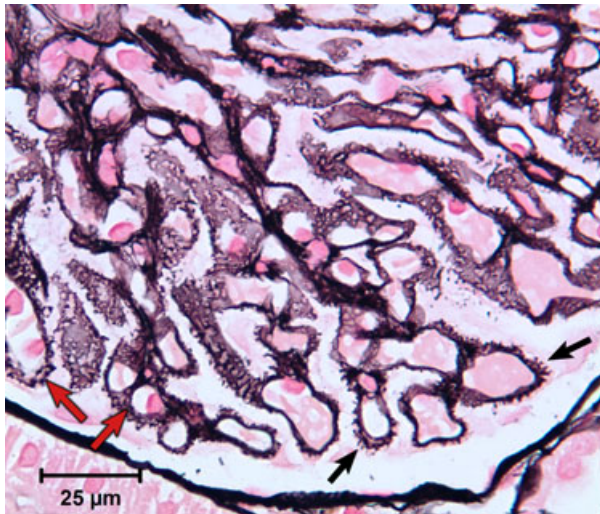


Fig 4. Light histologic image of a Jones Methenamine silver-stained canine glomerulus (100 $\times$) demonstrating numerous prominent spikes of glomerular basement membrane (GBM) material (black arrow) and holes (white arrow) within the GBM on the subepithelial surface. These features represent remodeling of the GBM around deposited immune deposits on or in the GBM (Courtesy of Dr George E. Lees, Texas Veterinary Renal Pathology Service).

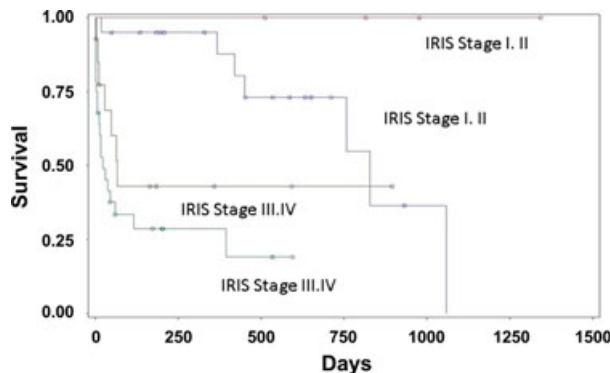


Fig 5. Plot of survival in dogs with unclassified proteinuric glomerular diseases with IRIS Stage I, II CKD, or IRIS Stage III, IV CKD, respectively, who were treated or untreated with immunosuppressive therapy. Although there were no statistical differences between the treated and untreated groups at each IRIS Stage, the trends suggest a survival benefit in dogs managed with immunosuppressive (and potentially anti-inflammatory) therapy (Courtesy, Drs Beth Orchutt and David Polzin, University of Minnesota, 2011).

filtration path of some lesions without an immunopathogenesis, ie, amyloidosis.

The judgment as to the involvement and active status of an immune process in the genesis of the glomerular disease is the purview of the reviewing nephropathologist(s) as the interpretative coherence of these lesions is beyond the scope and experience of the attending clinician. Care must be exercised not to over-interpret out-of-context descriptors provided in patho-

logic reports necessarily as evidence or proof of immunologic injury. This is especially true when descriptions are derived singularly from limited staining and review of light microscopic preparations.

Drug Options for Immunosuppressive Therapy of Glomerular Disease in Dogs

Recommendation 3:

The immunosuppressive strategy selected to manage dogs with documented glomerular disease of an apparent immune-mediated pathogenesis should be selected on the basis of the severity of the disease and its rate of progression. For diseases associated with high magnitude proteinuria, attendant hypoalbuminemia, nephrotic syndrome, or rapidly progressive azotemia, single drug, or combination therapy consisting of rapidly acting immunosuppressive drugs is recommended.

100% of voting consensus members agreed with Recommendation 3, and 60% of these voters expressed "strong agreement."

In the absence of controlled clinical trials of specific immunosuppressive drugs performed in dogs with glomerular disease of defined immune-mediated pathogenesis, no single drug recommendation can be made at this time. The use and selection of any immunosuppression protocol must be founded on an understanding of the actions, adverse effects, and costs of the selected agent(s). The following list of potential agents that may be effective in dogs with acute or rapidly progressive glomerular disease is based on their mode of action, their use in human patients with glomerular disease, uncontrolled or anecdotal experience of their use in dogs with spontaneous glomerular disease, or their use in dogs with other immune-mediated diseases.

Glucocorticoids

Glucocorticoids have an established basis in veterinary therapeutics for a variety of immune-mediated diseases in dogs, but they have been largely avoided in the management of glomerular disease unless associated with concurrent conditions known to be glucocorticoid responsiveness (eg, dogs with canine glomerular disease and immune-mediated polyarthritis). The avoidance of glucocorticoid therapy likely stems from the characteristic adverse effects (relative to benefits) of these agents in dogs compared to humans and other species. Expected adverse effects include polydipsia/polyuria, polyphagia, pendulous abdomen, panting, increased proteinuria, negative nitrogen metabolism, hypercatabolism and muscle loss, increased hypercoagulable and thromboembolic potential, sodium and fluid retention, systemic hypertension, behavioral changes, and adrenal suppression. Furthermore, the use of glucocorticoids should be considered carefully in the

presence of systemic or urinary tract infections to prevent worsening of these conditions (eg, pyelonephritis). These clinical perceptions are weakly supported by an observational study demonstrating worse outcomes in dogs with glomerulonephritis when managed with prednisone.⁸

In contrast with management schemes in dogs, management of glomerular disease in humans often relies heavily on the use of daily or pulse glucocorticoid therapy as a first-line treatment or in conjunction with other immunosuppressive agents for the management of virtually all types of primary glomerular disease.^{5,9} This difference in therapeutic practice may be predicated on the tendency of glucocorticoids to promote proteinuria in dogs, perceived species differences in the risk-to-benefit ratio in human versus dogs, respectively, possible differences in the immunopathogenesis of canine glomerular diseases, or the lack of documented evidence-based efficacy assessment for glucocorticoids in canine glomerular diseases. Clinicians should be aware of this difference, when seeking information, therapeutic guidance, or both from the human literature. A proportion of human patients within most glomerular disease categories may be categorized as “steroid-resistant” and switched to other protocols until a treatment effect is observed. The apparent species differences in the clinical and pathophysiological responses to corticosteroids should prompt careful selection of patients if they are to be used.

Recommendation 4:

Short-term administration of glucocorticoids may be appropriate in fulminate cases where immediate immunosuppression is required if their use is adjusted to minimize their adverse effects. Potential benefits of alternative treatment regimens, ie, pulse therapy versus continuous therapy, should be considered.¹⁰ However, on the basis of current practice perceptions and anecdotal experience, the use of glucocorticoid therapy should be tapered to the minimally effective dose as quickly as possible because of predictable adverse effects. Glucocorticoids may also be appropriate in the acute management of multisystemic immune-mediated diseases where their use has proven beneficial (eg, concurrent polyarthritis, immune-mediated hemolytic anemia).

100% of voting consensus members agreed with Recommendation 4, and 40% of these voters expressed “strong agreement.”

Mycophenolate

Mycophenolate is an immunosuppressive agent that inhibits inosine monophosphate dehydrogenase, the rate-limiting enzyme in the de novo synthesis of guanosine nucleotides. Its immunosuppressive activity results from selective inhibition of both T-cell and B-cell

proliferation that is dependent on the synthesis of purines. Mycophenolate appears to promote less toxicity (including myelotoxicity and hepatotoxicity) than alkylating agents, which affect all tissues with a high proliferative or mitotic activity.^{11–13} Mycophenolate has been used in veterinary therapeutics as a treatment for myasthenia gravis, aplastic anemia, and immune-mediated hemolytic anemia.^{14–16} Gastrointestinal adverse effects including anorexia, vomiting, and diarrhea are reported to be the main adverse effect of this drug; however, they usually are reversible upon dose reduction or withdrawal of the drug.¹⁷ Evidence for the effectiveness of mycophenolate in the management of glomerular diseases in human medicine is mounting¹⁸; however, the veterinary literature is limited to a single case report supporting its use in a dog with glomerulonephritis of uncharacterized pathology.¹⁹

Recommendation 5:

Although randomized, controlled clinical trial evidence is lacking, based on preliminary, uncontrolled clinical experience with mycophenolate and its low rate of severe complications, mycophenolate is recommended as the first choice for treatment of dogs with peracute or rapidly progressive glomerular disease of an apparent immune-mediated pathogenesis.

82% of voting consensus members agreed with Recommendation 5; however, only 35% of these voters expressed “strong agreement.”

Cyclosporine

The main activity of cyclosporine is exerted by its binding to the cytosolic protein, cyclophilin. The cyclosporine-cyclophilin complex inhibits calcineurin, which is essential for the transcription of interleukin 2 and interleukin 2 activation of T lymphocytes.¹¹ The effectiveness of cyclosporine in dogs with spontaneous glomerular disease has been evaluated in a randomized, controlled clinical trial, but was found to provide little apparent benefit.²⁰ A limitation of this study was the lack of precise pathologic characterization of the tested glomerular diseases for evidence of active immune-mediated injury potentially responsive to cyclosporine. It also is unknown if adequate blood levels of cyclosporine were achieved in the dogs in this study, as they were not measured. Further studies on the effectiveness of cyclosporine in dogs with glomerular disease have not been reported, and the use of cyclosporine cannot be recommended explicitly for the management of glomerular diseases in dogs until further evidence supporting its effectiveness emerges.

Cyclophosphamide

Cyclophosphamide is an alkylating agent that interferes with DNA replication, RNA transcription and

replication, and consequent disruption of nucleic acid function. In veterinary medicine, it has been used in the management of a variety of neoplastic and immune-mediated disorders¹¹; however, there are no studies in the veterinary literature to support or negate its use in dogs with glomerular diseases. Although shown to be effective in some glomerular diseases in humans, it is more likely to be associated with side effects than many other of the drugs discussed here. Potential adverse effects include gastrointestinal signs, myelosuppression, and hemorrhagic cystitis.²¹ Cyclophosphamide may be administered as pulse therapy in cases of rapidly progressive disease. In humans, it is considered to be fairly effective for a variety of immune-mediated diseases including glomerular disease, and it has a rapid onset of action.²²

Chlorambucil

Chlorambucil is an alkylating agent generally associated with minimal adverse effects, such as mild myelotoxicity.¹¹ In veterinary medicine, it has been used mainly in the management of neoplastic diseases. The efficacy of chlorambucil as monotherapy has not been demonstrated in veterinary literature for the management of glomerular disease.

Azathioprine

Azathioprine is a purine analog that interferes with normal cell DNA and RNA synthesis promoting cytotoxicity to lymphocytes and interference with cell-mediated immunity and antibody production.^{23,24} It is used routinely in the management of a variety of immune-mediated disorders in veterinary medicine (eg, immune-mediated hemolytic anemia, inflammatory bowel disease) including immune-mediated glomerular disease.²⁵ However, in dogs with glomerular disease requiring acute onset of drug action, azathioprine may not be an ideal selection because it reportedly has a delayed onset of action and may take up to 2–5 weeks or more of administration to be fully effective in dogs (reportedly 3–4 months in humans).²³ Although it may not be an appropriate choice when used alone for patients requiring a rapid initial response, it may have a role when combined with more rapidly effective agents.^{11,23} The predictable adverse effects of azathioprine include myelosuppression, gastrointestinal upset, hepatic disease or failure, and acute pancreatitis.^{26,27} Although these adverse effects are generally manageable, patients receiving azathioprine must be monitored closely for evidence of these adverse effects of treatment. Azathioprine may be an inappropriate drug selection for dogs with a history or predispositions to acute pancreatitis.^{23,28}

Although there are no randomized clinical trials to support or negate the use of azathioprine for the management of glomerular diseases in dogs, there is uncontrolled anecdotal clinical experience supporting its efficacy, usually in combination with chlorambucil.

Selecting an Immunosuppressive Protocol for Dogs with Glomerular Disease

Clinical presentations of canine glomerular diseases can be classified broadly as (1) peracute, rapidly progressive, or both; or (2) stable, slowly progressive, or both. Dogs with peracute, rapidly progressive glomerular diseases, or both are characterized by profound proteinuria, hypoalbuminemia (eg, albumin <2.0 g/dL), or overt clinical signs of nephrotic syndrome (eg, edema) or progressive azotemia. Dogs with stable or slowly progressive glomerular disease or both are characterized by protracted or minimally progressive proteinuria and azotemia with normoalbuminemia or minimal hypoalbuminemia (ie, serum albumin concentration >2.0 g/dL) and no evident edema or clinical signs of uremia. These presentations dictate the optimum characteristics of the immunosuppressive drug and protocol most appropriate for a specific patient; however, when selecting a drug protocol targeted to the presentation of the disease, the probability of therapeutic success should be balanced against the potential for toxic and adverse effects of the immunosuppressive agents or protocols. Regardless of the immunosuppressive protocol selected for management of a patient, it is assumed that it will be used in combination with recommendations for “standard therapy” (see “Consensus Recommendations for Standard Therapy of Glomerular Disease in Dogs” in this supplement).

Recommended Protocols for Peracute or Rapidly Progressive Glomerular Disease or Both

Recommendation 6:

Dogs with peracute or rapidly progressive glomerular disease should receive induction therapy with a potent immunosuppressive protocol characterized by a rapid onset of immunosuppression. The following immunosuppressive agents are suggested for these presentations (see Table 1 for dosages and adverse effects):

- Mycophenolate alone or in combination with prednisolone
- Cyclophosphamide (continuous or pulse therapy) alone or in combination with prednisolone

87% of voting consensus members agreed with Recommendation 6, and 45% of these voters expressed “strong agreement.”

The suggested use of mycophenolate is based on its rapidly growing application in the management of glomerular diseases in humans. In humans with glomerular disease, mycophenolate (alone or combined with glucocorticoids) has been found to be variably superior or equally effective to traditional immunosuppressive protocols incorporating cyclophosphamide, azathio-

prine, or cyclosporine (alone or combined with glucocorticoids) and generally provides a more favorable safety profile.^{18,29} Although the efficacy and safety of mycophenolate in dogs with glomerular disease is not well established, preliminary uncontrolled anecdotal and observational studies in dogs suggest that it has potential for effectiveness in management of canine glomerular disease and appears to have a low incidence of serious adverse effects.

Although cyclophosphamide is also a reasonable option for dogs with peracute or rapidly progressive glomerular disease, its use entails a notably greater risk of adverse effects and requires more aggressive (and expensive) monitoring. Nonetheless, cyclophosphamide may be a reasonable alternative therapy when mycophenolate appears to be ineffective.

Recommendation 7:

There is no clear evidence to support or reject a recommendation for glucocorticoids in induction therapy in dogs with glomerular disease. To minimize the adverse effects associated with glucocorticoid therapy in dogs, the use of glucocorticoids as a sole treatment is not recommended, and when used concurrently with mycophenolate or cyclophosphamide, glucocorticoids should be tapered as quickly as possible to the minimally effective dose with the goal to discontinue the corticosteroid component within as soon as possible.

100% of voting consensus members agreed with Recommendation 7, and 40% of these voters expressed "strong agreement."

Recommended Protocols for Stable or Slowly Progressive Glomerular Disease or Both

Dogs with stable or slowly progressive glomerular disease or both may be managed with protocols including drugs with either rapid onset or more protracted onset. This classification does not necessarily imply that drugs with less rapid onset (alone or in combination) are less valuable or less effective in dogs with more severe or progressive glomerular diseases. In fact, there is limited anecdotal evidence for the efficacy of these slower onset protocols in acute glomerular diseases, and they should be considered as an alternative to the rapidly acting protocols once an initial response is achieved or when initial protocols need alteration because of the emergence of adverse effects.

Recommendation 8:

Dogs with active but stable or slowly progressive glomerular diseases with an immune-mediated foundation or slowly progressive diseases that are only partially responsive to standard therapy may receive induction therapy with either drugs having a rapid onset or drugs with more delayed onset including:

- Mycophenolate
- Chlorambucil alone or in combination with azathioprine on alternating days
- Cyclophosphamide and glucocorticoids
- Cyclosporine

91% of voting consensus members agreed with Recommendation 8, and 40% of these voters expressed "strong agreement."

Monitoring Immunosuppressive Therapy

Systematic clinical and laboratory monitoring is essential to optimize the therapeutic management of dogs with glomerular disease. Regular follow-up evaluations should include (1) clinical signs (adverse drug effects), (2) dietary intake, (3), body weight, (4) body condition score or body composition assessment, (5) assessment of peripheral or pulmonary edema, ascites, pleural effusion, (6) blood pressure, and (7) quality-of-life assessment. Follow-up laboratory assessment should include^b (1) CBC, (2) serum chemistry profile including serum albumin, creatinine, urea nitrogen, phosphorus, electrolytes, hepatic enzymes, and cholesterol, (3) urine protein/creatinine ratio (UPC), and (4) assessment of identified underlying disease(s) (see "Consensus Recommendations for the Diagnostic Investigation of Dogs with Suspected Glomerular Disease" in this supplement).

Initial assessments should be performed no later than 1–2 weeks after initiation of the treatment and every 2 weeks thereafter for the first 4–6 weeks of treatment. Thereafter, assessments are recommended at least every 4 weeks for the next 3 months and then at quarterly intervals until resolution of the disease.

Criteria for Assessing Effectiveness of Treatment and Re-evaluation of the Treatment Plan

Clear expectations for therapeutic goals should be established before considering immunosuppressive therapy for use in canine glomerular disease. This is especially important in light of the predictable risks of the immunosuppressive approaches available. Although patient and renal survival times are the ultimate measures of outcome, changes in certain clinical and biochemical characteristics of canine glomerular disease may provide guidance as to the patient's response to treatment. The WSAVA Renal Standardization Project initiated in 2008 is a first attempt to assess the natural history for canine glomerular diseases, but outcome assessment of specific therapeutic strategies will require carefully designed prospective clinical trials to better understand the clinical benefits versus therapeutic risks of current and future recommendations.

Until more evidence-based and conclusive data are available to validate therapeutic recommendations, the therapeutic approach should be directed by surrogate clinical markers that are likely to correlate with patient

and renal survival. Therapeutic effectiveness can be assessed by the impact of treatment on (1) proteinuria (as measured by UPC), (2) renal function (as measured by serum creatinine concentration), and (3) serum albumin concentration. Responses to treatment may be characterized as (1) complete response, (2) partial response, or (3) therapeutic failure. Although performed uncommonly, the response to treatment may also be assessed by changes in renal biopsies obtained before and after the course of standard or immunosuppressive therapy or both.

Change in the Magnitude of the Proteinuria

Reduction in the magnitude of proteinuria is a marker of response to treatment and should be assessed to document improvement beyond standard therapy alone. It also should be recognized that day-to-day variance in the UPC ratio may be substantial in dogs with glomerular disease and should be considered when determining whether a measured change represents a clinically significant response (beneficial or adverse) to the therapeutic intervention. The absolute change in UPC necessary to predict a therapeutic improvement or worsening differs depending on the baseline magnitude of the proteinuria. At low UPC values (near 0.5), a minimum change of 80% is required to demonstrate a significant difference, whereas at high UPC values (near 12), a minimum change of 35% is necessary.³⁰ In humans with glomerular disease, reduction in proteinuria has been shown to be useful in predicting renal survival and the rate of progression of renal dysfunction. The magnitude of reduction in proteinuria in humans is related to the likelihood for favorable outcomes in renal survival and patient quality-of-life. Based on these observations, it seems likely that the therapeutic goal should be to achieve the greatest nadir in the proteinuria possible (as assessed by the UPC) while minimizing adverse effects of treatment. To achieve this goal, UPC should be assessed serially and treatment adjusted, extended, or both (within the constraints of therapeutic risks) to promote the maximal sustained reduction in proteinuria. Although UPC may normalize in some patients, this magnitude of response is unlikely an achievable goal in all dogs, especially if the disease affecting the glomerulus has promoted permanent structural alterations to the glomerular filtration barrier or treatment incompletely controls the pathogenesis.

When glucocorticoids are included as part of the immunotherapy, it may promote a transient or persistent increase in proteinuria. Although this effect appears to be contrary to the therapeutic goals defined above, these effects may be transient and reversible. It remains unclear whether this effect is a contraindication for corticosteroid therapy in dogs with glomerular disease. If a treatment fails to achieve the expected response goals, an alternative drug strategy should be considered.

Recommendation 9:

Response to treatment as measured by changes in UPC is defined as follows^c:

- A complete response is defined as a reduction in the UPC to less than 0.5.
- A partial response is defined as a reduction in the UPC by greater than 50% of the highest pretreatment UPC after standard therapy or with standard therapy if both were initiated simultaneously.
- Therapeutic failure is defined as a reduction in UPC of less than 50% of the highest pretreatment UPC after standard therapy or with standard therapy if both were initiated simultaneously.

91% of voting consensus members agreed with Recommendation 9, and 45% of these voters expressed "strong agreement."

One has to consider that the magnitude of proteinuria may decrease in animals in late stages of progressive kidney disease because of a reduction in the number of remaining nephrons through which protein loss can occur.

Improvement or Stabilization of Renal Function

In patients with acute presentation, renal function and markers of azotemia (serum creatinine and urea nitrogen) may return to reference ranges spontaneously or in response to standard therapy, immunosuppressive therapy (complete response), or both. By contrast, in dogs with long-standing, persistent proteinuria and chronic kidney disease or in dogs with acute presentations, the glomerular damage may not repair and may remain abnormal, progress, or resolve incompletely (partial response). For partial responders, some degree of sustained azotemia may be expected, and the maintenance of stable renal function can be regarded as a therapeutic goal.

Response to treatment also should be assessed by changes in kidney function and measured by changes in serum creatinine concentration in normally hydrated animals. Serum creatinine should be compared to the baseline serum creatinine concentration defined as the mean value of documented serum creatinine concentrations during the 30 days preceding initiation of immunosuppressive therapy.

Recommendation 10:

The response to treatment as measured by changes in serum creatinine concentration is defined as follows^d:

- A complete response is defined as reduction in serum creatinine concentration to less than

1.4 mg/dL (124 μ mol/L) (or the patient's last known serum creatinine concentration before onset of the glomerular disease).

- A partial response is a sustained reduction in serum creatinine concentration by 25% or greater than baseline serum creatinine concentration most proximate to starting treatment.
- Therapeutic failure is defined as a reduction in serum creatinine concentration less than 25% baseline serum creatinine concentration most proximate to starting treatment.

91% of voting consensus members agreed with Recommendation 10, and 45% of these voters expressed "strong agreement."

Improvement in Serum Albumin Concentrations (improvement in serum globulins, total plasma protein, antithrombin, and fibrinogen may also represent improvement in glomerular permselectivity)

Although small increases in serum albumin concentration may reflect a response to treatment, the therapeutic goal is to achieve a serum albumin concentration greater than 2.0 gm/dL (20 g/L). Presumably, an increase in serum albumin concentration reflects a decrease in urinary loss; however, it may also occur consequent to decreased protein filtration resulting from progressive excretory failure independent of improvement in glomerular permselectivity. Consequently, improvement in serum protein concentrations should be compared to simultaneous measurement of serum creatinine and UPC values.

Recommendation 11:

Response to treatment as measured by changes in serum albumin concentration from baseline (defined as the mean of serum albumin concentration values during the 30 days preceding immunosuppressive therapy) is defined as follows:

- A complete response is a sustained increase in serum albumin concentration to greater than 2.5 g/dL (25 g/L)
- A partial response is either (1) a sustained increase in serum albumin concentration to 2.0–2.5 g/dL (20–25 g/L) or (2) a sustained increase of 50% or more in serum albumin concentration from the baseline serum albumin concentration.
- A therapeutic failure is defined as failure to increase serum albumin concentration to greater than 2.0 g/dL (20 g/L) or by less than 50% increase from baseline serum albumin concentration.

77% of voting consensus members agreed with Recommendation 11, however, only 30% of these voters expressed "strong agreement."

Improvement in Glomerular Histology

Resolution or improvement in the pathologic alterations of glomerular architecture may be the "gold standard" for evaluating therapeutic efficacy and outcome. However, obtaining sequential kidney biopsies to assess responses to treatment is not a conventional practice pattern in veterinary medicine. Nevertheless, serial kidney biopsy should be considered as a monitoring option in patients with suboptimal or incomplete responses to treatment. Similarly, serial kidney biopsy may provide a more objective and sensitive outcome marker in clinical trials testing therapeutic efficacy and recommendations.

Secondary Therapeutic Goals

Secondary therapeutic goals must be assessed in concert with standard therapy as the goals of standard therapy may not be achievable (eg, resolution of nephrotic signs, improvement in body composition, or correction of blood pressure) until there is maximal therapeutic resolution of the proteinuria and renal dysfunction, which standard therapy may not address.

Recommendation 12:

Secondary therapeutic goals include:

- *Improvement in blood pressure.* In conjunction with standard therapy, systolic blood pressure should be maintained at <150 mmHg.³¹ (See, "Consensus Recommendations for Standard Therapy of Glomerular Disease in Dogs" in this supplement.)
- *Resolution of nephrotic signs-peripheral and/or pulmonary edema, ascites, pleural effusion.*
- *Stabilization (or improvement) of body weight and body condition or body composition as measures of nitrogen metabolism.* Establish body weight to historical reference or body composition score to "ideal".

100% of voting consensus members agreed with Recommendation 12, and 45% of these voters expressed "strong agreement."

Termination of Immunosuppression Therapy

Immunosuppressive treatment should be discontinued when clinically intolerable (eg, GI signs) or life-threatening (eg, pancreatitis, systemic bacterial or fungal infection, hepatotoxicity) adverse effects (compared to the underlying disease) can be attributed to the treatment per se. If neutropenia (<3000 cells/ $10^3/\mu$ L) is present, immunosuppression should be discontinued or modified transiently until the leucopenia is improved. Treatment should be discontinued or adjusted if any other adverse effects are reported by the owner that can be attributed to the immunosuppressive therapy.

Table 1. Recommendation 14—Recommended dosages and adverse effects of representative immunosuppressive drugs for the management of glomerular disease with a known immune basis.

Generic Name	Main Mechanism of Action	Recommended Dose	Main Adverse Effects
Azathioprine	Antagonizes purine metabolism	2 mg/kg PO q24h for 1–2 weeks, then 1–2 mg/kg q48h	GI upset, myelosuppression, acute pancreatitis, GI disorders, hepatotoxicity, infection, malignancy
Chlorambucil	Alkylating agent	0.2 mg/kg PO q24–48h	GI upset, myelosuppression
Cyclophosphamide	Alkylating agent	Pulse therapy 200–250 mg/m ² every 3 weeks Continuous therapy—50 mg/m ² PO 4 days/week	Myelosuppression, GI upset, hemorrhagic cystitis, infection
Cyclosporine	Calcineurin inhibitor	5–20 mg/kg PO q12h (consider tapering dose upward from low to higher doses to avoid gastrointestinal complications)	GI upset, gingival hyperplasia,
Mycophenolate	Antagonizes guanosine metabolism	10 mg/kg PO q12h	GI upset
Prednisolone (or other glucocorticoids at an appropriate dosage and route of administration)	Inhibition of PLA2, reduction in cytokines release, inhibition of neutrophils migration, down regulation of the Fc receptor	1 mg/kg PO q12h initially. Taper down as soon as possible	PU/PD, polyphagia, muscle wasting, panting, haircoat changes, weight gain, induction of liver enzymes, GI ulceration, lipidemias, infection, adrenal suppression, thromboembolism

86.4% of voting consensus members agreed with Recommendation 14, and 35% of these voters expressed “strong agreement.”

Recommendation 13:

In the absence of overt adverse effects, at least 8 weeks of the rapidly acting non steroidal drug therapy (eg, mycophenolate) and 8–12 weeks of slowly acting drug therapy (eg, azathioprine) should be provided before altering or abandoning an immunosuppressive trial. If no response is evident or therapeutic goals are not achieved within these time intervals, consideration should be given to an alternative drug or dosing protocol. If no therapeutic response is noted after 3–4 months, consideration should be given to discontinue immunosuppressive therapy. Immunosuppressive therapy should be continued in dogs demonstrating a complete or partial response to initial treatment for a minimum of 12–16 weeks. Thereafter, consideration should be given to tapering the treatment to a dose/schedule that maintains the response without worsening of the proteinuria, azotemia, or clinical signs.

82% of voting consensus members agreed with Recommendation 13; however, only 25% of these voters expressed “strong agreement.”

Footnotes

^a Enhanced histopathologic studies include use of “thin-section” (3 micron) tissue sections stained by an array of stains (including H&E, PAS, Jones methenamine silver, Masson’s Trichrome and Congo Red).

^b Frequency of evaluation of these tests will depend on the treatment protocol as well as the clinical response of the patient.

^c Pooled UPC determinations may be a more reliable means of assessing true changes in the magnitude of proteinuria.

^d Although creatinine and albumin are identified in the recommendations as “serum creatinine or albumin,” it is intended to be interchangeable with “plasma creatinine or albumin.”

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Conflict of Interest: Authors disclose no conflict of interest.

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Consensus Guidelines for Immunosuppressive Treatment of Dogs with Glomerular Disease Absent a Pathologic Diagnosis

IRIS Canine GN Study Subgroup on Immunosuppressive Therapy Absent a Pathologic Diagnosis, B. Pressler, co-chair, S. Vaden, co-chair, B. Gerber, C. Langston, and D. Polzin

Background: In certain situations, veterinarians must decide whether or not to recommend immunosuppressive therapy for dogs with suspect glomerular disease in the absence of renal biopsy-derived evidence that active immune mechanisms are contributing to glomerular injury. The purpose of this report is to provide guidelines for the use of immunosuppressive drugs under these conditions.

Animals: Animals were not used in this study.

Methods: Recommendations were developed by a formal consensus method.

Results: Four recommendations were developed and accepted at a high level of consensus (median 92.5% agreement). Renal biopsy should not be performed when contraindications are present or when results will not alter treatment or outcome. Immunosuppressive drugs should not be given when the source of proteinuria is unknown, they are otherwise contraindicated, or a familial nephropathy or amyloidosis is likely. However, they should be considered when dogs are already being given standard therapy and the serum creatinine is >3.0 mg/dL, azotemia is progressive, or hypoalbuminemia is severe. Thorough client communication regarding pros and cons of such treatment as well as close and careful patient monitoring is required.

Conclusion and Clinical Importance: These recommendations can help guide the decision about renal biopsy in patients with proteinuria as well as the use of immunosuppressive drugs in those patients where the decision was made not to perform renal biopsy.

Key words: Canine; Glomerulonephritis; Glomerulopathy; Proteinuria.

There is no substitute for a pathologic diagnosis in the formulation of therapeutic plans for dogs with glomerular diseases. However, there are times when renal biopsy cannot be performed because of medical, practical, or financial limitations. In these situations, veterinarians may have to decide whether or not to recommend immunosuppressive/anti-inflammatory therapy for dogs with glomerular diseases in the absence of firm knowledge of whether an active immune mechanism is contributing to glomerular injury. Herein, we provide guidance on when to recommend immunosuppressive/anti-inflammatory therapy for dogs with a presumptive diagnosis of glomerular disease absent pathologic findings from a renal biopsy.

Proteinuric dogs suspected of having glomerular disease, but absent a renal pathologic diagnosis, should generally be managed initially using standard therapy and regular monitoring. However, standard therapy for canine glomerular disease rarely leads to complete resolution of the renal injury. Furthermore, adverse effects of drugs used for standard therapy may limit their use in some dogs. If the targeted reduction in proteinuria is not achieved and the addition of immunosuppressive/anti-inflammatory therapy is being considered, it is prudent to reconsider recommending a renal biopsy. Nonetheless, in some situations, it may not be possible to obtain a renal biopsy (see the section entitled “Recommendations for When Not to Perform Renal Biopsy”). Furthermore, in some situations, it is clear that immunosuppressive/anti-inflammatory therapy is inappropriate (see the section entitled “Recommendations for Exclusion Criteria for Using Immunosuppressive/Anti-inflammatory Therapy in Dogs with Glomerular Disease”). When use of immunosuppressive/anti-inflammatory therapy is not contraindicated, the risk-to-benefit assessment for the patient should consider the arguments for and against using immunosuppressive drugs in dogs with glomerular disease (see the section entitled “Arguments for and Against Immunosuppressive/Anti-inflammatory Therapy in Dogs with Glomerular Disease Absent a Renal Pathologic Diagnosis”).

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Recommendation 1:

Renal biopsy should not be performed in dogs (1) with IRIS CKD Stage 4; (2) when other medical contraindications are present and cannot be mitigated (including coagulopathy, renal cystic disease, moderate-to-severe hydronephrosis, pyelonephritis, perirenal abscess, uncontrolled hypertension, severe anemia, and pregnancy); or when results of renal

biopsy are deemed unlikely to alter treatment, outcome, or prognosis.

95% of voting consensus members agreed with Recommendation 1 and 75% of these voters expressed "strong agreement."

Relative contraindications to renal biopsy include severe azotemia (ie, creatinine >5 mg/dL, which is associated with increased risk of bleeding and other postbiopsy complications), coagulopathy, cystic disease or moderate-to-severe hydronephrosis, pyelonephritis or perirenal abscess, uncontrolled hypertension (systolic blood pressure consistently >160 mmHg), severe anemia, pregnancy, and lack of access to a renal diagnostic pathology center equipped and qualified to perform and interpret electron and immunofluorescent microscopy as well as light microscopy using an appropriate array of special stains (including H&E, PAS, trichrome, Congo red and Jones methenamine silver stains) performed on 3 micron sections.¹

If results of renal biopsy are deemed unlikely to alter treatment, outcome, or prognosis, then renal biopsy should not be recommended. If the kidneys are small, the damage present is likely irreversible, and it would be unlikely that the renal biopsy will contribute to patient care more so than less invasively obtained biochemical parameters, renal biopsy should not be recommended. When chronic azotemia is present, renal changes may be irreversible and histopathology less likely to alter treatment. However, if the duration of azotemia cannot be established, renal histopathology may establish chronicity and predict potential reversibility. An exception to these guidelines may include suspected chronic glomerular disease in which the disease process may still be active and modifiable, despite azotemia being mild to moderate (ie, serum creatinine <5.0 mg/dL). Lastly, if a rational presumptive diagnosis of acute kidney injury can be made noninvasively (eg, exposure to ethylene glycol without observed ingestion, recent hypotensive episode), then renal biopsy is unlikely to substantially change the therapeutic approach. Other factors that may preclude performing a renal biopsy include financial constraints of the owner, ethical concerns of the owner, or lack of available experienced personal to perform renal biopsy.

Recommendation 2:

Immunosuppressive/anti-inflammatory therapy should not be administered to dogs with proteinuria before renal biopsy when (1) proteinuria is not definitively glomerular in origin; (2) immunosuppressive therapy is otherwise contraindicated; (3) the dog breed and age of disease onset suggest that a nonimmune-mediated familial nephropathy is likely; or (4) amyloidosis is the most likely histopathologic diagnosis.

95% of voting consensus members agreed with Recommendation 2 and 60% of these voters expressed "strong agreement."

Verifying Proteinuria Is of Renal Origin

Proteinuria may be prerenal, renal, or postrenal, and renal proteinuria may be caused by glomerular disease, tubular disease, or inflammatory/exudative disease of the kidneys. When the source of proteinuria has not been definitively localized, immunosuppressive drugs are not indicated because, in general, immunosuppressive drugs are only indicated for treatment of glomerular disease.

Urinary tract infection, inflammation, or macroscopic hematuria may increase UPC above reference range, and must be excluded as causes for proteinuria.² Overload proteinuria, which occurs when large amounts of low molecular weight plasma proteins within the glomerular ultrafiltrate exceed tubular resorptive capacity (eg, neoplastic production of paraproteins or Bence Jones proteinuria and excessive hemolysis or rhabdomyolysis), must also be excluded before empiric treatment with immunosuppressive medications.

Mild proteinuria is common in dogs with acute or chronic tubular injury. Collective anecdotal experience of this consensus panel suggests that dogs with chronic tubulointerstitial disease rarely have UPC values >2.0–3.0, although occasionally, acute kidney injury may transiently be associated with greater UPC results (ie, ≥5.0). However, it is believed that UPC ranges from dogs with glomerular disease versus acute or chronic tubulointerstitial disease overlap, and therefore results must be interpreted in conjunction with other clinicopathologic findings when predicting type of disease. Thus, dogs with UPC values <2.0 in conjunction with increased serum creatinine concentration and persistent isosthenuria or absence of proteinuria at the time of initial diagnosis of kidney disease should not receive treatment with immunosuppressive/anti-inflammatory therapy without biopsy-supported evidence of active immune-mediated glomerular injury.³

Contraindications to Immunosuppressive Therapy

Immunosuppressive therapy should not be administered to dogs with concurrent illnesses for which immunosuppression is contraindicated. Common diseases where these drugs should be avoided include diabetes mellitus, hyperadrenocorticism, and fungal or bacterial infectious diseases. In addition, specific immunosuppressive drugs may be contraindicated with particular conditions (eg, glucocorticoids in dogs with pancreatitis or uncontrolled hypertension, azathioprine in dogs with bone marrow suppression, hepatic dysfunction, or pancreatitis). Dogs from geographic regions, where infectious diseases associated with glomerular damage and proteinuria are more prevalent, should be appropriately evaluated for possible occult infection before initiating immunosuppressive therapy.

Familial Nephropathy

Familial disease should be suspected when multiple related dogs are diagnosed with similar proteinuric renal disease or when a dog is diagnosed with proteinuric renal disease that is characteristic of a familial disease reported to occur in that breed. Familial proteinuric renal disease has been reported in Bull Terriers, English Cocker Spaniels, Dalmatians, Samoyeds, Rottweilers, Bernese Mountain Dogs, Newfoundlands, Doberman Pinschers, Pembroke Welsh Corgis, Bullmastiffs, French Mastiffs, Chinese Shar Peis, Beagles, English Foxhounds, and Soft-Coated Wheaten Terriers.⁴ The pathogenesis, clinical findings, and progression of disease vary among these breeds. Veterinarians are cautioned, however, that while dog breed, age of onset of proteinuria, and suspected or confirmed proteinuric kidney disease in related dogs may increase suspicion of a familial nephropathy, this diagnosis should be considered presumptive until confirmed by renal biopsy.

Familial nephropathies in most of the above-listed breeds are steroid-resistant.⁴ Treatment of dogs with hereditary or spontaneous familial glomerular diseases with immunosuppressive drugs, particularly prednisone, has thus far been unrewarding and morbidity attributable to drug side effects outweighs the likelihood of a responsive disease phenotype. Exceptions to this guideline may be glomerular disease in Soft-Coated Wheaten Terriers and Bernese Mountain Dogs. Soft-Coated Wheaten Terriers with concurrent glomerular disease and enteropathy may benefit from immunosuppressive drugs.⁵ Bernese Mountain Dogs are predisposed to developing membranoproliferative glomerulonephritis, which may be responsive to some immunosuppressive protocols.⁶

Amyloidosis

Although indirect evidence suggests that reactive amyloidosis in dogs is associated with a dysregulated immune response, immunosuppressive therapy in people and dogs is either of no benefit or may contribute to more rapid progression of disease.^{7,8} Although glomerular amyloidosis in dogs can be associated with very high UPC values, there is too much overlap in UPC values between dogs with amyloidosis and dogs with other glomerulopathies to reliably use the UPC value to predict histopathologic diagnosis.⁹ Renal amyloidosis may be more likely in dog with glomerular disease when (1) the affected dog is of a breed known to be predisposed to amyloidosis (eg, Shar Pei); (2) additional clinical signs associated with hereditary amyloidosis in Shar Peis are present, including cyclical fever or distal joint effusion¹⁰; or (3) amyloid deposition has been confirmed in other organs, particularly the liver.¹¹

standard therapy and do not have a biopsy-confirmed renal pathologic diagnosis when (1) serum creatinine is >3.0 mg/dL, or azotemia is progressive; or (2) hypoalbuminemia is severe (ie, <2.0 g/dL).

90% of voting consensus members agreed with Recommendation 3 and 60% of these voters expressed "strong agreement."

Currently, the evidence that immunosuppressive therapy ameliorates glomerular disease in dogs is weak.¹² However, clinical studies of canine glomerular disease based on renal pathologic diagnosis have not yet been performed. In human beings, immunosuppressive agents may be effective for the treatment of membranous- and membranoproliferative glomerulonephritis.¹³ These forms of glomerulonephritis in humans are typically characterized by the presence of immune complexes, evidence of an immune process affecting the glomerulus. These same renal disorders are common glomerular diseases in dogs. As reported elsewhere in this supplement,¹⁴ 241 of 501 (48.1%) renal biopsies obtained from dogs suspected of having clinical evidence of glomerular disease had evidence of immune complex glomerular disease and thus would be candidates for immunosuppressive/anti-inflammatory therapy. Thus, approximately 1 of every 2 dogs with clinical evidence of glomerular disease would likely be candidates for immunosuppressive/anti-inflammatory therapy.

Reported survival times for dogs with azotemia, nephrotic syndrome, or both as a result of glomerular disease are short, with a median of <60 days.⁹ However, survival of nonazotemic dogs without nephrotic syndrome can be substantially longer, with a reported median of 605 days.⁹ Because survival can be short in dogs with glomerular disease characterized by either azotemia or nephrotic syndrome and nearly 50% of dogs with clinical evidence of glomerular disease have immune complex glomerular disease possibly responsive to immunosuppressive drugs, a therapeutic trial might be warranted.

There are select situations where immunosuppressive therapy might be considered in dogs with glomerular range proteinuria without the benefit of a renal biopsy. One such situation would be if renal biopsy is not possible and the glomerular disease is rapidly progressive in spite of standard therapy. In some cases, the dog may have an acute crisis and need to be stabilized via emergency treatment before performing a biopsy. Immunosuppressive therapy might also be indicated in dogs with rapidly progressive disease immediately after renal biopsy, but before results of the biopsy are not yet available. Specifically, aggressive immunosuppression may be considered if (1) azotemia is acutely severe, progressive, or both (ie, creatinine >5 mg/dL, IRIS AKI Stages 4 or 5) at the time of diagnosis and there is no evidence of chronic disease; or (2) hypoalbuminemia is severe (serum albumin <2.0 g/dL). In these situations, the protocols for peracute and rapidly

Recommendation 3:

Immunosuppressive drugs should be considered in dogs with glomerular disease that are being given

progressive diseases should be followed (see “Guidelines for Immunosuppressive Treatment of Dogs with Glomerular Disease Based on Established Pathology” in this supplement).

Likewise, immunosuppressive therapy might be indicated if biopsy is not possible, neither age nor breed is indicative of familial renal disease, and other contraindications to immunosuppressive therapy are not present. In this situation, the protocols for more protracted disease should be followed (see “Guidelines for Immunosuppressive Treatment of Dogs with Glomerular Disease Based on Established Pathology” in this supplement).

In the absence of a renal biopsy to help predict the likelihood of response, immunosuppressive therapy should be considered a therapeutic trial. If there is no response to treatment after 8–12 weeks, it is recommended that immunosuppressive/anti-inflammatory therapy be discontinued and the previous decision to not perform a renal biopsy be revisited.

Recommendation 4:

Immunosuppressive drugs should be administered to dogs in the absence of a renal pathologic diagnosis only after thorough client communication regarding the arguments for and against the use of these drugs in this setting. These agents should be administered cautiously, with close and careful patient monitoring.

90% of voting consensus members agreed with Recommendation 4 and 74% of these voters expressed “strong agreement.”

Veterinarians are fond of reminding each other to “above all do no harm.” Another way of saying this is “given an existing problem, it may be better not to do something, or even to do nothing, than to risk causing more harm than good.” In this case, the question is whether we do harm by recommending an unproven immunosuppressive/anti-inflammatory treatment for a dog with clinical evidence of glomerular disease that may not benefit from the treatment. However, it is important to remember that canine glomerular disease can lead to serious complications or death. Failure to provide a potentially helpful treatment in this setting may result in more harm than the potential risks of the treatment. Thus, it is important to consider both the potential risks and potential benefits of recommending immunosuppressive/anti-inflammatory treatment for dogs with clinical evidence of glomerular disease absent the findings of a renal biopsy.

As reported elsewhere in this supplement, there is approximately a 50 : 50 chance that we would appropriately recommend immunosuppressive/anti-inflammatory therapy for a dog with clinical evidence of glomerular disease absent a renal biopsy.¹⁴ Likewise, there is a 50 : 50 chance that recommending such treatment could be inappropriate for the patient without a

biopsy. However, the risk of adverse effects of immunosuppressive/anti-inflammatory treatment in dogs with clinical evidence of glomerular disease may depend on many factors including some that remain unknown. The decision to proceed with treatment requires a case-by-case consideration of the risks of treatment.

Potential Benefits of Immunosuppressive/Anti-Inflammatory Treatments in Dogs with Clinical Evidence of Glomerular Disease

The view that immunosuppressive/anti-inflammatory therapy may be effective in improving clinical outcomes in dogs with some forms of glomerular disease is based on observations in humans with glomerular disease as well as recent anecdotal evidence. Cyclosporine, the only drug that has been studied prospectively in dogs with glomerular disease, was found to be of no detectable benefit, although the dose of cyclosporine used in this study may have been too low to be effective against glomerular disease.¹ However, dogs were included in this study regardless of renal pathologic diagnosis and thus it can be argued that some portion of the study dogs probably had glomerular disease that was not likely to respond to immunosuppressive/anti-inflammatory therapy. We hypothesize that a treatment effect would have been found if only dogs with documented immune complex-mediated glomerular disease were studied. No other studies of the use of immunosuppressive agents in dogs with glomerular disease have been published. As a consequence, it is difficult to predict a positive treatment effect with any accuracy.

Potential Risks of Immunosuppressive/Anti-Inflammatory Treatments in Dogs with Clinical Evidence of Glomerular Disease

Immunosuppressive drugs are widely recognized as having many potential adverse effects.¹⁵ The principal adverse effect of mycophenolate in dogs appears to be diarrhea that is largely dose-dependent, whereas vomiting and anorexia are less common adverse effects. Cyclophosphamide may cause bone marrow suppression as well as vomiting and diarrhea, and occasionally sterile hemorrhagic cystitis.¹⁵ Glucocorticoids have numerous adverse effects including polyuria, polydipsia, polyphagia, weight gain, weakness, exacerbation of hypertension, increased susceptibility to urinary tract infections, hepatopathy, and promote proteinuria.^{15–17} In addition, approximately 15% of dogs with glomerular proteinuria will likely have amyloidosis, and corticosteroids are contraindicated in the management of amyloidosis in humans.⁷ Azathioprine, a commonly used immunosuppressive drug, can produce bone marrow suppression in dogs and has also been associated with the development of pancreatitis.¹⁵ Cyclosporine can cause vomiting, diarrhea, and anorexia and has multiple recognized drug interactions.¹⁵ Methotrexate causes bone marrow suppression and may be hepatotoxic.¹⁵

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Consensus Recommendations for Treatment for Dogs with Serology Positive Glomerular Disease

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Key words: *Borrelia burgdorferi*; Formal consensus; *Leishmania*; Lyme.

Implications of Seropositivity in Dogs with Glomerular Disease

The finding of glomerular proteinuria concurrent with a positive titer for an infectious agent with recognized potential to incite glomerular disease is often interpreted to support a cause–effect relationship between these two findings. Such a conclusion is risky and may erroneously influence clinical decisions including the diagnostic approach, diagnosis, therapeutic strategies and risks, and outcomes of the disease. Although any infection can be causally associated with glomerular diseases, infections with *Borrelia burgdorferi* and *Leishmania* in dogs are especially problematic because of the seemingly common association of these infections with proteinuric conditions (Tables 1, 2). However, in some areas of the United States, common exposures to heartworm, *Borrelia burgdorferi*, *Ehrlichia*, *Anaplasma*, or *Babesia* spp. lead to seroconversion without development of proteinuria; and in Mediterranean countries exposure to *Leishmania* promotes a positive IFAT result that indicates exposure to the infection but not necessarily active infection. Thus, serology alone is not conclusive evidence that an infectious agent is the cause of a coexisting glomerular disease (Tables 1, 2).

If a patient has clinical evidence of an active infectious disease associated with proteinuria, it is compelling to presume the linkage between the serology and glomerular disease may be real. This is clearly evident for Leishmaniasis in which serology has a 98% predic-

Abbreviations:

ELISA	enzyme-linked immunosorbent assay
FDA	Federal Drug Administration
IFAT	indirect fluorescent antibody technique
NGAL	neutrophil gelatinase-associated lipocalin
PCR	polymerase chain reaction

tive sensitivity, but is less compelling for dogs with *Borrelia*-associated glomerular disease in which less than 30% of dogs with concurrent seropositivity and glomerular disease have nonrenal signs consistent with active Lyme disease.¹ Proteinuria is an uncommon finding in dogs with Lyme-seropositivity,² and may be recognized in less than 2% of seropositive dogs.³ Furthermore, clinical signs ascribed to an infectious agent like *Borrelia* may be the result of a coinfection with another organism (ie, *Rickettsia rickettsii*, *Ehrlichia* spp., etc) that also can cause proteinuria and similar clinical signs (Table 2). Lyme seropositivity, at a minimum, is a marker for tick and wildlife exposure, and co-infections which cause proteinuria should be considered. Similarly depending on the phase of disease and magnitude of the titer, up to 55% of seropositive dogs with titers for Leishmaniasis do not manifest evident clinical signs of the disease. The response to anti-infective treatment also is not reliable proof of causation because other infectious agents may be sensitive to the treatment, and therapies like doxycycline may have anti-inflammatory and antiarthritic properties sufficient to resolve clinical signs unassociated with an infectious cause.^{4,5}

Renal histopathology can be helpful to support an association between the seropositivity and the glomerulopathy if lesions characteristic for the disease are identified in the biopsy, but this too cannot prove a cause–effect relationship unless agent-specific antigens associated with active immune deposits in glomerular lesions are proven to be from the infectious agent. However, currently this testing is not available clinically. In a study of Lyme-positive dogs with glomerular disease, 84% had Lyme-specific immune deposits in their glomeruli.⁶ Agent-specific immune deposits, however, may have been deposited nonspecifically in abnormal glomeruli. Nonetheless, a properly processed and evaluated renal biopsy provides a basis for guiding decisions to intervene with immunotherapy. (See “Consensus Recommendations for Immunosuppressive Treatment of Dogs with Glomerular Disease Based on Established Pathology” and “Pathologic Evaluation of

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Table 1. Tick-borne and nontick-borne infectious agents associated with glomerulonephritis.

Infectious Agent	Class of Infection	Type of Infected Cell	Major Vector	Tests Other Than cytology	Need Paired Tests	Bite or Blood-borne	Rx	Carrier Status
<i>Anaplasma phagocytophilum</i>	Rickettsial	Granulocyte	Ixodes	ELISA PCR	X	X	D	X
<i>Anaplasma platys</i>	Rickettsial	Platelet	Rhipi	ELISA PCR	X	X	D	X
<i>Bartonella</i> spp.	Bacterial	Endothelial cells, Epi-rbc, Macrophages	Ixodes, fleas, other ticks	PCR WB IFA culture	?	X	DF DR Z	X
<i>Borrelia burgdorferi</i> (Lyme disease)	Spirochete	Extracellular near fibroblasts	Ixodes	ELISA C6*, WB IFA	No	Unlikely	D (A,EZ)	X
<i>Babesia canis, rossi, vogeli</i> , large B.	Protozoan	Rbc	Rhipi	PCR IFA	X	X	Im	X
<i>Babesia gibsoni, conradae</i> , small B.	Protozoan	Rbc	Rhipi	PCR IFA	X	X	Z/Q (Clin)	X
<i>Babesia microti</i>	Protozoan	Rbc	Ixodes	PCR IFA	X	X	Z/Q	X
<i>Brucella canis</i>	Bacterial	Lymphocyte Varied	Venereal Varied	RSAT ELISA PCR	X	X	MD D	X
<i>Cytauxzoon felis</i>	Protozoan	Rbc, schizonts in macr	Ambly Derma	PCR	No	X	Z/Q Im	X
<i>Ehrlichia canis</i>	Rickettsial	Monocytic wbc	Rhipi	ELISA IFA PCR	X	X	D	X
<i>Ehrlichia chaffeensis</i>	Rickettsial	Monocytic wbc	Ambly Derma	ELISA PCR	X	X	D	X
<i>Ehrlichia ewingii</i>	Rickettsial	Granulocyte	Ambly	PCR	X	X	D	X
<i>Hepatozoon americanum</i>	Protozoan	Myocyte, Lymphoid, Liver, Wbc	Ambly (dog eats tick)	PCR Muscle biopsy	No		spc	X
Leishmaniasis	Protozoan	Extracellular Macrophages	Sandfly, Vertical	IFA ELISA HISTOLOGY	No	X	PA	X
Leptospirosis	Spirochete	Extracellular	Urine	PCR MAT	X	Unlikely	D (A)	X
<i>Mycoplasma</i> spp.	Bacterial	Epi-rbc	Fleas, Ticks	PCR ELISA	?	X	DF	X
<i>Rickettsia rickettsii</i> (Rocky Mountain Spotted Fever, RMSF)	Rickettsial	Endothelial cells	Derma Rhipi	IFA DFA	X	Unlikely	D F	No

A, amoxicillin; C6*, C6 peptide antigen in SNAP-3Dx, SNAP-4Dx, SNAP-4DxPLUS or Lyme Quant C6 (IDEXX)/Or other Lyme antibody tests: Abaxis, AccuPlex4, Multiplex4, ospA, ospC, ospF tests; Clin, clindamycin; D, doxycycline; Derma, *Dermacentor*; E, erythromycin; F, fluoroquinolones; Im, imidocarb; MAT, microagglutination test; MD, minocycline and dihydrostreptomycin; PA, pentostam (sodium stibogluconate), amphotericin B, allopurinol; Q, atovaquone; R, rifampin; Rhipi, *Rhipicephalus*; RSAT, rapid slide agglutination test; Rx, treatment; spc, sulfas, pyrimethamine and clindamycin; decoquinat; UPC, urine protein/creatinine ratio; X, yes; Z, azithromycin (zithromax).

Adapted from Littman⁸ and Littman MP: Cytopenias, Vasculitis, Polyarthropathy, and Proteinuria. *Atlantic Coast Veterinary Conference Proceedings*, April 2011.

Canine Renal Biopsies: Methods for Identifying Features That Differentiate Immune Mediated Glomerulonephritides from Other Categories of Glomerular Diseases” in this Supplement.)

If results of a renal biopsy are not available to guide this decision, the cautious approach is to assume that there is a role for the infectious agent in the origin of

glomerular injury until proven otherwise. As mentioned above, of particular concern in the United States is the glomerular disease associated with *Borreliosis*, which may be rapidly progressive and fatal.^{7,8} On this basis, treatment for both proteinuria and the organism may be justified in these patients. Dogs with chronic suppurative infections may have positive serol-

Table 2. Infectious diseases associated with proteinuria, polyarthropathy, or cytopenias.

Infectious Agent	Lameness	↑UPC or ↑BUN	Vasculitis or Epistaxis	Oculo-neural signs	RBC ↓	WBC ↓	Platelets ↓	Alb ↓	Glob ↑
<i>Anaplasma phagocytophilum</i>	X	X	X	X	X	X	X	X	
<i>Anaplasma platys</i>		?	?	?	X	X	X	?	
<i>Bartonella</i> spp.	X	X	X	X	X	?	?	?	?
<i>Borrelia burgdorferi</i> (Lyme disease)	X	X	?	Rare			X if PLN	X	
<i>Babesia canis, rossi, vogeli</i> , large B		X		X	X	Varied	X	X	?
<i>Babesia gibsoni, conradae</i> , small B		X			X	Varied	X	?	?
<i>Babesia microti</i>		X?			X?		X?	X?	?
<i>Babesia canis</i>	X	X		X				X	X
<i>Cytauxzoon felis</i>					X	Varied	X	X	
<i>Ehrlichia canis</i>	X	X	X	X	X	X	X	X	X
<i>Ehrlichia chaffeensis</i>	X	X	X	X	X	X	X	X	X
<i>Ehrlichia ewingii</i>	X	X	X	X	X	X	X	X	X
<i>Hepatozoon americanum</i>	Muscle	X			X	Up		X	X
Leishmaniasis	X	X	X	X	X	X	X	X	X
Leptospirosis	Muscle	Tubular	X	X	X	Varied	X	X	
<i>Mycoplasma</i> spp.	?	?		X	X		X	?	?
<i>Rickettsia rickettsii</i> (Rocky Mountain Spotted Fever)	X	X	X	X	X	X	X	X	Down

Adapted from Littman⁸ and Littman MP: Cytopenias, Vasculitis, Polyarthropathy, and Proteinuria. *Atlantic Coast Veterinary Conference Proceedings*, April 2011.

ogy for specific fungal or bacterial agents associated with a concurrent glomerular disease that either could be immune-mediated or secondary to amyloidosis. Under these circumstances, renal histopathology is essential to diagnose the associative disease and direct appropriate treatment.

Recommendations for the Management of Glomerular Disease in Dogs with Serologic Positive Test Results^a for an Infectious Agent Absent the Results of a Renal Biopsy or before Obtaining Results of a Renal Biopsy

Recommendation 1: Serology alone cannot confirm a cause-effect relationship between an infectious agent and causation of glomerular disease. Nonetheless, a properly processed and evaluated renal biopsy may provide a basis for guiding decisions to intervene (or not) with immunotherapy. If results of a renal biopsy are not available to guide this decision, the cautious approach is to assume that there is a role for the infectious agent in the origin of glomerular injury until proven otherwise.

90% of voting consensus members agreed with Recommendation 1, and 55% of these voters expressed "strong agreement."

Recommendation 2: In serologically positive dogs with glomerular disease, Standard Treatment is recommended when the urine protein:creatinine ratio (UPC) is persistent and ≥ 0.5 .

86% of voting consensus members agreed with Recommendation 2, and 55% of these voters expressed "strong agreement."

Once glomerular disease has been identified in a dog with known exposure to an infectious agent suspected to be associated with glomerular pathology, Standard Treatment should be instituted as early in the course of the disease as possible in addition to treatment for the infectious disease. (See "Consensus Recommendations for Standard Therapy of Glomerular Disease in Dogs" in this Supplement.) The threshold advised to trigger standard treatment is a persistent UPC of 0.5 or above. The efficacy of Standard Treatment should be assessed by the changes in UPC, serum creatinine and potassium concentrations, and systemic blood pressure, every 1–2 weeks after initiation or modification of Standard Treatment depending on the stage of disease and stability of the patient. The response to both anti-infective treatment and Standard Treatment should be monitored every 4 weeks thereafter for the first 3 months or until the disease has stabilized, and then accordingly as recommended for Standard Treatment (see above). The recommendation to initiate the treatment at this level of proteinuria, even in nonazotemic dogs, is justified by the potentially dire consequences should the dog develop progressive kidney disease associated with the infection. The prognostic value of the UPC for the extent of glomerular injury and response to Standard Treatment may be difficult to interpret during the active phase of the infection (especially for leishmaniasis) and should be reassessed after the course of treatment for the infection.

Use of Anti-Infective Treatment against Serology Predicated Pathogens

The recommendation for starting Standard Treatment does not imply a priority of management relative to that of the attending infection suggested by the positive serology.

Recommendation 3: The possibility that an infectious agent induced the glomerular injury highlights the rationale to initiate specific anti-infective treatment immediately when the coexistence of glomerular disease and seropositivity is recognized, despite the lack of evidence for causality of the infection and the proteinuria.

95% of voting consensus members agreed with Recommendation 3, and 50% of these voters expressed “strong agreement.”

The use of anti-infective medications that are potentially nephrotoxic (eg, amphotericin B) should be avoided if possible. When potentially nephrotoxic anti-infective drugs are the most appropriate or the only therapeutic option, they should be used cautiously with heightened efforts to monitor renal function and minimize additional renal injury. Monitoring of renal biomarkers [like neutrophil gelatinase-associated lipocalin (NGAL)] that have potential to signal the early onset of kidney injury may facilitate dosing of nephrotoxic drugs. Quantitative serology (eg, quantitative C6 antibody for *Borrelia burgdorferi* infection or quantitative serology for fungal infections such as coccidioidomycosis or cryptococcosis) should be utilized for disease surveillance where serology is thought to correlate with elimination of the infectious agent or resolution of clinical disease.

Use of Anti-Inflammatory or Immunotherapy in Seropositive Patients Absent Biopsy Findings

Recommendation 4: Clinically stable dogs without azotemia, progressive increases in serum creatinine or UPC, or clinical consequences associated with their proteinuria^b should be managed initially only with specific anti-infective treatment against the suspected infectious agent plus Standard Treatment for glomerular disease.

91% of voting consensus members agreed with Recommendation 4; however, only 30% of these voters expressed “strong agreement.”

Recommendation 5: For seropositive dogs that are azotemic (IRIS CKD Stage II or greater), or have progressive nonazotemic (IRIS AKI Grade I) to azotemic (IRIS AKI Grade II or greater) increases

in serum creatinine, or have evidence of rapidly progressive glomerular disease based on clinical and/or laboratory assessments, the following are recommended:

- (1) A thorough search for possible geographic-specific coinfections (eg, for *Babesia microti*-like infection in Spain).⁹
- (2) A renal biopsy evaluated by a nephropathologist using appropriate light, immunofluorescent, and electron microscopic imaging^c to support potential intervention with immunosuppressive treatment.
- (3) In the absence of a renal biopsy, immunosuppressive treatment should be considered to manage potential immune causes of the glomerular injury.^d

86% of voting consensus members agreed with Recommendation 5; however, only 30% of these voters expressed “strong agreement.”

Timing of Immunosuppressive Treatment

The decision to initiate anti-infective treatment and Standard Treatment may precede the decision to recommend renal biopsy. Renal biopsy may or may not be recommended in nonazotemic dogs with mildly increased UPCs and a normal albumin concentration that do not show evidence of progressive kidney disease. The glomerular injury in dogs with stable renal function and UPCs may no longer be active and progressive and may represent an earlier insult. The recommendation for renal biopsy should be advocated more strongly in patients with more severe clinical and laboratory abnormalities or those whose disease is progressive. Standard and anti-infective therapies should not be delayed while deciding whether or not to perform a biopsy.

Recommendation 6: If the results of renal biopsy document an active immune component to the glomerular disease, appropriate immunosuppressive treatment is indicated.^c At this stage, the glomerular disease should be considered as having an immune-mediated component, and not infectious alone, and it is not likely to respond merely to anti-infective treatment and irradiation of the infection if severe or progressive.

95% of voting consensus members agreed with Recommendation 6, and 55% of these voters expressed “strong agreement.”

For some seropositive infections, dogs may show no active clinical evidence of the infection despite overt manifestations of active and progressive glomerular disease. Control of the immune events directed at the

glomerulus are warranted and required to control this component of the disease. Similarly, it may not be possible to delay treatment with immunosuppressive treatment for the glomerular disease while the infection is being controlled. Initiation of immunosuppressive treatment directed at the glomerular disease poses the likelihood to compromise control of the predisposing infection. This is clearly a therapeutic dilemma, but the glomerular disease and the infection must be considered and managed as separate comorbid diseases. Proceeding with immunosuppressive or anti-inflammatory treatment without biopsy confirmation for their specific indications imposes the potential for greater therapeutic risk in dogs with active or sequestered infection based on positive serology compared to dogs treated without an infectious comorbidity. Conversely, dogs with peracute and rapidly progressive disease associated with seropositivity (ie, Lyme-associated nephritis) often die from consequences of their glomerular disease within days of diagnosis and require immediate and aggressive immunomodulatory treatment in addition to supportive treatment for renal failure.

Assessment and Management of Specific Infection-Associated Glomerular Diseases

Lyme-Associated Glomerular Disease

In North America, glomerular disease associated with seropositivity for *Borrelia burgdorferi* (Lyme nephritis) may be seen and may be severe and progressive.⁴ There is no experimental model for Lyme nephritis, and the exact pathogenesis and natural progression of this entity is unknown. Most (95%) Lyme-seropositive dogs show no signs of illness (neither Lyme arthritis nor Lyme nephritis). Nonclinical seropositive dogs may have very high quantitative C6 antibody levels with circulating immune complexes and it is unknown why or how a subset of dogs with Lyme-specific circulating immune complexes develop Lyme nephritis. Because certain breeds are predisposed to Lyme nephritis (eg, Labradors and Golden Retrievers), it may be that an underlying genetic podocytopathy¹⁰ or immunodysregulatory defect causes illness with deposition of Lyme-specific antigen-antibody complexes in glomeruli, which in severe forms is accompanied by tubular degeneration and necrosis. There may be earlier stages or milder forms of the disease that are treatable with antimicrobials and standard treatment alone. However, as indicated above, progressive or severe disease does not respond to these general therapies, and additional aggressive immunotherapy appears warranted. Prospective clinical trials need to be performed to find the best therapeutic approaches for these cases.

Definitive evidence for the length of doxycycline treatment needed to eliminate *Borrelia burgdorferi* is lacking. One month of doxycycline at 20 mg/kg/d only cleared 85% of experimentally infected dogs, based on culture and PCR of skin biopsies taken from the area where ticks were placed on the dogs.^{11–14} The criterion for effective treatment usually includes response to

treatment based on improving clinical signs and laboratory findings. For Borreliosis, baseline and 6-month posttreatment assessment of the quantitative C6 titer may help assess the effectiveness of treatment. Successful treatment may be defined as a $\geq 50\%$ decrease from pretreatment values.¹⁵ However, anecdotally when the initial Lyme C6 titer is relatively low, the decline in titer observed with treatment may be small. Additional anecdotal observations indicate the qualitative SNAP-4Dx results often remain positive for years, despite treatment, but the Lyme quantitative C6 titers usually wane, sometimes to less than 30, with treatment even when the SNAP-4Dx C6 peptide antibody test still documents a positive result. Anti-infective treatment for *Borrelia burgdorferi* with doxycycline is typically initiated at 10–20 mg/kg/d; however, the optimal dose and length of treatment for *Borrelia burgdorferi* infection is unknown, but may be greater than 1 month of doxycycline at 20 mg/kg/d per os. Consequently, in cases of suspected Lyme-associated nephropathy, doxycycline treatment perhaps should be continued until clinical resolution or until the C6 quantitative titer is determined to have declined to less than 50% from pretreatment values, which may take up to 6 months.

Leishmaniasis-Associated Glomerular Disease

Diagnosis of leishmaniasis in dogs should be based on consideration of signalment, history, clinical findings, and results of testing to detect the parasite or the immune response of the host. Generally, leishmaniasis can be confirmed by cytologic, serologic, or PCR analyses in dogs with overt clinical signs or severe alterations of relevant clinicopathologic variables.¹⁶ Any breed can be affected by leishmaniasis, which occurs with a bimodal onset at <3 years of age and from 8–10 years of age. The distribution of leishmaniasis varies widely throughout the world, and it is essential to determine if suspect cases have a known travel history to an endemic area of phlebotomine vectors. Consistent clinical features include lymph node enlargement, mucocutaneous lesions, poor nutritional state, hepatosplenomegaly, lameness, joint swellings, and fever.

Specific diagnosis is confirmed by direct cytologic, histologic, or PCR identification of the agent in aspirates or biopsies of affected sites or indirectly by serologic testing and evaluation of cellular immune responses. When injured tissues (including bone marrow) have direct evidence of *Leishmania* infection, dogs should be considered as affected by leishmaniasis, regardless of serologic results. When there is no direct evidence of *Leishmania* spp. infection, serologic testing such as IFAT and ELISA is crucial for diagnosis. An antibody titer above the cutoff of reference value for the laboratory with concurrent proteinuria confirms the present of disease.

Anti-*Leishmania* drugs include antimonial compounds, aminosidine (paromomycin), miltefosine, and allopurinol. For dogs merely exposed to infection, no specific treatment is necessary. Dogs should be monitored serologically for 2–4 months after the finding

antibodies against *Leishmania* spp. at low titers. If characteristic clinical findings develop, additional evaluation is indicated, including parasitological testing performed with a direct method. Dogs with documented infection on the basis of direct detection of the organism or a positive antibody titer require specific antileishmania treatment. Significant comorbidities associated with infection require appropriate ancillary treatment in addition to anti-leishmania drugs. Combination treatment with meglumine antimoniate and allopurinol is the most widely used anti-leishmania treatment protocol and can be administered to dogs with confirmed infection regardless of the severity or clinical stage of the disease.¹⁷ Meglumine antimoniate is given at 100 mg/kg SC once daily for 4 weeks; and allopurinol is provided at 10 mg/kg PO every 12 hours for at least 6 months. The dosage of meglumine antimoniate can be divided into 2 equal doses of 50 mg/kg administered SC every 12 hours for 4–8 weeks.

Rarely, *Leishmania* infections are recognized in the United States where meglumine antimoniate is not FDA approved for use in dogs, and alternative treatment protocols may be required. Allopurinol is well tolerated and seems to slow the deterioration of renal function in nonazotemic dogs with proteinuria but can predispose to xanthine urolithiasis.¹⁸ For most dogs, the combination of meglumine antimoniate and allopurinol promotes a clinical cure that is sustained for longer than 1 year. This combined treatment decreases the parasite load for several months, which is necessary to reduce transmission of the parasite by phlebotomine vectors. Clinical improvement is highly contingent on the degree and severity of comorbid complications.^{17,19}

For dogs that remain unresponsive to treatment after a confirmed diagnosis and appropriate administration of meglumine antimoniate and allopurinol alternative anti-leishmania treatment should be considered. The combination of miltefosine administered PO at 2 mg/kg every 24 hours for 28 days and allopurinol, at the same dosage as recommended with meglumine antimoniate, appears promising and as effective as the combination of meglumine antimoniate and allopurinol.²⁰ However, miltefosine also is not FDA approved for use in dogs in the United States.

Footnotes

^aAbsent clinical or other evidence supporting active disease in the patient (serologic positivity failing to differentiate between infection and exposure).

^bTier I, see “Consensus Recommendations for the Diagnostic Investigation of Dogs with Suspected Glomerular Disease” in this Supplement.

^cSee “Consensus Recommendations for Immunosuppressive Treatment of Dogs with Glomerular Disease Based on Established Pathology” and “Pathologic Evaluation of Canine Renal Biopsies: Methods for Identifying Features That Differentiate Immune Mediated Glomerulonephritides from Other Categories of Glomerular Diseases” in this Supplement.

^dSee “Recommendations for Immunosuppressive Treatment in Dogs with Glomerular Disease Absent a Renal Pathologic Diagnosis” in this Supplement.

Acknowledgment

Conflict of Interest Declaration: The authors disclose no conflict of interest.

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Pathologic Evaluation of Canine Renal Biopsies: Methods for Identifying Features that Differentiate Immune-Mediated Glomerulonephritides from Other Categories of Glomerular Diseases

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Background: Human renal biopsies are routinely evaluated with light microscopy (LM) using a panel of histologic stains, transmission electron microscopy (TEM), and immunofluorescence (IF) microscopy to obtain a diagnosis. In contrast, the pathologic evaluation of glomerular disease in veterinary medicine has relied mostly on LM and was of limited utility. To address this problem, recently established veterinary renal diagnostic centers have adopted methods used in human nephropathology for evaluation of renal biopsies. Three broad categories of disease, which have the greatest implications for clinical management of proteinuric dogs, have been established and include amyloidosis, immune complex-mediated glomerulonephritis (ICGN), and non-ICGN.

Objective: To demonstrate histopathologic, ultrastructural, and IF findings in renal biopsy specimens that experienced veterinary nephropathologists utilize to make accurate and clinically useful diagnoses in dogs with proteinuric glomerular disease and to provide guidelines for the proper evaluation of renal biopsies.

Methods: Renal biopsy specimens were routinely examined by LM, IF, and TEM. Samples were reviewed by members of the World Small Animal Veterinary Association Renal Standardization Study Group to identify lesions that were diagnostic for, or suggestive of, the presence of immune complexes (IC) or amyloidosis in all modalities. Ten guidelines for renal biopsy evaluation were formulated.

Results: Each method of investigation contributed important findings that were integrated to make an accurate final morphological diagnosis. The guidelines were validated by an independent group of veterinary pathologists.

Conclusions and Clinical Importance: Routine evaluation of renal biopsies with LM, TEM, and IF is feasible and necessary for making accurate, morphologic diagnoses that can be used to guide clinical management of dogs with glomerular disease.

Key words: Canine; Glomerulonephritis; Immune complex; Immunofluorescence; Proteinuria; Transmission electron microscopy.

Although human renal biopsies are routinely evaluated with light microscopy (LM) using a panel of special stains, transmission electron microscopy (TEM), and immunofluorescence (IF) to obtain a specific and clinically useful diagnosis, the evaluation of glomerular disease in veterinary medicine has often relied on LM alone, which has been of limited utility.

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Abbreviations:

CR	Congo red
GBM	glomerular basement membrane
HE	hematoxylin and eosin
ICGN	immune complex glomerulonephritis (nephritides)
IC	immune complex(es)
IF	immunofluorescence microscopy
IVRPS	International Veterinary Renal Pathology Service
JMS	Jones methenamine silver
LM	light microscopy
MT	Masson's trichrome
Non-ICGN	nonimmune complex glomerulonephropathy (ies)
PASH	periodic acid Schiff hematoxylin
TEM	transmission electron microscopy
UVNS	Utrecht Veterinary Nephropathology Service
WSAVA-RSSG	World Small Animal Veterinary Association – Renal Standardization Study Group

Since 2008, two collaborating centers, one located in North America and the other in Europe, have routinely performed these specialized diagnostic evaluations (LM, TEM, and IF) on all adequate samples of renal tissue submitted for pathologic evaluation. The North American center was initially organized at Texas A&M University and was known as the Texas Veterinary Renal Pathology Service. Recently, it has been reorganized as a cooperative effort between the Ohio State University and Texas A&M University and is now known as the International Veterinary Renal Pathology Service (IVRPS). The European center is

located at Utrecht University in the Netherlands and is known as the Utrecht Veterinary Nephropathology Service (UVNS). By the middle of 2013, the combined cumulative experience of both centers exceeded 960 canine cases. One goal of these centers is to evaluate a sufficient number of specimens annually to develop and maintain the technical and interpretive expertise needed to examine renal specimens with these specialized methods and to reliably make correct diagnoses. In human nephropathology, it has been suggested that the number of cases required for this purpose is 200 per year.¹

The veterinary pathologists associated with these centers have developed expertise in the evaluation of each modality. These diagnostic centers have not only created a strong foundation for the diagnosis of canine glomerular disease but they also have revealed the deficiency of each modality when used alone. Illustrations of lesions that are definitively diagnostic for, suspicious for, or inconsistent with 3 main categories of glomerular disease (ICGN, amyloidosis, and non-ICGN) are the main goals of this publication. These 3 broad diagnostic categories were created because they have the greatest implications for prognosis and treatment.

Materials and Methods

Cases were evaluated retrospectively by the World Small Animal Veterinary Association Renal Standardization Study Group (WSAVA-RSSG) from the pool of more than 900 canine renal tissue specimens to identify representative cases that clearly demonstrate features, which are definitively diagnostic for, suggestive of, or inconsistent with ICGN, amyloidosis, or non-ICGN. Only specimens that had sufficient tissue for evaluation by all modalities (LM, TEM, and IF) were examined, except for some cases in which amyloidosis was diagnosed via LM. Tissues for LM were sectioned at 3- μ m thickness and stained with hematoxylin and eosin (HE), Masson's trichrome (MT), or periodic acid Schiff hematoxylin (PASH). Most tissue sections were also stained with Jones methenamine silver (JMS) and Congo red (CR) the latter of which was performed on 8-micron-thick sections.

Transmission electron microscopy specimens were fixed in chilled 3% buffered glutaraldehyde and then postfixed in 1% osmium tetroxide, serially dehydrated, infiltrated in an acetone/epoxy plastic, and then embedded in a plastic mold. Plastic blocks were cut with an ultramicrotome^a and thick sections were stained with toluidine blue to identify optimal areas for thin sectioning. Thin sections were cut at 55–60 nm and placed on copper grids^b and stained with uranyl acetate and lead citrate. They were examined in a transmission electron microscope^c and representative digital images were captured by the authors (FJC, CAB, and LA) at the Texas Heart Institute, University of Georgia, and University of Padova, respectively.

For IF, unfixed tissue samples were embedded in Optimal Cutting Temperature compound^d and snap-frozen in liquid nitrogen vapor. Tissues were cryosectioned at 4- μ m thickness, and sections were labeled with an appropriate dilution of fluorescein isothiocyanate-labeled anti-IgG,^e anti-IgM,^e anti-IgA,^e anti-C1q,^f anti-C3,^e anti-Lambda light chains,^f and anti-Kappa light chains^f antibodies. Labeling patterns and fluorescent intensity were evaluated by the authors (GEL and JvdL) with an epifluorescence microscope^g using appropriate filters and digital imaging.

High resolution, whole-section digital images of all LM slides were obtained with a slide scanning instrument.^h The scanned LM sections and the digital images of the TEM and IF evaluations were stored on a server where the digital images were managed by digital pathology softwareⁱ that delivers the images via the internet to participating pathologists for viewing^{i-k} either independently or jointly during online^l case-review conferences. In each modality, features diagnostic for, suggestive of, or inconsistent with the presence of IC deposits or amyloid were identified, discussed, and annotated as such. These features were used for the categorization of ICGN, amyloidosis, or non-ICGN.

In addition, the WSAVA-RSSG formulated guidelines for the pathologic evaluation of renal tissue. These guidelines were then distributed among the RSSG and also sent to 5 additional non-RSSG members for external validation. This procedure involved voting (strongly agree, agree, neutral, disagree, and strongly disagree) on each guideline using an anonymous ballot and allowed validators to provide written comments. After ballot submission, the guidelines with greater than 75% agreement were considered to be validated.

Results

ICGN

Definitive diagnosis of ICGN depends on ultrastructural findings of electron-dense immune deposits in subepithelial, subendothelial, intramembranous, or mesangial locations, with many cases also demonstrating IC in multiple locations within the glomerular capillary tuft (Fig 1A,B). Other electron-dense material that is not IC can also be identified with TEM (eg, hyalinosis), but the appearance of these substances differs from IC (Fig 3E,F). Care must be taken by the pathologist to correctly interpret these findings. In our experience to date, glomerular IC deposition is diffuse (ie, affecting every glomerulus) in dogs. Specifically, in cases of canine ICGN where multiple glomeruli were available for evaluation by EM and IF, IC were present in all patent glomerular tufts.

Immune deposits are further characterized by immunofluorescent labeling with antibodies against immunoglobulins and complement components (Fig 1H). A granular, global staining pattern of glomeruli clearly associated with capillary walls, the mesangium, or both with antibodies against C3 and at least 1 class of immunoglobulin is diagnostic of IC, whereas splotchy, segmental staining is considered nonspecific (Fig 3H). Inexperience in the evaluation of IF is cited as the main cause of diagnostic dilemmas in the evaluation of renal biopsies by physician nephropathologists and, therefore, staining patterns should be interpreted with care.²

Light microscopy can reveal a wide variety of lesions in ICGN; some of which provide a presumptive diagnosis before TEM or IF results become available. The presence of uniform, prominent remodeling (spikes, holes, or double and irregular contours) of the glomerular basement membrane (GBM) in most glomeruli is highly suggestive of ICGN. Jones Methenamine Silver is the preferred technique for the evaluation of these features (Fig 1E,F), although they can also be observed with PASH (Fig 1C,D). Notably,

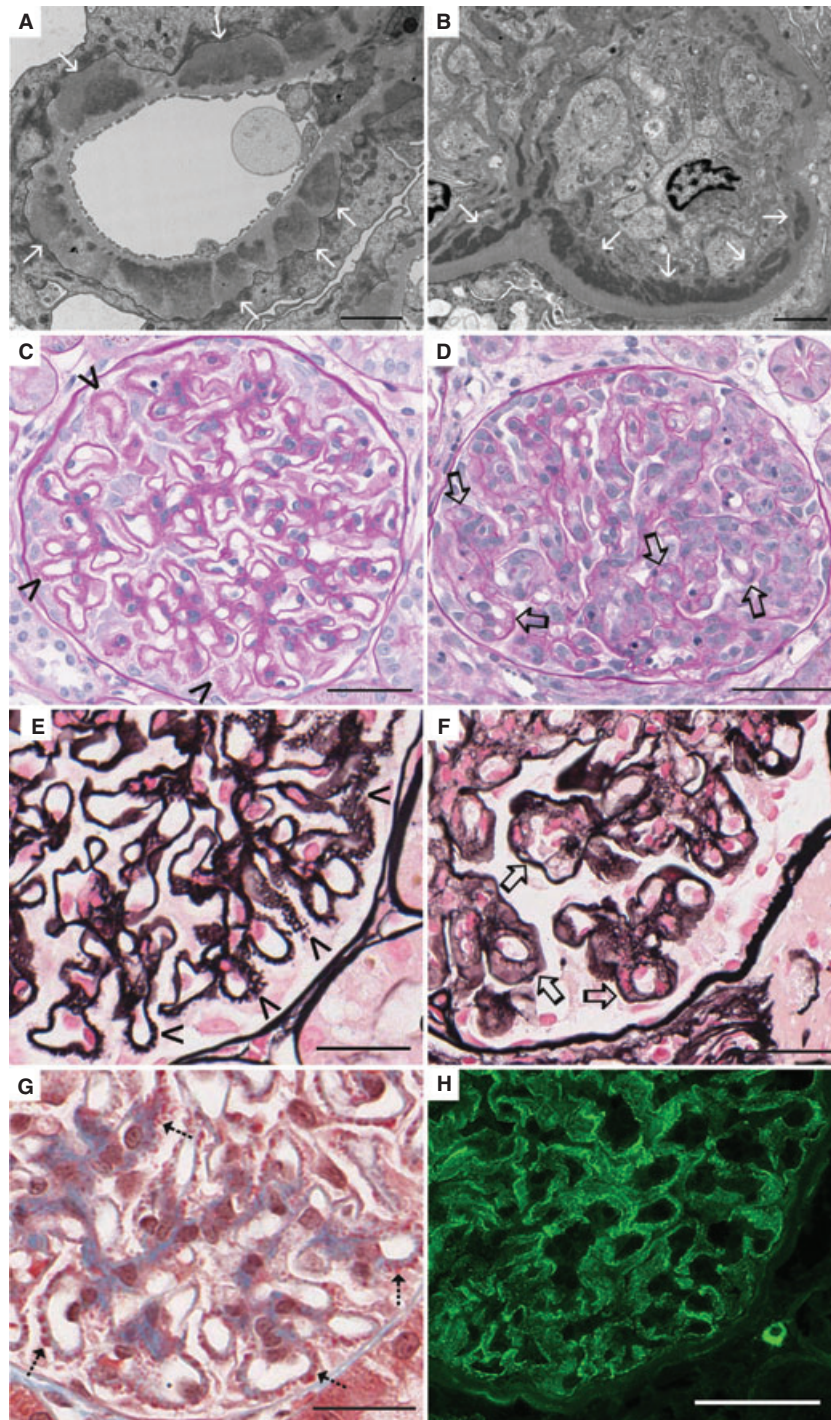


Fig 1. Histologic, ultrastructural, and immunofluorescent findings in cases of ICGN. **(A, C, E, G, H)** Renal biopsy from a 6-year-old castrated male English Setter dog with proteinuria (UPC = 6.6), but not azotemia. **(A)** Electron microscopy reveals immune complexes (IC) (white arrows) in the abluminal side (subepithelial) of the capillary wall. Scale bar = 2 μ m. **(C, E)** Periodic acid Schiff hematoxylin (PASH) and Jones methenamine silver (JMS) stains of a glomerulus, respectively, reveals remodeling of the glomerular basement membrane (GBM) characterized by spike-like projections (arrowheads) on the subepithelial surface. Scale bar = 50 μ m. **(G)** Masson's trichrome stain demonstrates numerous IC on the subepithelial surface (dotted arrows). Scale bar = 50 μ m. **(H)** Immunofluorescence staining with an antibody against Lambda light chain demonstrates the presence of immunoglobulins along capillary walls. Scale bar = 50 μ m. Staining for C3 and IgG revealed a similar pattern (not shown). **(B, D, F)** Renal biopsy from a 3-year-old castrated male Boxer dog with proteinuria and azotemia (UPC = 12.4, serum creatinine 4.4 mg/dL). **(B)** Electron microscopy reveals IC (white arrows) along the luminal side (subendothelial) of the capillary wall. Scale bar = 2 μ m. **(D, F)** PASH and JMS stains of a glomerulus, respectively, reveals remodeling of the GBM characterized by double contours of the GBM (open arrows). Glomeruli also have global endocapillary and mesangial hypercellularity. Scale bar = 50 μ m.

uniform thickening of the GBM alone without evidence of spikes, holes, or double or irregular contours is not sufficient for a presumptive diagnosis of ICGN, because it can be seen in cases of non-ICGN. Likewise, the lack of GBM remodeling viewed by LM does not rule out ICGN, especially in cases of relatively recent onset of proteinuria, as those cases have had less time for modification in the GBM. In those cases, TEM and IF are required for the correct diagnosis. Masson's trichrome may reveal the presence of large immune complexes, especially those in a subepithelial location (Fig 1G). Occasionally, IC can also be observed on the luminal side of the GBM with this stain.

Glomerular endocapillary hypercellularity is suggestive of, but not diagnostic for, ICGN. A diffuse and global increase in numbers of nucleated cells, either circulating polymorphonuclear leukocytes or native cell types (endothelial and interposed mesangial cells) within capillaries (ie, endocapillary hypercellularity), is often caused by the presence of subendothelial or intramembranous IC (Fig 1D). Focal or segmental endocapillary hypercellularity and mesangial hypercellularity can have multiple etiologies that may or may not be associated with IC and are, therefore, non-specific (Fig 3G). Importantly, ICGN with predominantly subepithelial deposits is usually not associated with endocapillary hypercellularity.

Findings inconsistent with a diagnosis of ICGN are the absence of IC on TEM, IF, or both. There are no LM findings that definitively rule out ICGN. Furthermore, LM is required for assessment of the distribution and severity of the disease, as well as for evaluation of the degree of glomerular and tubulointerstitial scarring.

Amyloidosis

Definitive diagnosis of amyloidosis requires the presence of green birefringent material in CR-stained sections (cut at 8- to 10- μ m thickness) when viewed with polarized light (Fig 2C). The CR-stained material is peach to orange when viewed with regular nonpolarized LM (Fig 2B). Although TEM is usually not required for definitive diagnosis, ultrastructural evaluation reveals distinct fibrillar material (9–11 nm in diameter) in the mesangium and GBM (Fig 2A). The degree of effacement of glomerular capillary lumens is the basis for determining the severity of the lesion. In addition, similar Congoophilic material may be present in the interstitium (often medullary) or in arterial/arteriolar walls.

Light microscopy lesions suggestive of amyloidosis include expansion of mesangial zones with or without thickening of the walls of capillary loops by material that is eosinophilic on HE, pale pink on PASH (Fig 2D), a mix of pale blue to pale orange on MT (Fig 2F), and does not take up the silver on the JMS (Fig 2E). The HE stain can be unreliable for this diagnosis because there is significant overlap in the eosinophilia of amyloid and sclerotic collagen; how-

ever, foci of sclerosis will take up silver on the JMS enabling differentiation between these two processes. Furthermore, nonamyloid, noncongoophilic fibrillary deposits, with similar tinctorial qualities on PASH, MT, HE, and JMS stains, have been identified in canine glomeruli,^{3,4} and therefore a CR should always be performed for a definitive diagnosis.

There are no LM features, which rule out a diagnosis of amyloid. Importantly, cases of early or mild amyloidosis may have very small or subtle amyloid deposits, which are only visible by TEM.

Non-ICGN

Non-ICGN is a broad category of glomerular lesions in which neither IC nor amyloid can be identified and is therefore often a diagnosis of exclusion. The non-ICGN group includes, but is not limited to, cases of focal segmental to global glomerulosclerosis, abnormalities of the GBMs without associated sclerosis, glomerular lipidosis, and congenital or developmental nephropathies that involve glomeruli. Segmental glomerulosclerosis is solidification of a portion of the peripheral capillary loops by extracellular matrix, which is positively stained by MT, PASH, and JMS (Fig 3A–D,G).

Definitive diagnosis of non-ICGN requires ultrastructural and IF evaluation of the tissue to rule out the presence of IC. As mentioned previously, the pathologist's experience in these advanced diagnostic modalities is essential to making the correct diagnosis. The presence of electron-dense material without the ultrastructural characteristics of ICs in the mesangium or GBM may be confusing. For example, lipid can be electron-dense and is present in various types of glomerular injury (Fig 3E).⁵ Likewise, hyalinosis, which is the result of plasma insudation into the GBM, is also electron-dense, but usually well circumscribed and finely granular or glassy (Fig 3F). Furthermore, hyalinosis may stain nonspecifically with IF (so-called trapping) in areas of glomerulosclerosis (Fig 3H).

As this is a broad category, there are a wide variety of LM lesions in these patients. Importantly, smooth outer contours of the GBM and lack of double contours with the JMS method (Fig 3C) suggests the absence of IC deposition in the GBM, but this is only valid in cases of chronic, sustained proteinuria where one would expect some degree of GBM remodeling. However, there are few studies examining serial biopsies from dogs with ICGN,^{6,7} so the duration of time required for GBM remodeling to occur is currently unknown. Glomerular hypercellularity and expansion of the mesangial zones are highly variable in dogs with non-ICGN (Fig 3G) and their presence or absence does not assist in the categorization of these cases.

The presence of IC precludes a diagnosis of non-ICGN. In addition, glomeruli that are normal via LM, EM, and IF should not be classified as non-ICGN. In those rare cases, an alternate etiology or explanation of the proteinuria should be investigated.

Validation of Guidelines for the Evaluation of Renal Biopsies

Based on our experience evaluating canine renal biopsies, the WSAVA-RSSG created 10 guidelines for the processing and evaluation of such biopsies (Box 1). All 10 guidelines surpassed the 75% cut-off for validation by the RSSG and 5 external (non-RSSG) veterinary pathologists. Specifically, 4 guidelines received 100% agreement and 6 received 90% agreement.

Box 1

Consensus Recommendations for the Pathologic Evaluation of Renal Biopsies from Dogs with Suspected Glomerular Disease

- 1 For optimal evaluation, renal biopsy specimens should be sent to a diagnostic center capable of evaluating the tissue by the 3 modalities of light microscopy (LM), transmission electron microscopy (TEM), and immunofluorescence (IF) microscopy with interpretation by veterinary pathologists experienced in nephropathology.
- 2 All specimens must be sectioned at 3 μ m or less, except for the Congo Red section, which should be 8–10 μ m thick. Special stains must be performed to highlight the glomerular basement membrane (Periodic acid schiff hematoxylin [PASH] and Jones methenamine silver or Periodic acid-methenamine silver) and to identify immune deposits and matrix collagen deposition (Masson's trichrome).
- 3 Tissue should be sectioned serially (2–3 sections per slide), so that focal, segmental lesions are not missed. All glomeruli should be evaluated in a core biopsy and at least 50 glomeruli should be evaluated in wedge specimens.
- 4 The glomerular basement membrane (GBM) is best visualized in the peripheral capillary loops of the glomerular tufts and should be carefully examined for a normal smooth outer contour. Signs of GBM remodeling include spike-like projections, holes, double contours, irregular outer contours, or generalized thickening without the previously mentioned features. Thickened capillary loops alone are not pathognomonic for immune complex glomerulonephritis (ICGN). If there is no LM evidence of GBM remodeling or immune complex deposition, a comment regarding the insensitivity of this method should be included in the preliminary interpretation, especially for cases with a recent onset of proteinuria.
- 5 Glomerular hypercellularity should be specified as to location (mesangial, endocapillary, or both), cell type, and severity.
- 6 Synechiae (adhesions between the tuft and Bowman's capsule) and GBM hyalinosis (insudation of plasma into the capillary wall) should be evaluated.
- 7 Severity of interstitial fibrosis, inflammation, and tubular atrophy must be evaluated. Arterio- and arteriosclerosis, as well as arterial and arteriolar hyalinosis (changes indicative of hypertension) should be assessed.
- 8 Immunofluorescence staining is preferable to immunohistochemistry because of its greater sensitivity, and IF should be performed for immunoglobulin heavy chains (IgG, IgM, IgA) as well as the complement component, C3. Additional IF stains that can support a diagnosis of ICGN include lambda light chains and C1q. Evidence of ICGN is provided by finding granular staining with C3 and at least 1 immunoglobulin along peripheral capillary loops, mesangium, or both. The staining pattern (granular, linear, spotty), staining intensity, and location for each immuno-reactant should be assessed. IF findings are sometimes equivocal and must be verified using TEM.

Segmental, spotty staining indicates trapping of plasma constituents in areas of segmental sclerosis or hyalinosis and therefore is not diagnostic of ICGN even if it appears to be strongly positive. As there may be nonspecific staining patterns, experience in the evaluation of IF is essential for proper interpretation of the findings.

- 9 Transmission electron microscopy should be performed on either formalin- or glutaraldehyde-fixed tissue. If patent glomeruli cannot be identified in samples processed for TEM, tissue with glomeruli can be removed from the paraffin block for reprocessing for TEM, although there will be mild artifacts associated with this procedure. Ultrastructural evaluation should be performed on glomeruli that are not undergoing atrophy or global glomerulosclerosis. The peripheral capillary loops and mesangium should be examined for the presence or absence of IC, amyloid, or other fibrils. The GBM should be assessed for remodeling or splitting. Depending on the stage of the disease, ICs range from electron dense to electron lucent. Location of deposits (subendothelial, subepithelial, intramembranous, mesangial) should be determined. Podocyte injury and foot process effacement should also be evaluated. As with IF evaluation, experience in the interpretation of TEM findings is crucial to the diagnostic workup of canine renal biopsies.
- 10 The written report of the findings should include the number of glomeruli evaluated, the number (or percent) of globally sclerotic glomeruli, whether the distribution of glomeruli with lesions is less than 50% (focal) or greater than 50% (diffuse), and the degree of lesion involvement in the glomerulus as either occupying a portion of the glomerular tuft (segmental) or all of the tuft (global). A description of salient glomerular features as evidenced by all 3 diagnostic modalities (as specified in the previously stated guidelines) should be included in the report. Finally, an assessment of pathologic changes in the tubulointerstitial and vascular compartments should also be included as the condition of these compartments is important in formulating an overall prognosis.

Discussion

The accumulation of prognostic information and the development of optimal treatment strategies in humans with glomerular disease has relied upon the integration of clinical history, clinicopathologic data, and accurate, detailed morphologic diagnoses achieved through the routine use of LM, IF, and TEM. To facilitate treatment and monitoring of canine and feline patients, veterinary nephrologists currently use the International Renal Interest Society staging system for chronic kidney disease. This system utilizes serum creatinine, urine-specific gravity, and renal palpation or diagnostic imaging, and patients are then substaged based on the degree of proteinuria or hypertension. Notably, the definitive pathologic diagnosis is not factored into this system. Furthermore, these guidelines only apply to patients with chronic, stable renal disease.⁸ Therefore, four sets of recommendations for treatment of dogs with proteinuric kidney disease have been recently formulated and validated by veterinary nephrologists.⁹ One of these sets of recommendations is for dogs that have undergone renal biopsy, and determination of the presence or absence of IC is vital to the algorithm. Separation of ICGN from

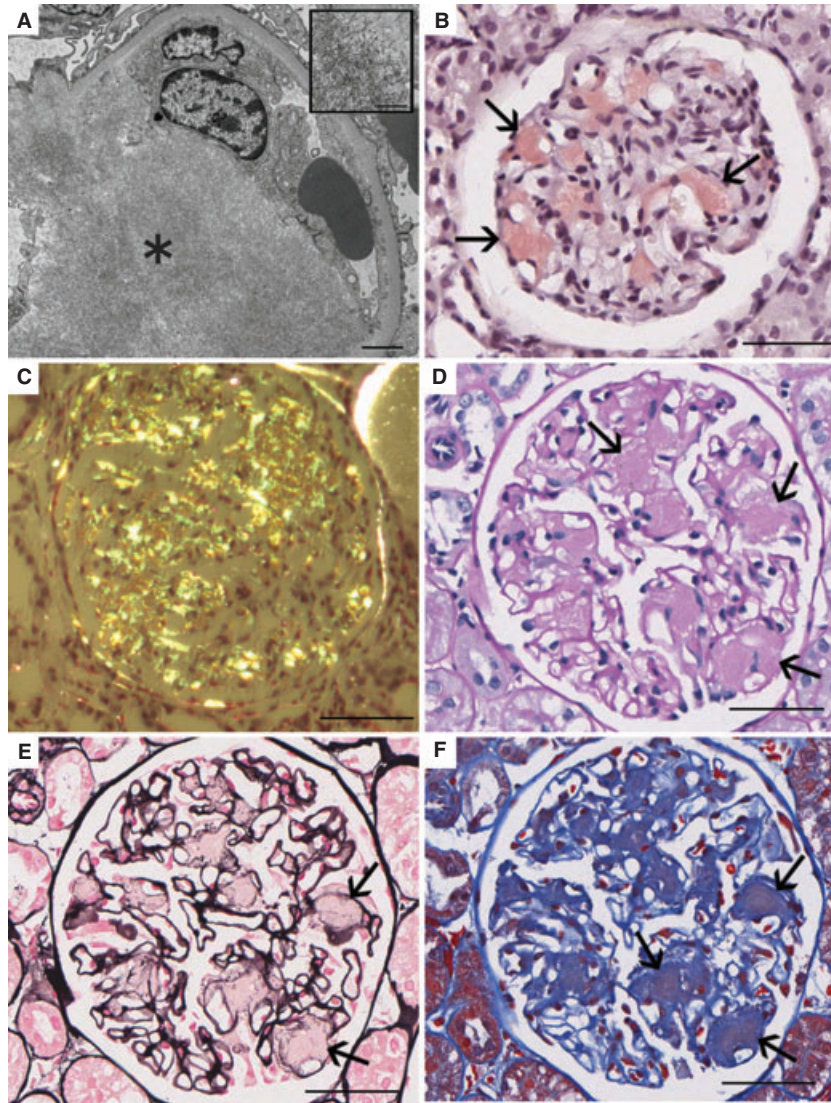


Fig 2. Histologic, ultrastructural, and immunofluorescent findings in amyloidosis. A renal biopsy from a 13-year-old castrated male small mixed breed dog with proteinuria (UPC = 9.8), but not azotemia. **(A)** Electron microscopy reveals expansion of the mesangial zone by haphazardly arranged fibrils. Scale bar = 2 μ m. Inset: Higher magnification of fibrils. Scale bar = 250 nm. **(B–F)** Histologic appearance of a glomerulus. Mesangial zones are expanded by Congophilic peach material **(B)** that demonstrates apple green birefringence when viewed with polarized light **(C)** and is consistent with amyloid (arrows). This material (arrows) is pale pink on periodic acid Schiff hematoxylin **(D)**, does not take up silver with the Jones methenamine silver method **(E)**, and is a mix of blue-orange on Masson's trichrome **(F)**. Scale bar = 50 μ m.

non-ICGN by LM can be difficult, as demonstrated above, and both are common causes of proteinuria in dogs.¹⁰ Even so, the pathologist must provide an accurate diagnosis, so that future evaluations of treatment efficacy are possible.

Before the establishment of the IVRPS and UVNS, the advanced modalities of renal biopsy evaluation were difficult to coordinate for a feasible diagnostic workflow. Because of this, proteinuric animals were generally diagnosed via LM alone by veterinary pathologists without advanced training in EM or IF. In many instances, clinicians felt that the expense and the potential risks of the renal biopsy procedure outweighed the benefits of having a definitive pathologic

diagnosis, especially when LM alone can be misleading. Our adoption of—and familiarity with—methods used by our physician nephropathology colleagues has enabled us to establish 3 broad categories of canine proteinuric kidney disease: ICGN, amyloidosis, and non-ICGN. As stated previously, these categories were created because they have the greatest implications for treatment and prognosis.

A diagnosis of amyloidosis can often be made solely by LM; however, the distinction between ICGN and non-ICGN is difficult because their LM lesions overlap or can be subtle depending on the stage of the disease. Specifically, cases of ICGN with recent onset proteinuria might not have had enough time for the GBM to

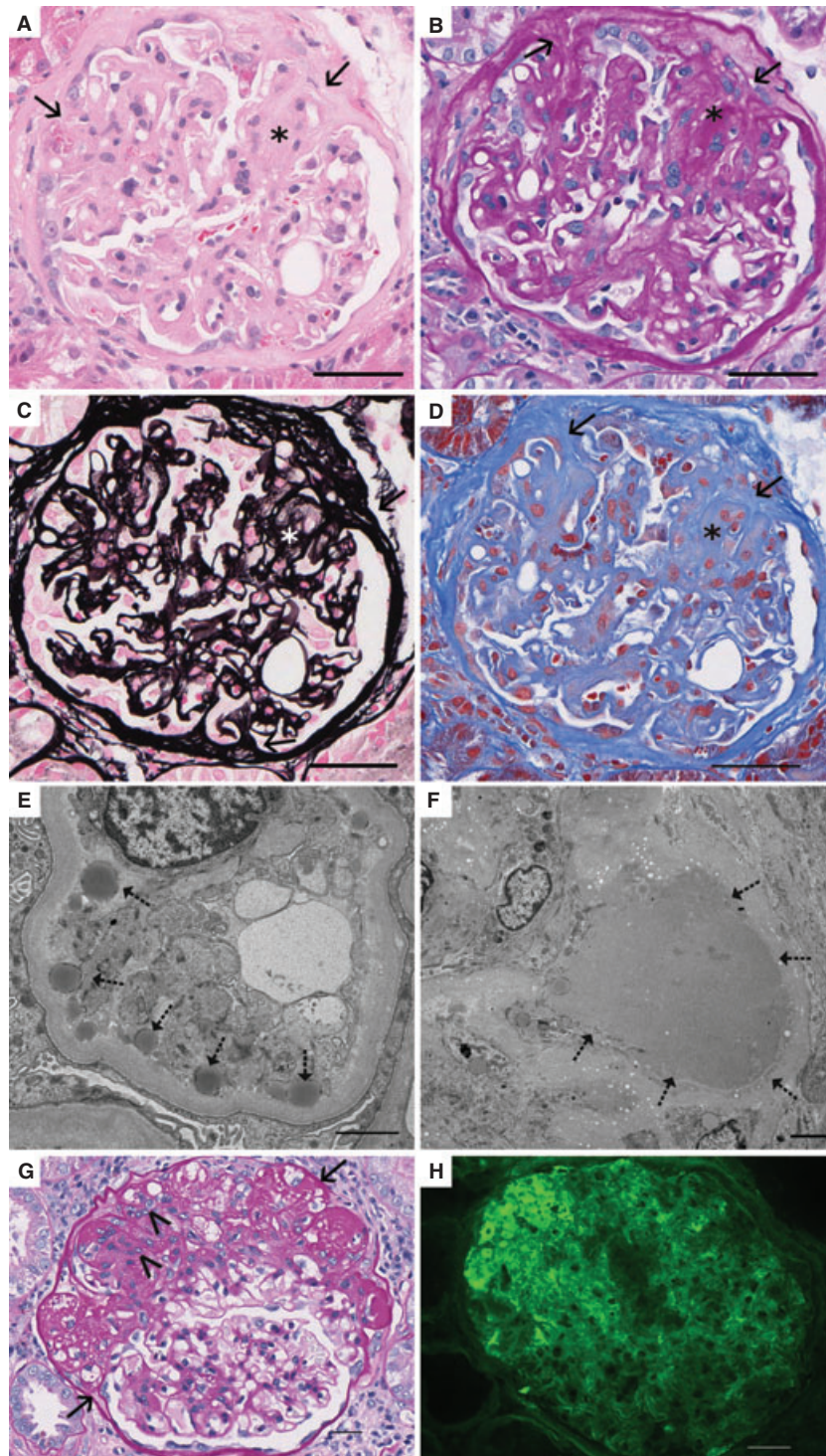


Fig 3. Histopathologic, ultrastructural, and immunofluorescence findings in non-ICGN. (A–D) hematoxylin and eosin, periodic acid Schiff (PAS), Jones methenamine silver (JMS), and Masson's trichrome stained serial sections of a glomerulus from a 12-year-old Yorkshire Terrier with severe proteinuria (UPC: 12.7). Segments of the tuft are effaced by extracellular matrix (segmental sclerosis) (*) and are adhered to Bowman's capsule (synechiae) (arrows). Glomerular basement membrane is thickened, but has a smooth outer contour. Scale bar = 50 μ m. Electron microscopy (E–F) of the same dog reveals thickened GBM with multifocal electron-dense material (dotted arrows) that is consistent with lipid (E) and hyalinosis (F). Scale bar = 2 μ m. Immune deposits were not identified via electron microscopy (EM) or immunofluorescence (IF). (G) PAS-stained glomerulus from a 4-year-old Rhodesian Ridgeback dog with proteinuria (UPC: 2.0) associated with a congenital anomaly of the kidney reveals moderate mesangial hypercellularity (arrowhead), segmental sclerosis, and a broad synechia (arrows). EM and IF ruled out the presence of immune complexes in this case. Scale bar = 50 μ m. (H) Immunofluorescence staining for IgG in a glomerulus from a 14-year-old standard poodle with proteinuria (UPC: 3.2) reveals strong, splotchy (as opposed to granular) staining in a portion of the tuft, indicative of nonspecific trapping within a sclerotic segment. EM did not reveal immune complex deposition. Scale bar = 50 μ m.

remodel in response to the deposits. In those cases, the glomerulus would look completely normal by LM. Likewise, some cases of non-ICGN demonstrate glomerular hypercellularity or thickened GBMs. Therefore, a definitive diagnosis necessitates verification of IC deposition by TEM, IF, or both. As illustrated above, these advanced modalities require specialized training for correct interpretation. For example, electron-dense IC have a specific ultrastructural appearance, which differs from that of hyalinosis, and circulating immunoglobulins can be nonspecifically trapped in areas of glomerulosclerosis. It is essential that the final morphologic diagnosis is an integration of all 3 modalities.

We recognize that some veterinary pathologists may argue against our decision to establish the category of non-ICGN because it includes such a wide range of lesions, ranging from segmental glomerulosclerosis to ultrastructurally abnormal GBM to developmental renal anomalies that affect glomeruli. Notably, the significance of glomerulosclerosis, specifically focal segmental glomerulosclerosis, as a cause of proteinuria is under-recognized in veterinary medicine, even though it is considered to be a common cause of proteinuria in humans.¹¹ Multiple lines of evidence point to podocyte injury as an inciting factor in the development of glomerulosclerosis, and mutated podocyte genes, toxic insults, viral infections, and cytoskeletal stress from cell hypertrophy have all been demonstrated to injure this cell lineage.¹² This diagnostic entity is mentioned rarely in veterinary literature, although recent research has identified candidate genes associated with this disease in certain breeds of dogs.¹³ Furthermore, IC deposition can also lead to podocyte damage and secondary sclerosis, further confounding the pathologist's ability to categorize a case by LM alone. The significance of many of the other lesions in the category of non-ICGN is currently unknown. For instance, multifocal glomerular lipidosis has been viewed as an incidental lesion, because it has been identified rarely in sections of kidneys from "normal" dogs at necropsy. The lack of recent urinalysis results from these "normal" dogs and the presence of this lesion as the only diagnostic abnormality in renal biopsy specimens from proteinuric dogs suggest that it should be investigated further.

The validated guidelines presented herein provide a starting point for the proper examination of renal biopsy tissue. Guidelines 4 through 9 mention lesions that, in our experience, are often encountered in canine renal biopsy specimens. Many of the listed lesions provide the basis for the classification of the case as ICGN, amyloidosis, or non-ICGN, whereas others (eg, synechiae) are less specific, but still carry prognostic weight. For aspiring veterinary nephropathologists and nephrologists, these features are merely the beginning of what can be gleaned from a renal biopsy specimen and studies to examine and categorize lesions in finer detail are currently under way (see below).

Importantly, the final interpretation of the lesions must be placed into the context of the patient's clinical presentation. The severity of interstitial fibrosis and

tubular atrophy must be assessed in every case, so that the clinician can determine the likelihood that renal function will deteriorate or improve. In azotemic, proteinuric dogs without histologic evidence of tubular injury, the clinician should be alerted to evaluate the areas not captured by the kidney biopsy (eg, renal medulla) using diagnostic imaging or to search for pre- and postrenal causes of azotemia. In that same vein, ICGN cases with significant scarring of the glomeruli need to be documented as such, as they may not respond to the same therapies as cases of recent onset ICGN. Lesions in arteries and arterioles can prompt a clinician to consider the role of hypertension in their patient.

Additional studies are currently under way to subdivide the 3 broad categories to aid in the identification of possible etiologies as well as to improve prognostication. In its ongoing studies, the WSAVA-RSSG examines and annotates more than 100 LM, EM, and IF features of each renal biopsy specimen. Subcategories of canine glomerular diseases exhibiting particular combinations of features are identified and associated with their respective clinical characteristics (ie, clinical findings at diagnosis, response to treatment, and outcome). Needless to say, these studies entail extensive, long-term clinical follow-up. In the meantime, dogs undergoing a renal biopsy need to be categorized correctly, so that therapeutic regimens are based on treatment of IC-mediated disease, amyloidosis, or the lack thereof. Incorrect categorization will skew efficacy and prognostic data. Furthermore, it will always be important for pathologists and nephrologists to have a systematic approach to harvest, submit, prepare, and evaluate kidney tissue. Therefore, the validated guidelines provided herein will help optimize the diagnostic workflow in veterinary nephropathology.

Footnotes

^a UltraCut S, Reichert Technologies, Depew, NY

^b Electron Microscopy Sciences, Hatfield, PA

^c JEM-1240; JEOL USA, Peabody, MA

^d Tissue-Tek OCT Compound, Sakura Finetek USA, Torrance, CA

^e Bethyl Labs, Montgomery, TX

^f Dako North America, Carpinteria, CA

^g Olympus, Center Valley, PA

^h ScanScope CS, Aperio, Vista, CA

ⁱ Spectrum, Aperio, Vista, CA

^j ImageScope, Aperio, Vista, CA

^k WebScope, Aperio, Vista, CA

^l GoToMeeting, Citrix Systems, Inc, Santa Barbara, CA

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